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- 1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
- 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
- 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K22

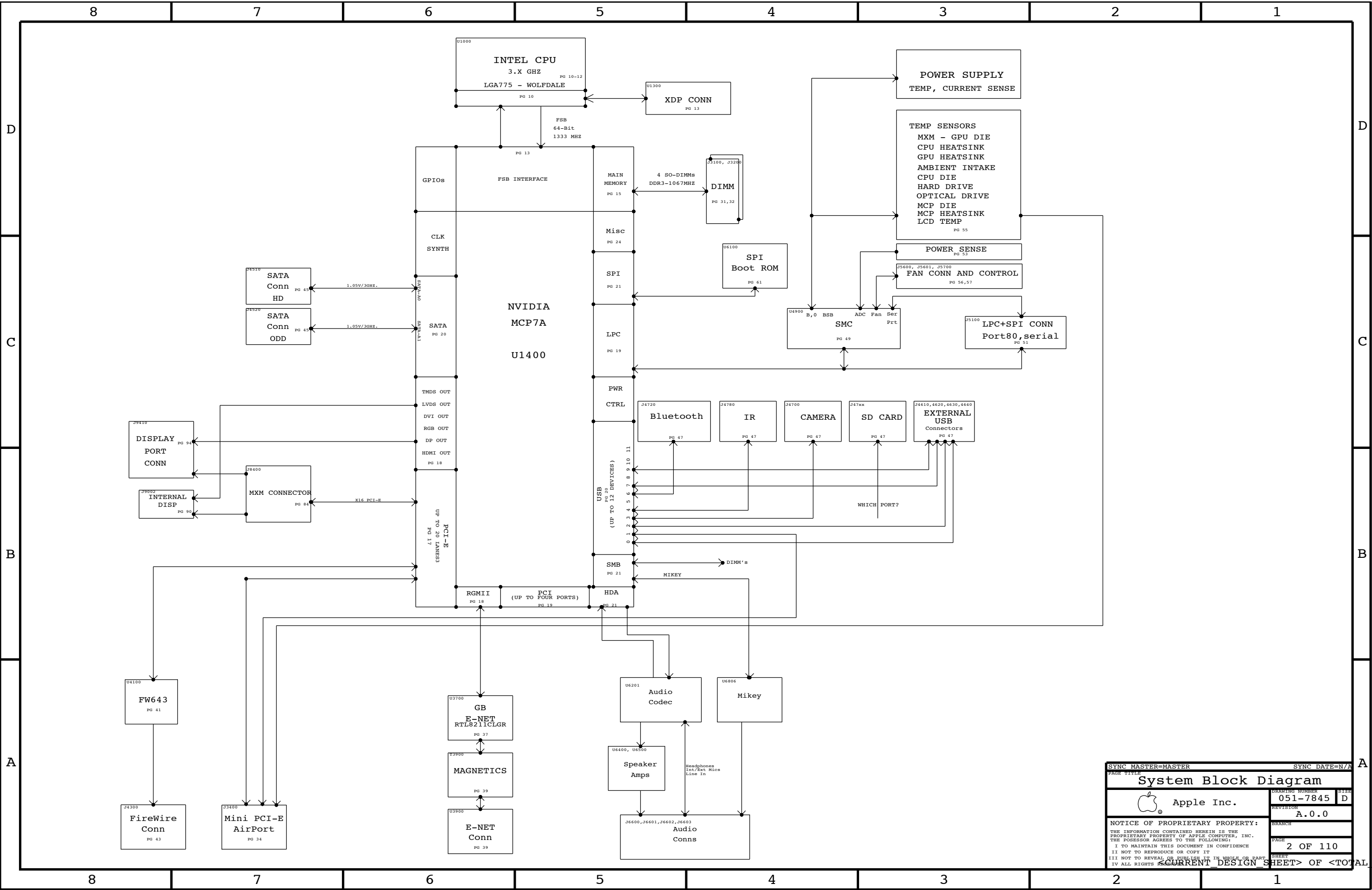
LAST_MODIFIED=Thu Aug 20 17:40:45 2009

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System Block Diagram

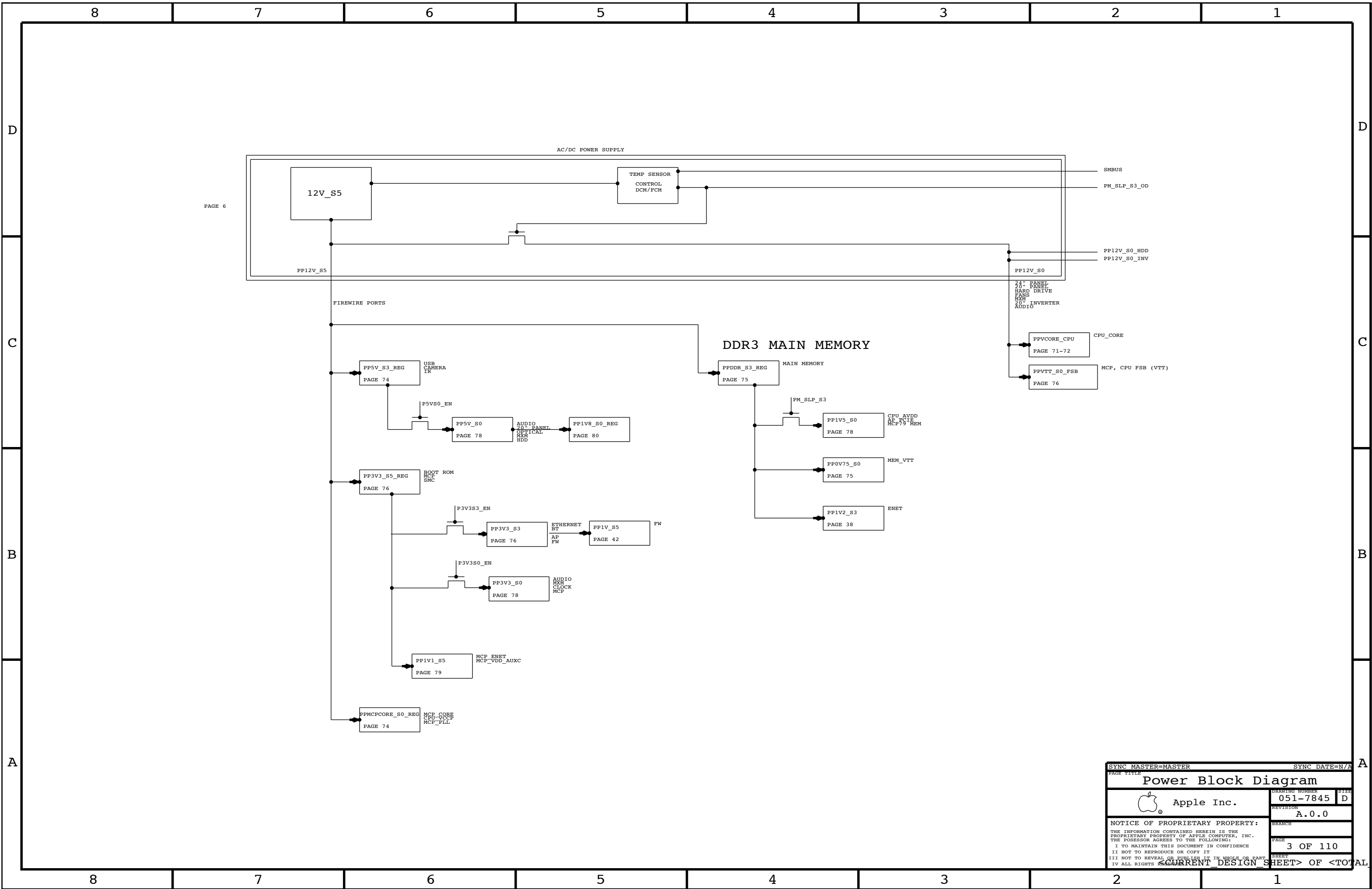
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Power Block Diagram			
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630-9768	PCBA,MLB,GOOD,K22	K22,2P80GHZ_CPU,BASIC,IG
639-0036	PCBA,2.8 GHZ CPU,MXM,K22	K22,2P80GHZ_CPU,BASIC,MXM,K22_MXM
639-0184	PCBA,2.93 GHZ CPU,IG,K22	K22,2P93GHZ_CPU,BASIC,IG
639-0186	PCBA,2.93 GHZ CPU,MXM,K22	K22,2P93GHZ_CPU,BASIC,MXM,K22_MXM
639-0037	PCBA,3.0 GHZ CPU,IG,K22	K22,3P0GHZ_CPU,BASIC,IG
630-9878	PCBA,MLB,CTO,K22	K22,3P0GHZ_CPU,BASIC,MXM,K22_MXM
639-0183	PCBA,3.06 GHZ CPU,IG,K22(Investigation)	K22,3P06GHZ_CPU,BASIC,IG
639-0324	PCBA,MLB,3.16GHZ,MXM,K22	K22,3P16GHZ_CPU,BASIC,MXM,K22_MXM
639-0206	PCBA,MLB,3.33GHZ,IG,K22	K22,3P33GHZ_CPU,BASIC,IG
639-0207	PCBA,MLB,3.33GHZ,MXM,K22	K22,3P33GHZ_CPU,BASIC,MXM,K22_MXM
639-0392	PCBA,2.8 GHZ-2M CPU,IG,K22	K22,2P80GHZ_2M_CPU,BASIC,IG
639-0393	PCBA,2.8 GHZ-2M CPU,MXM,K22	K22,2P80GHZ_2M_CPU,BASIC,MXM,K22_MXM
607-4426	PCBA,MLB,DEV,K22	DEVELOPMENT,DEV_GROUP

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC	COMMON,ALTERNATE,MCP7A,XDP,BETTER,MCP_ISL9563A,MLB_PNL_PWR,PRODUCTION
MCP7A	BOOT_MODE_USER,MEMRESET_HW,MEMRESET_MCP
DEV_GROUP	XDP_CONN,LPCPLUS,VREFMRGN,MCP_PWR_SENSE,MCP_CPU_TDIODE,PECI_SMB,MOJOMUX

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0731	1	IC,GMCP,MCP7A-2A,B03,35X35MM,BGA1437,DT	U1400	CRITICAL	IG
338S0732	1	IC,MCP,MCP7A-DA,B03,35X35MM,BGA1437,DT	U1400	CRITICAL	MXM
341T0170	1	IC,EFT_BOOTROM,K22/K23	U6100	CRITICAL	
338S0765	1	IC,XIO22112AY,1394B,167BGA	U4100	CRITICAL	
338S0694	1	IC,RTL8251CA,GIGE TRANSCEIVER, 48P TQFP	U3700	CRITICAL	
825-7122	1	MLB_LABEL,48.0X4.8	X14	CRITICAL	

MCP -J SKU HAS INTEGRATED GPU
MCP -D SKU DOES NOT

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S3745	1	WLF,QXXX,Q8,2.80G,65W,1066,R0,3M,LGA	CPU	CRITICAL	2P80GHZ_CPU
337S3742	1	WLF,SLB9J,PRQ,2.83G,65W,1333,E0,6M,LGA	CPU	CRITICAL	2P83GHZ_CPU
337S3726	1	WLF,SLB9J,PRQ,3.0G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P0GHZ_CPU
337S3715	1	WLF,SLB9K,PRQ,3.16G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P16GHZ_CPU
337S3727	1	WLF,SLB9L,PRQ,3.33G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P33GHZ_CPU
337S3807	1	WLF,SLB9L,PRQ,2.93G,65W,1333,E0,6M,LGA	CPU	CRITICAL	2P93GHZ_CPU
337S3766	1	WLF,SLB9L,PRQ,3.06G,65W,1333,E0,6M,LGA	CPU	CRITICAL	3P06GHZ_CPU
337S3804	1	WLF,SL0U9,PRQ,2.80G,65W,1066,R0,2M,LGA	CPU	CRITICAL	2P80GHZ_2M_CPU

K22 PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-7845	1	SCH,K22,MLB	SCH1		K22
820-2494	1	PCBF,K22,MLB	MLB1		K22
341T0168	1	IC,SMC,K22	U4900	CRITICAL	K22

(338S0563 - BLNK)

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0111	127S0060		C6211	AUDIO, NEED QUAL

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BOM Configuration

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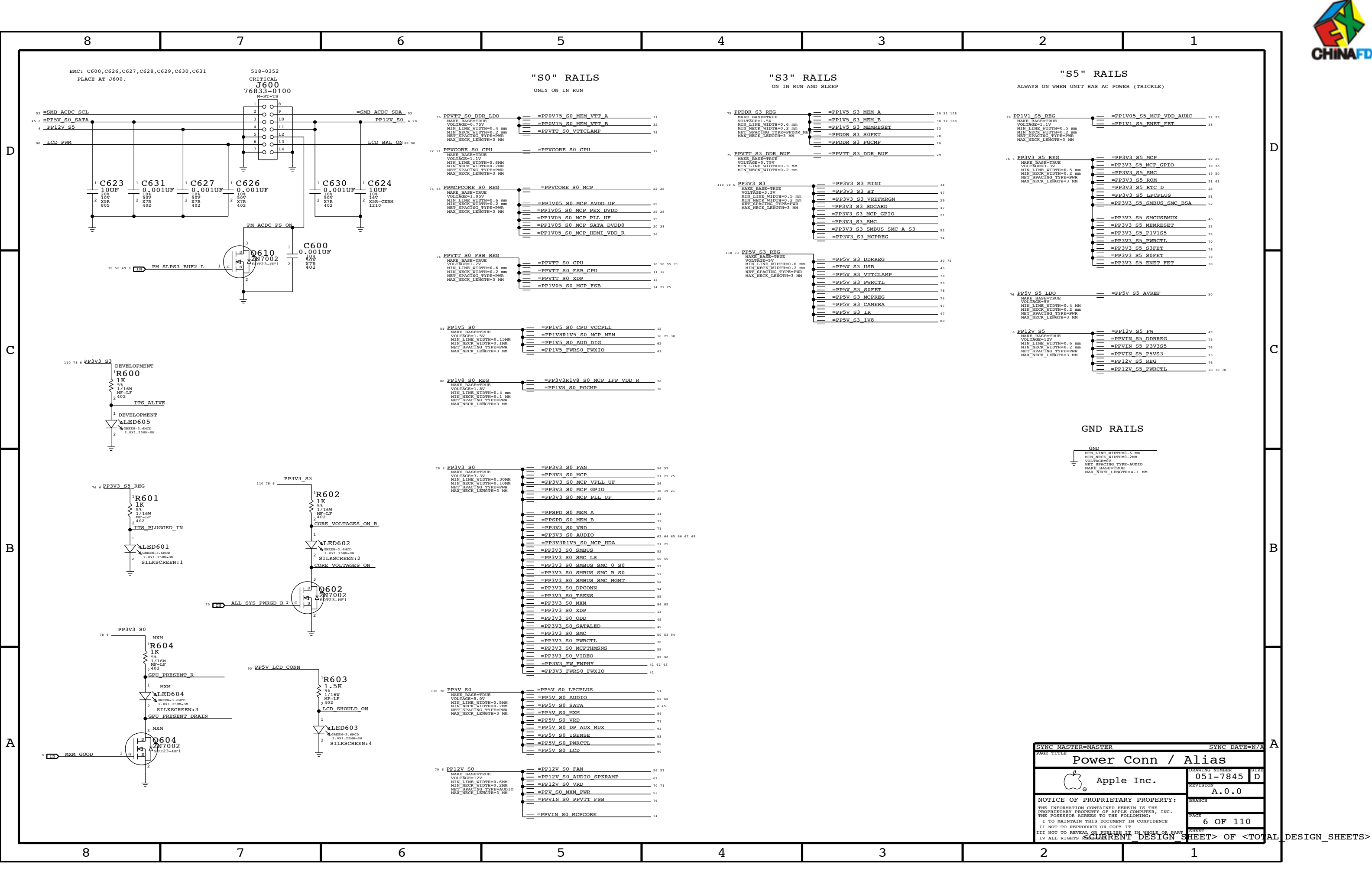
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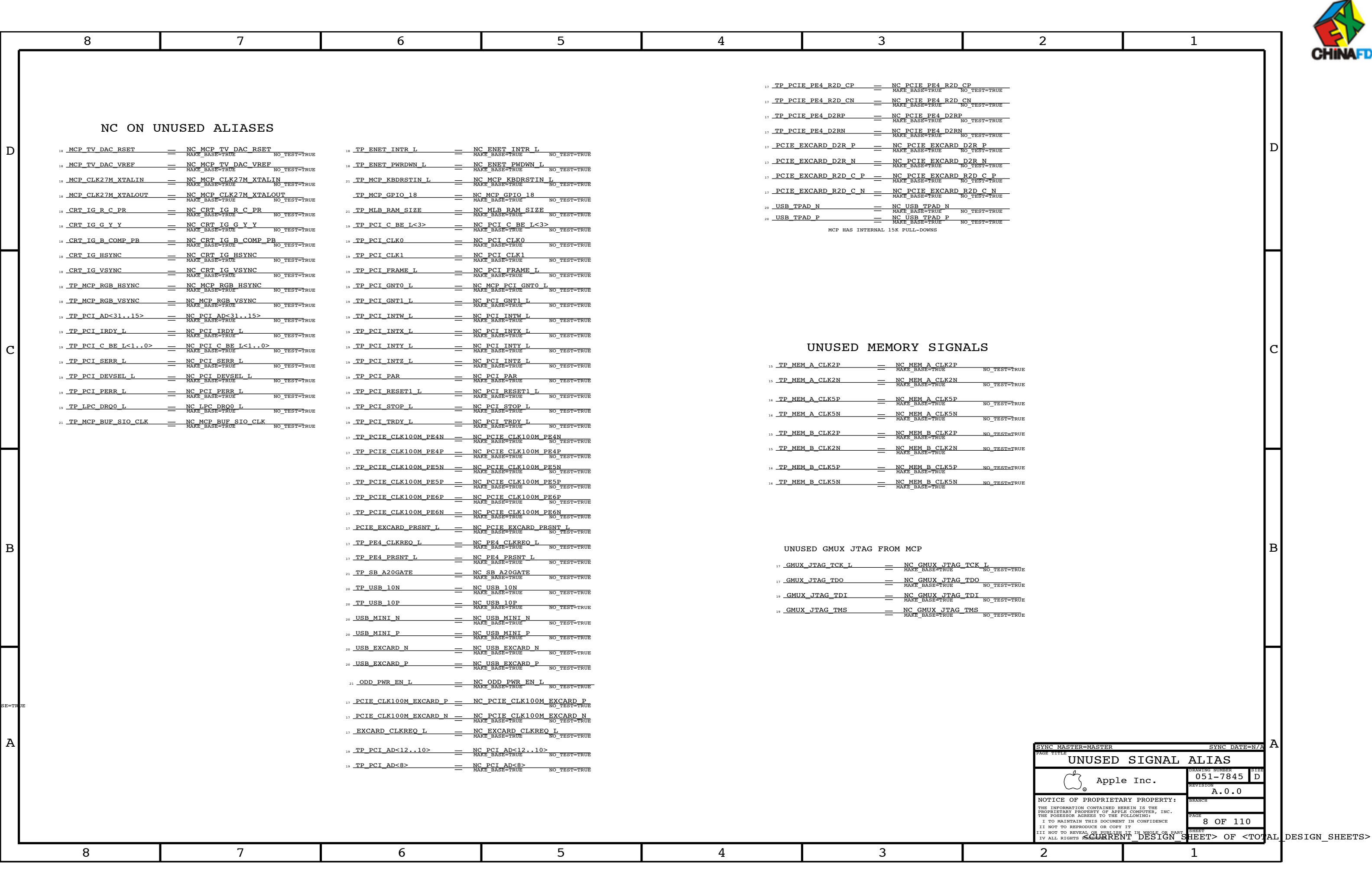
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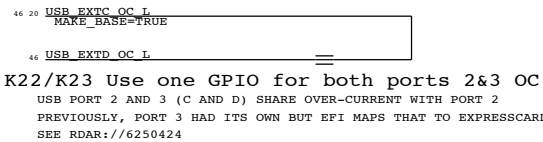
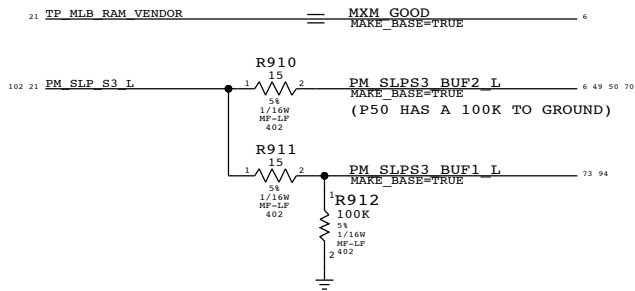
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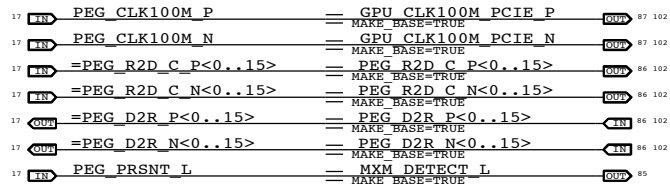




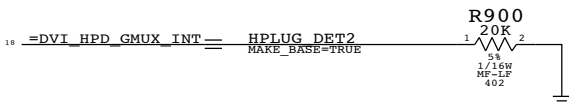
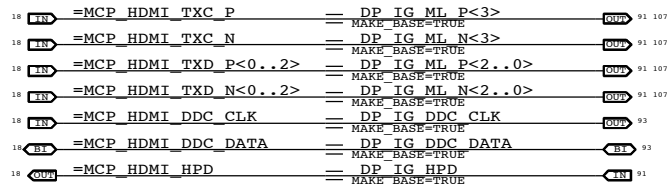
SIGNAL ALIAS



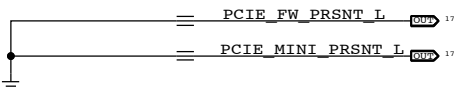
PEG Slot Support



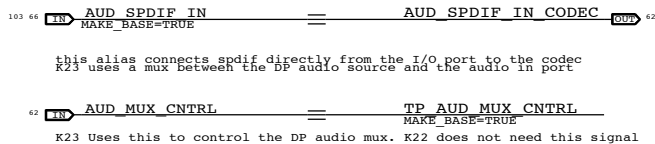
DisplayPort / TMDS Support



MCP79 PCIe PRSNT# Straps

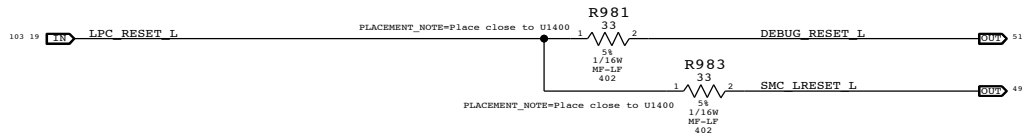


Audio Mux aliasing

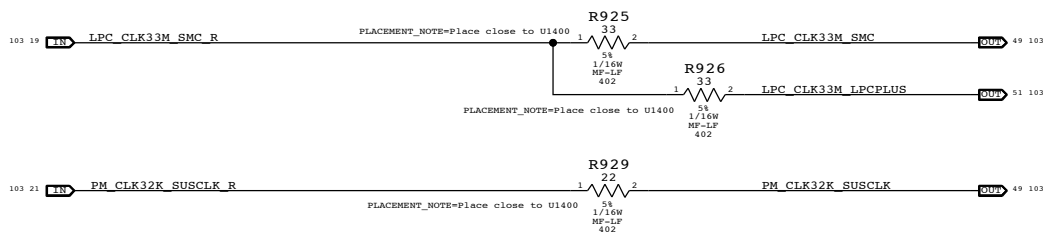
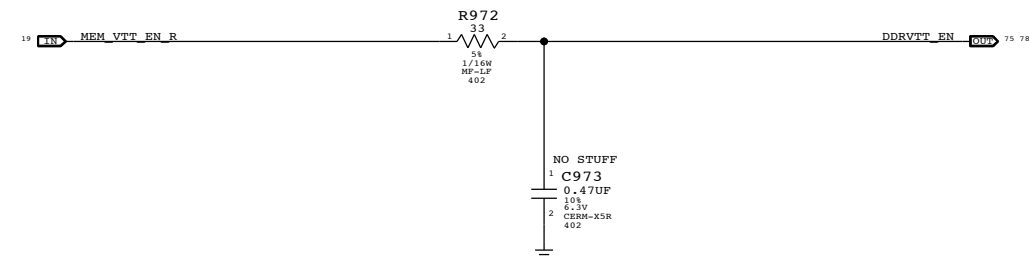
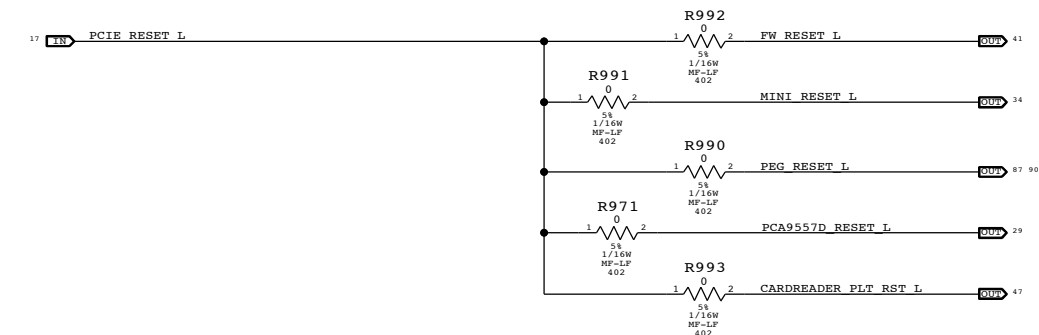


Platform Reset Connections

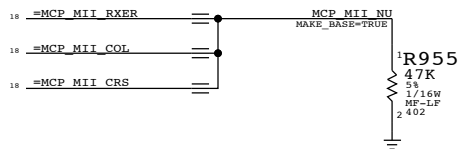
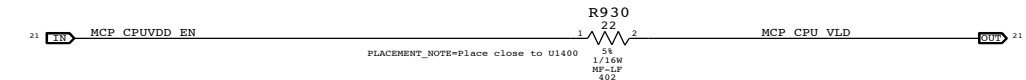
LPC Reset (Unbuffered)



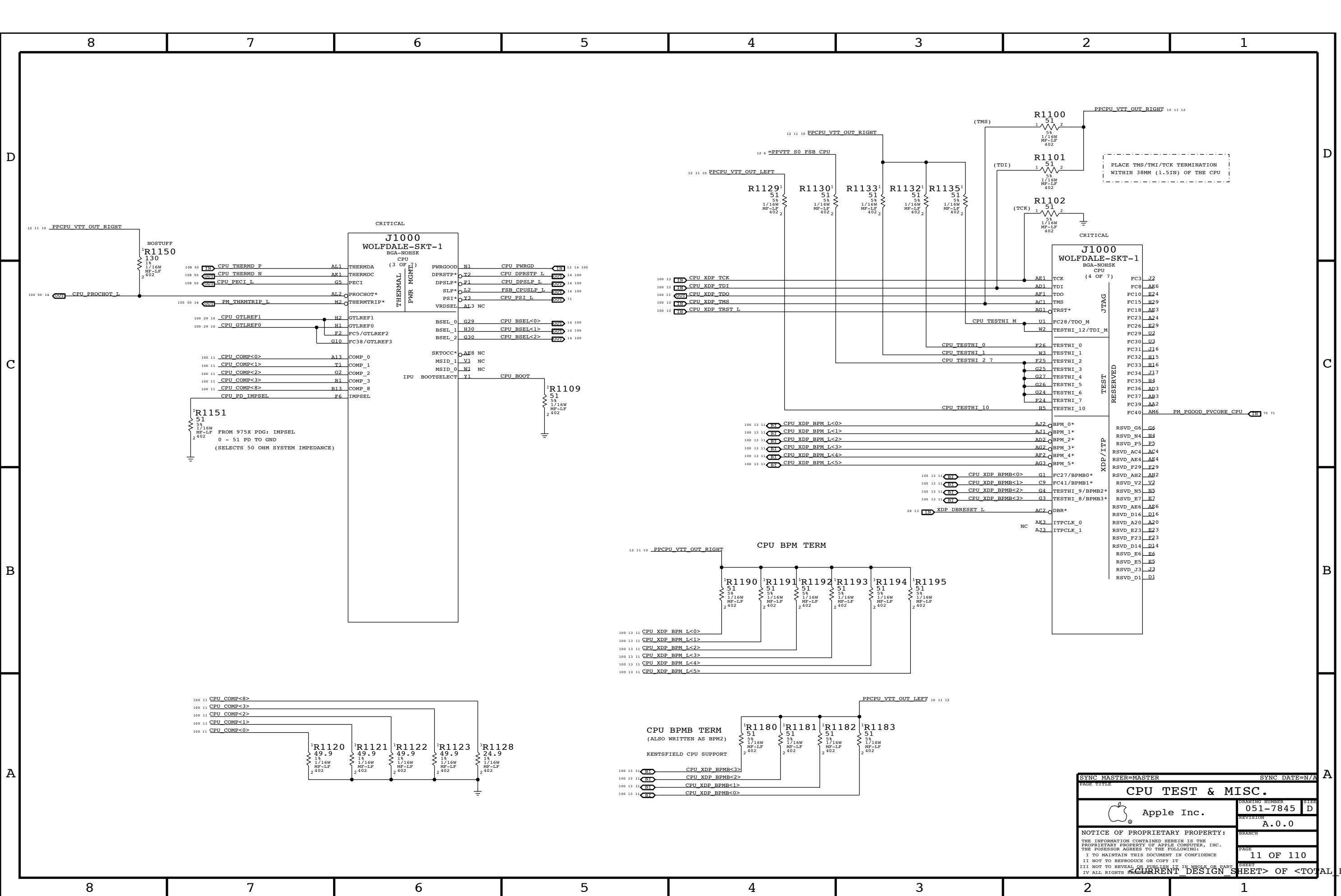
PCIE Reset (Unbuffered)



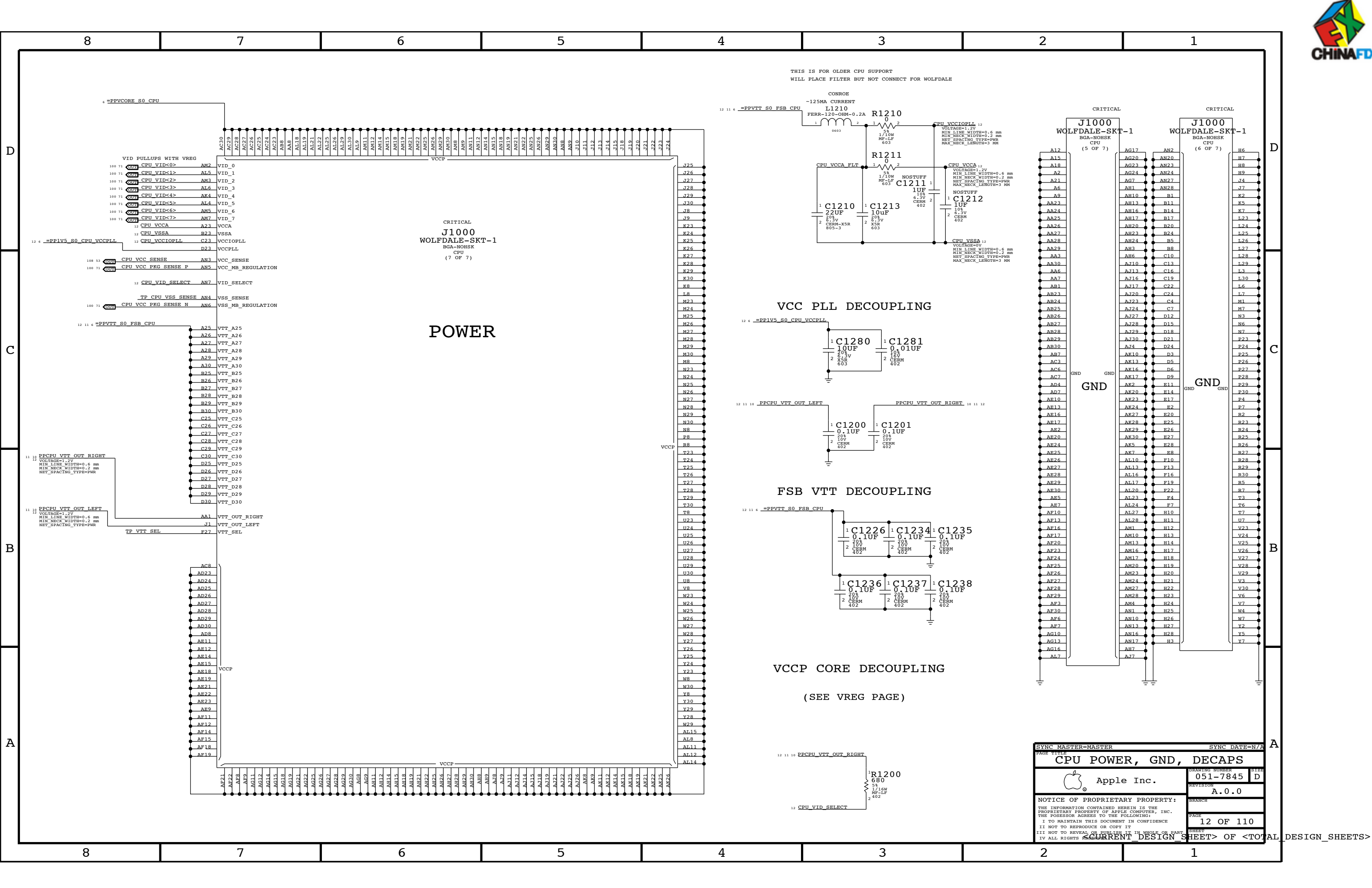
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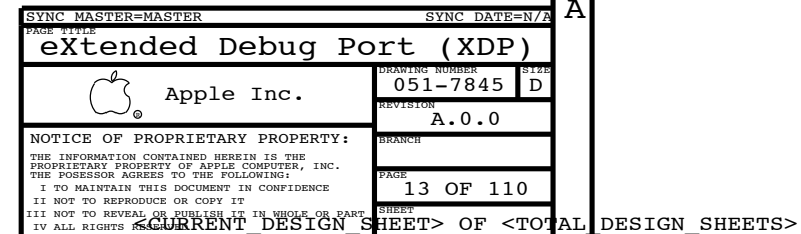


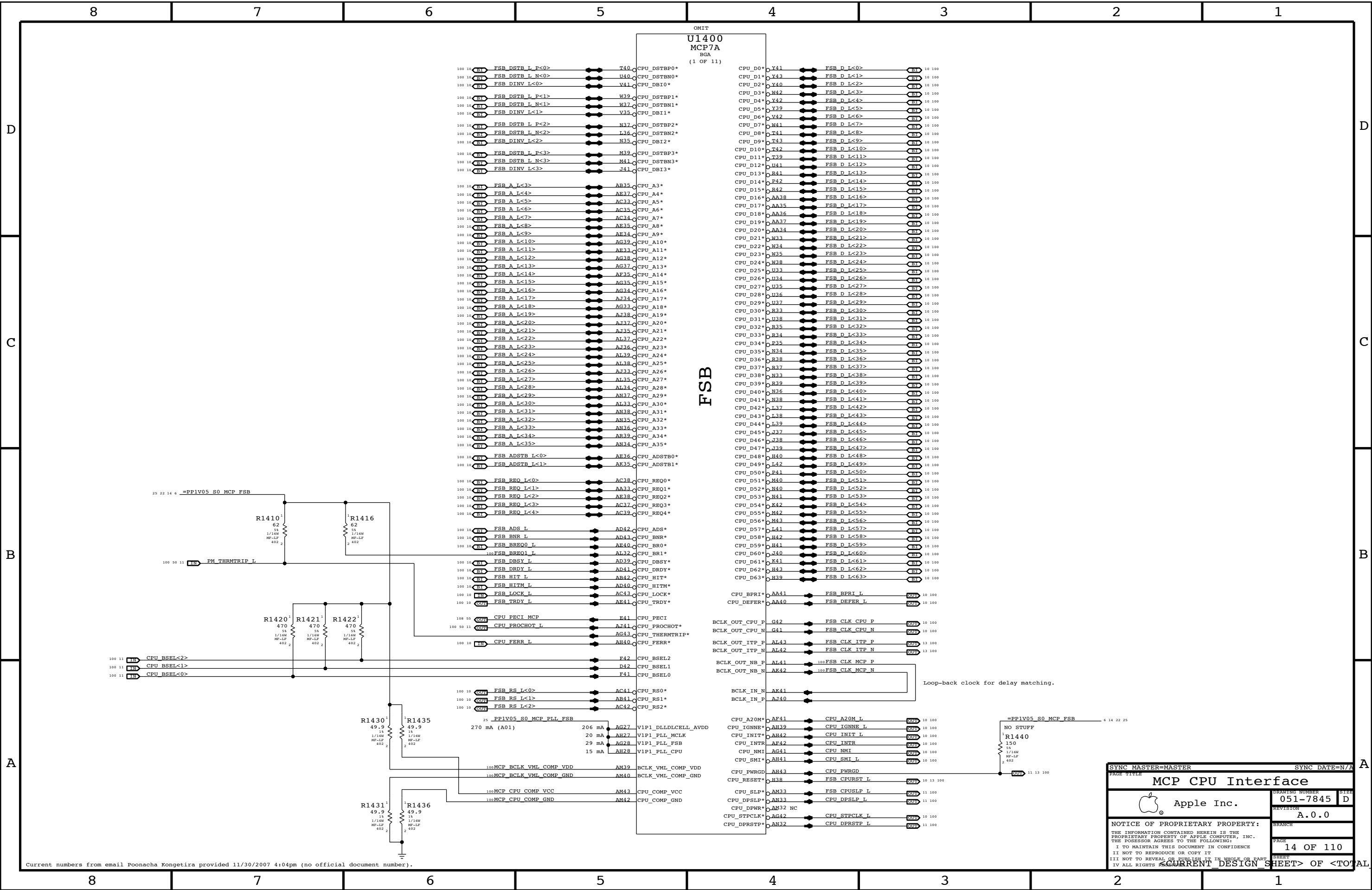
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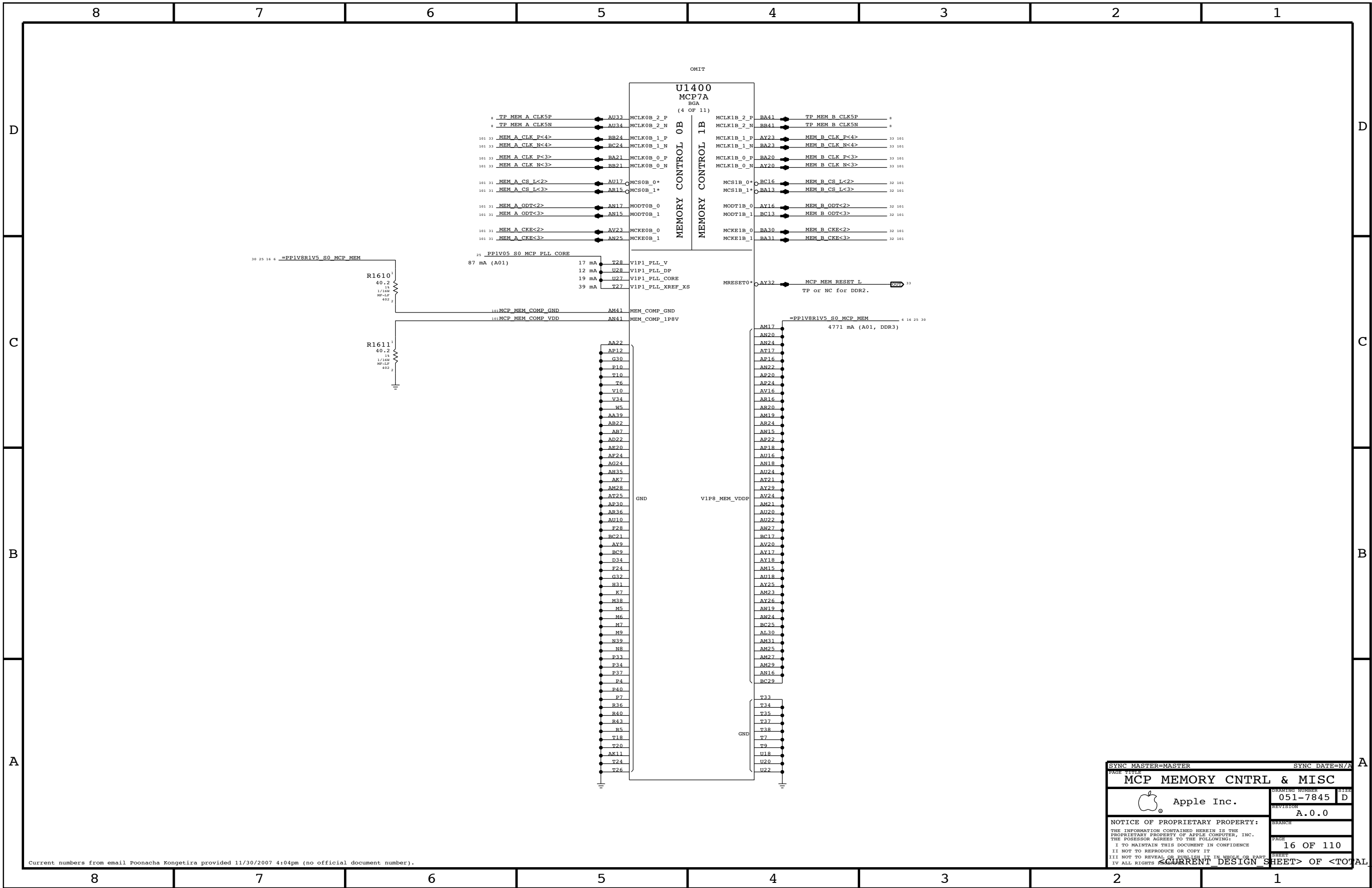




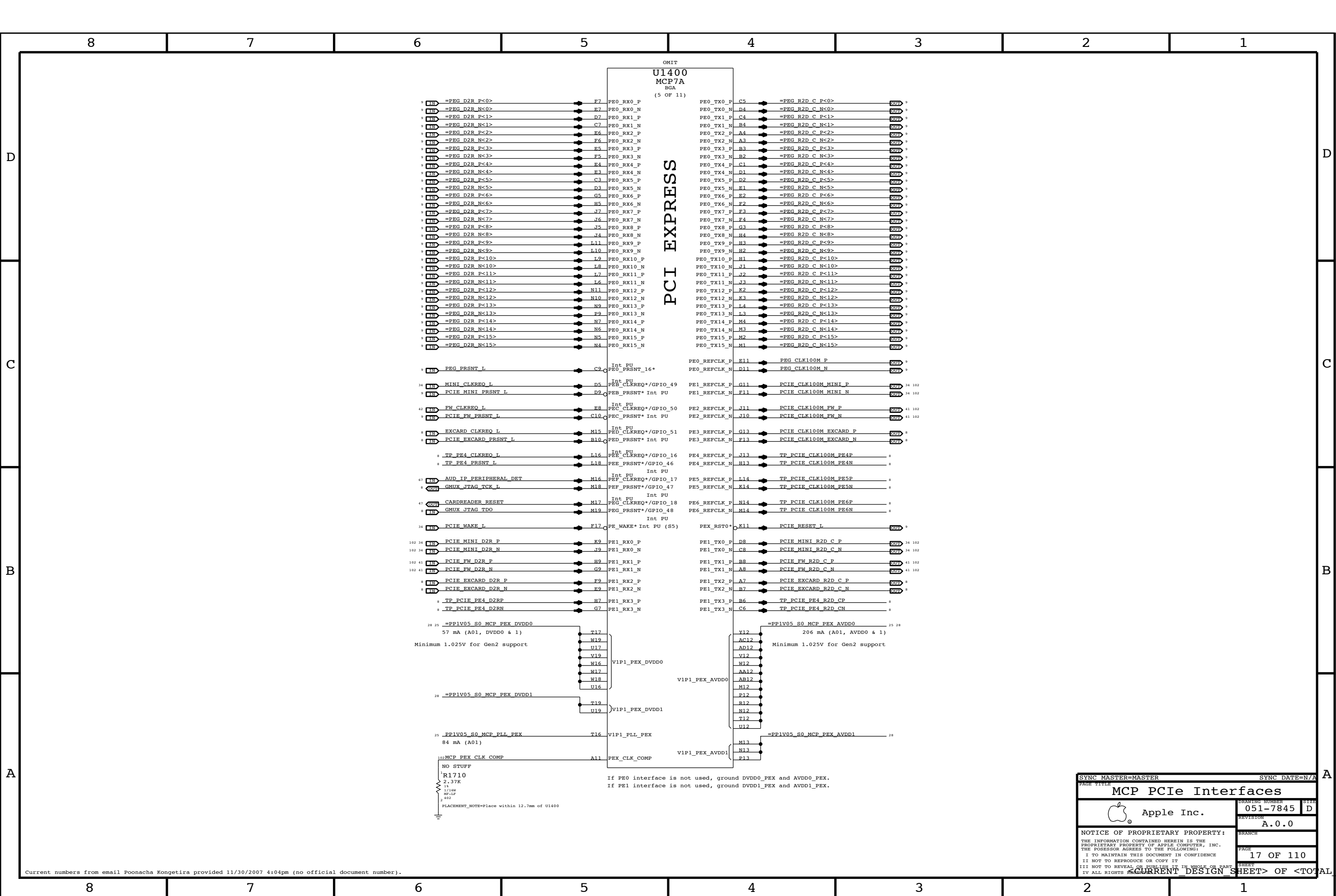


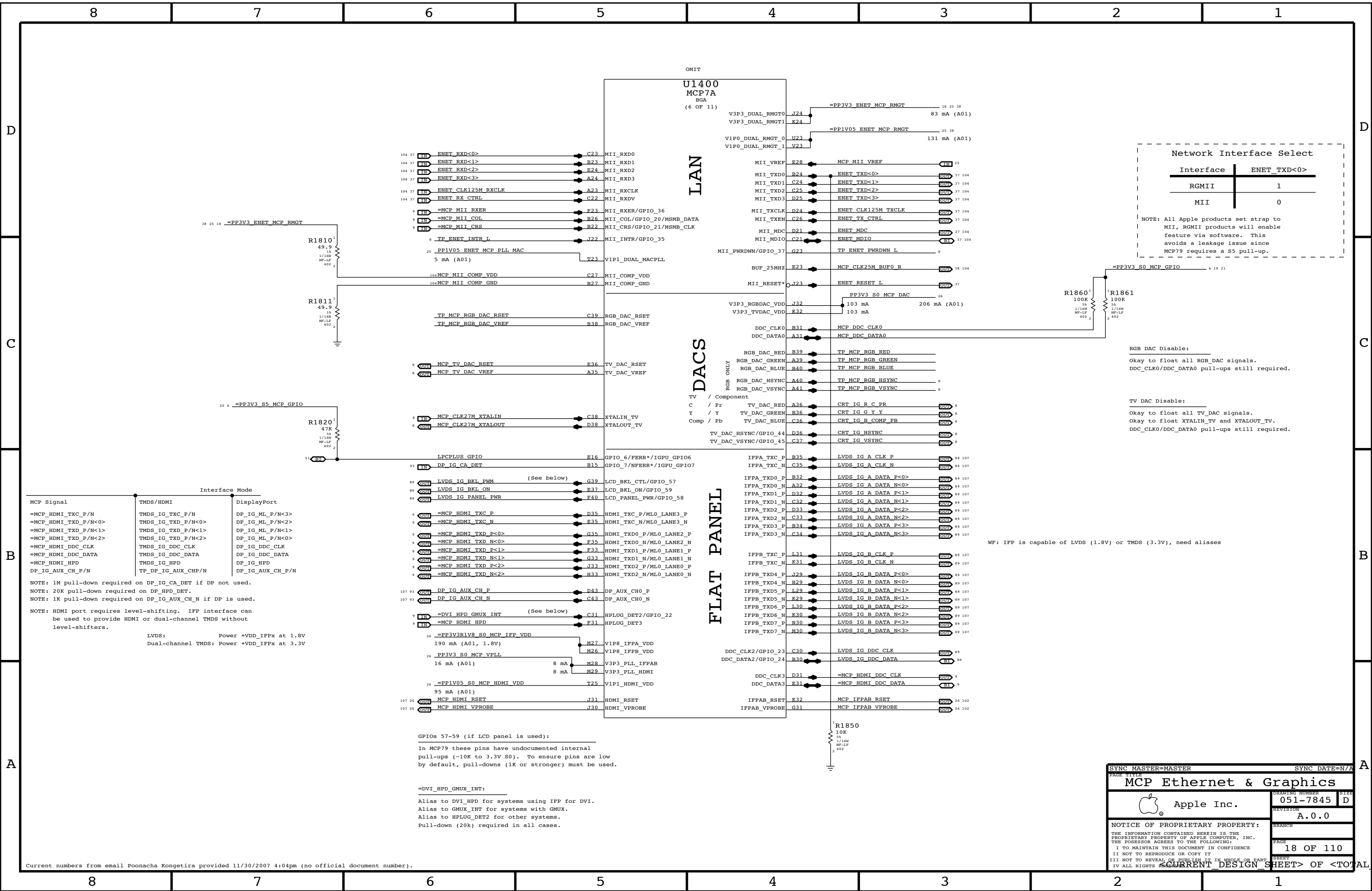
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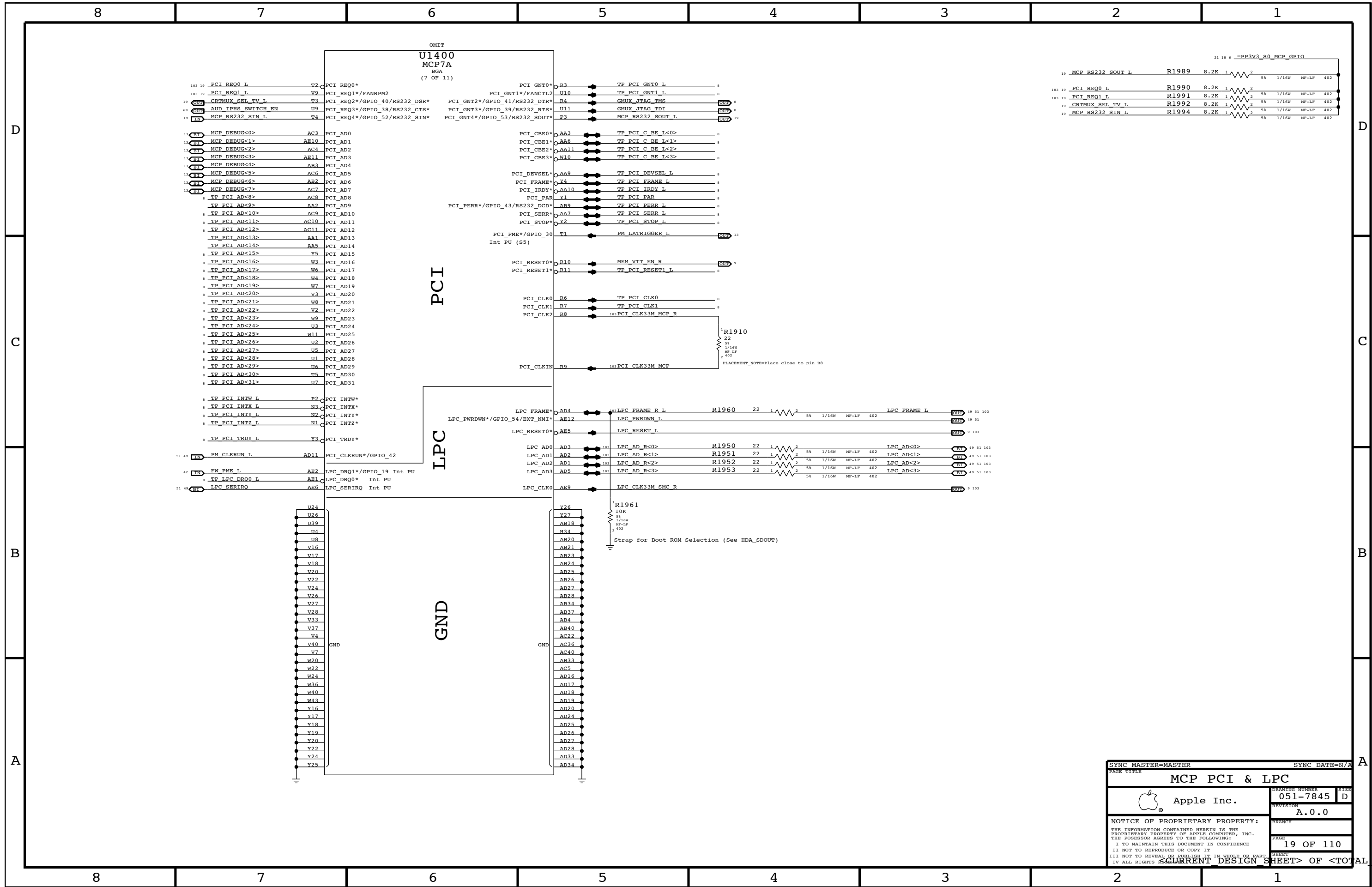


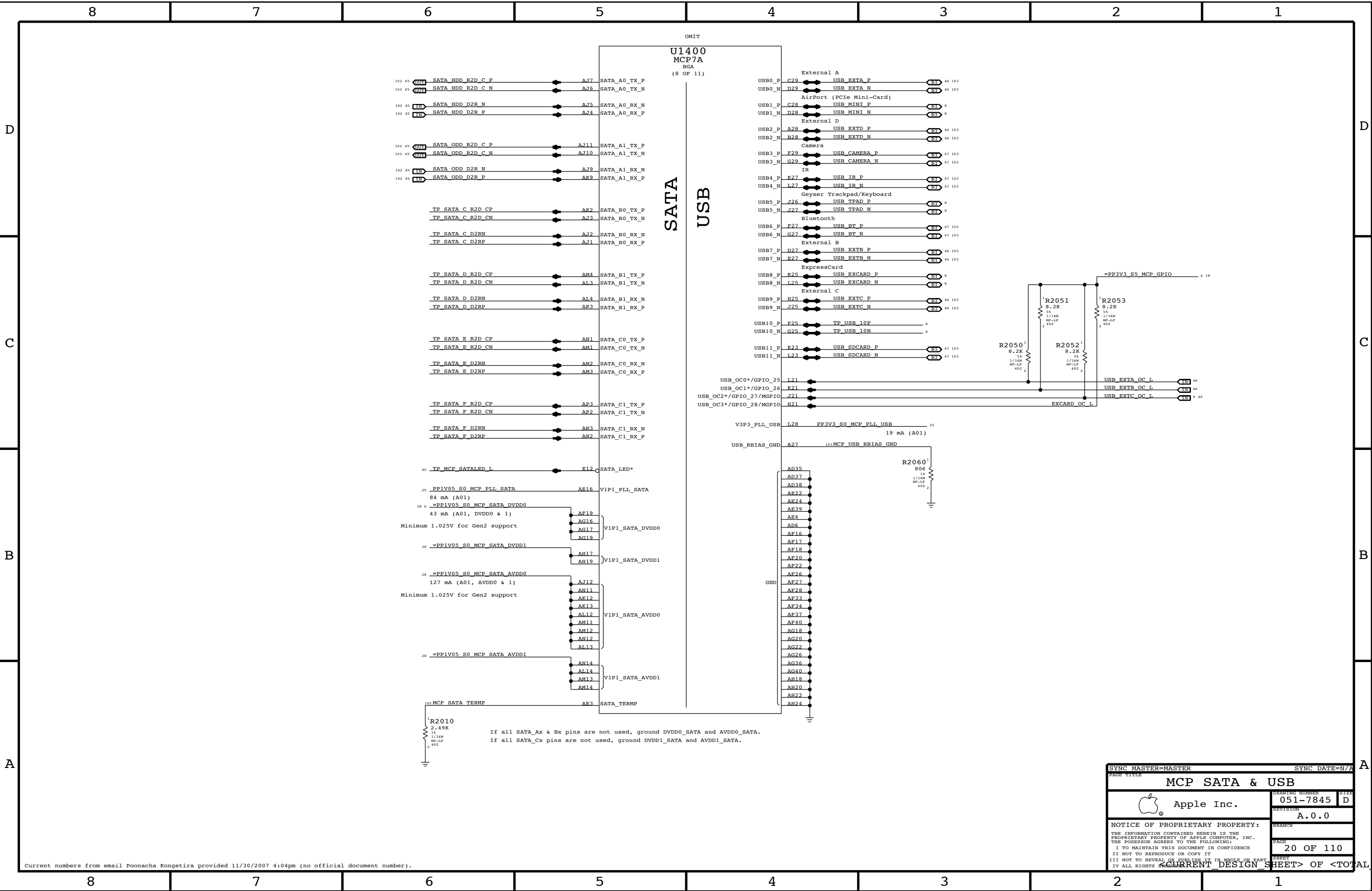


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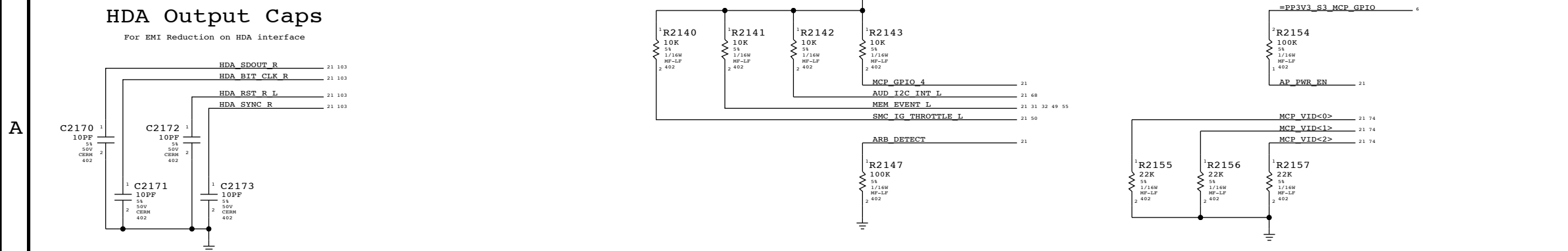






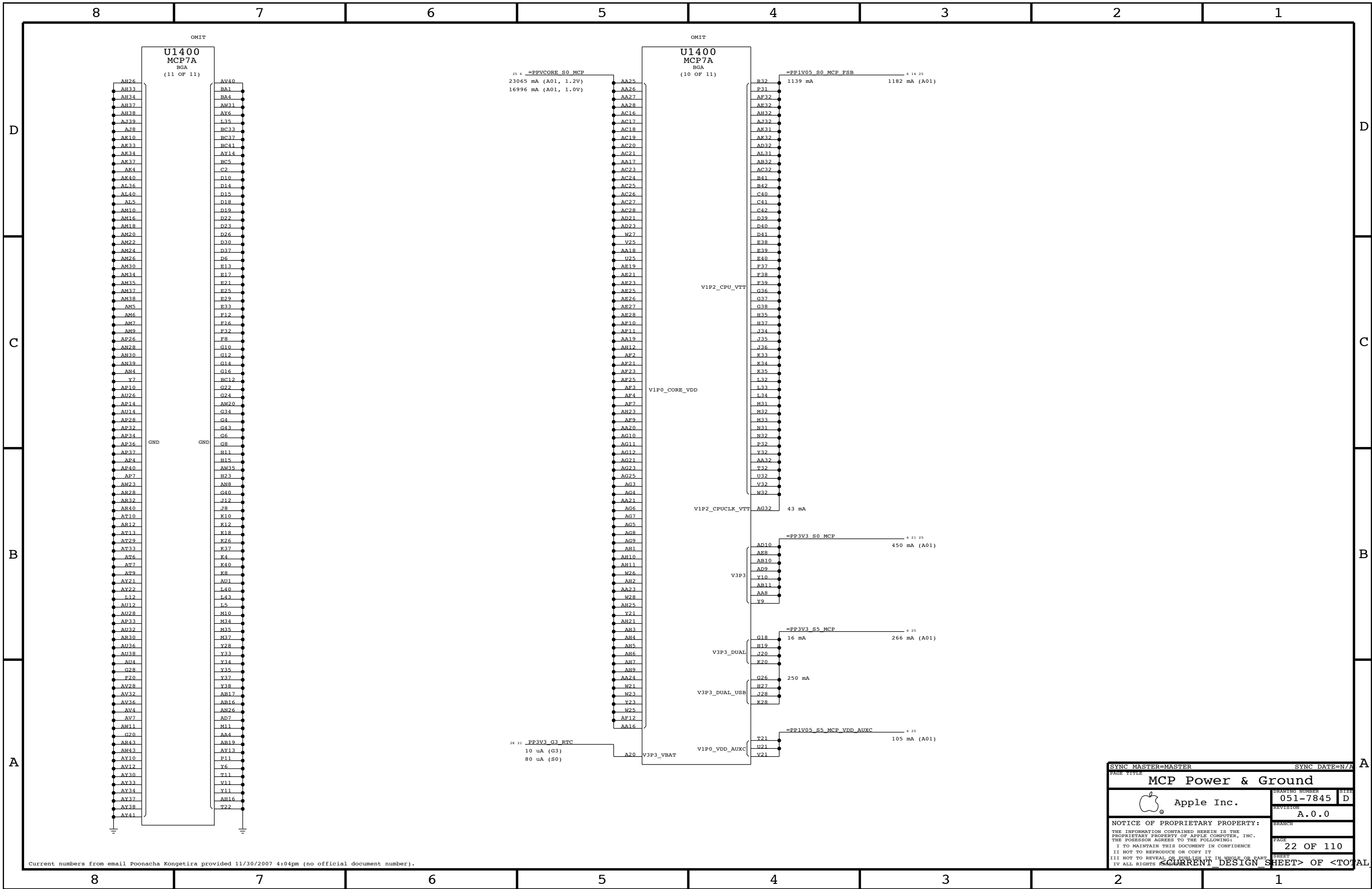


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Frequency	SPI_DO	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

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MCP HDA & MISC			
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SYNC MASTER=MASTER		SYNC DATE=N/A	
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MCP Power & Ground			
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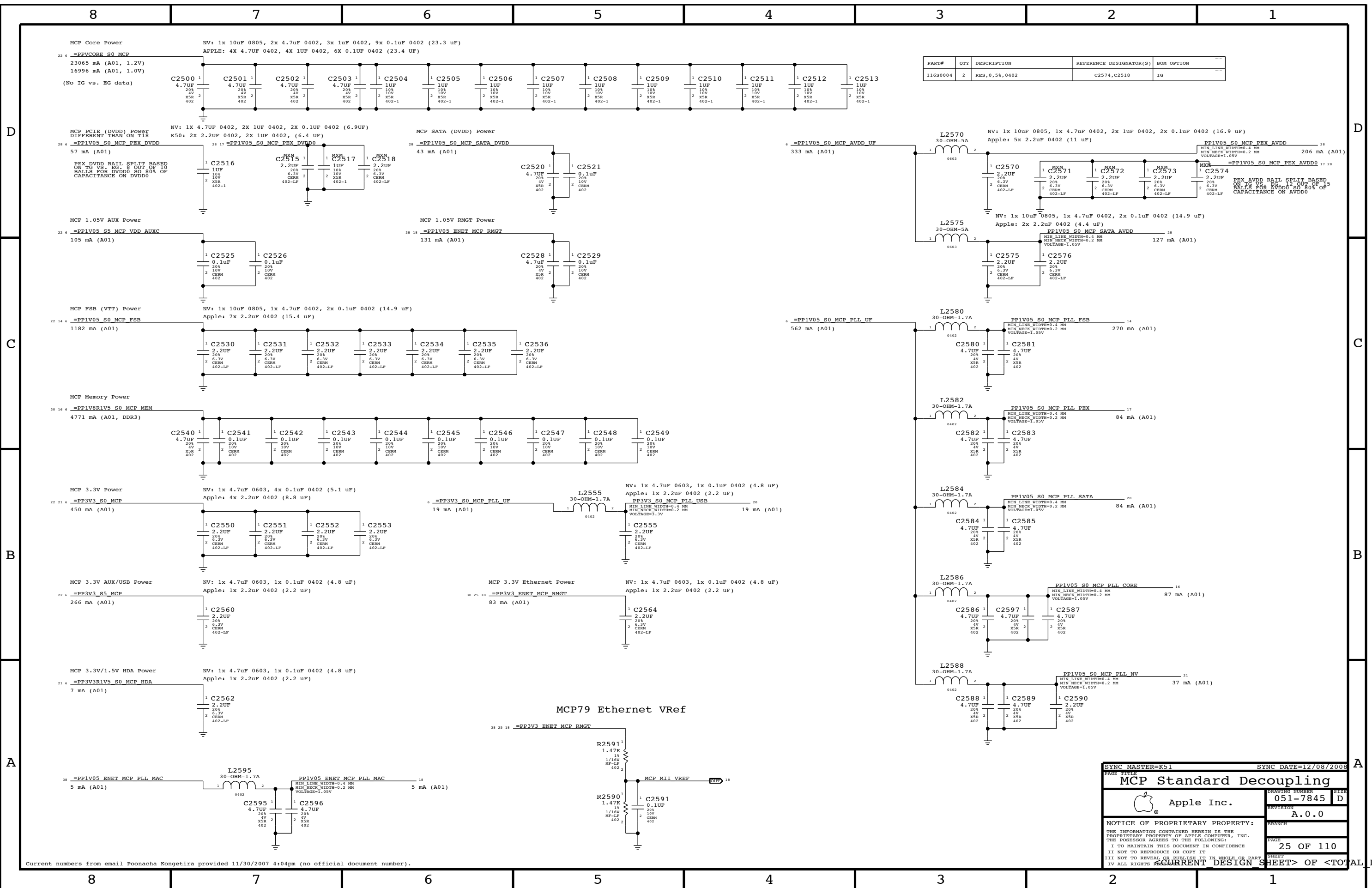
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<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0004	2	RES,0.5%,0402	C2574,C2518	IG

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MCP Standard Decoupling

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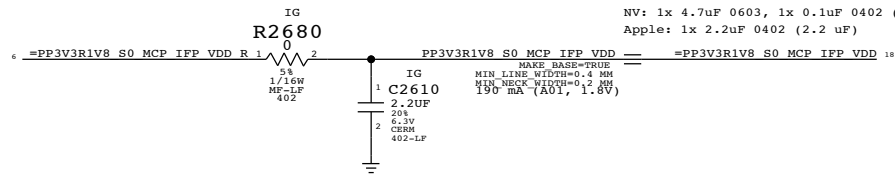
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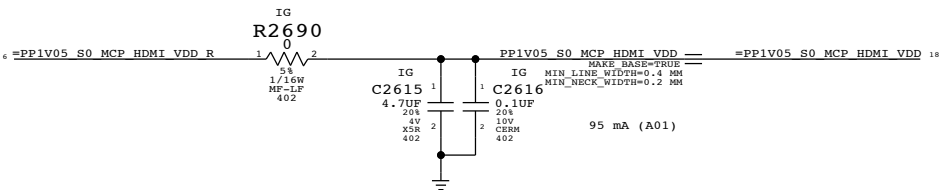
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WF: Checklist says 0-ohm resistor placeholder for ferrite bead.

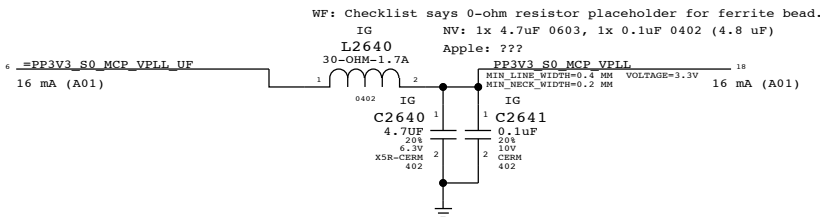
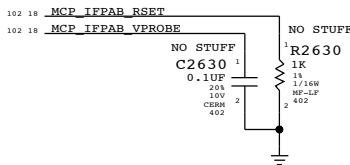
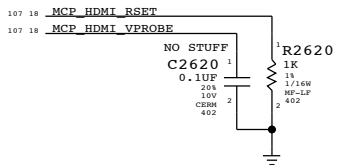
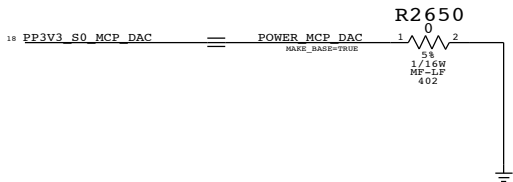
NV: 1x 4.7uF 0603, 1x 0.1uF 0402 (4.8 uF)
Apple: 1x 2.2uF 0402 (2.2 uF)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	1	RES,0,5%,402	C2610		MXM



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116S0004	1	RES,0,5%,402	C2616		MXM



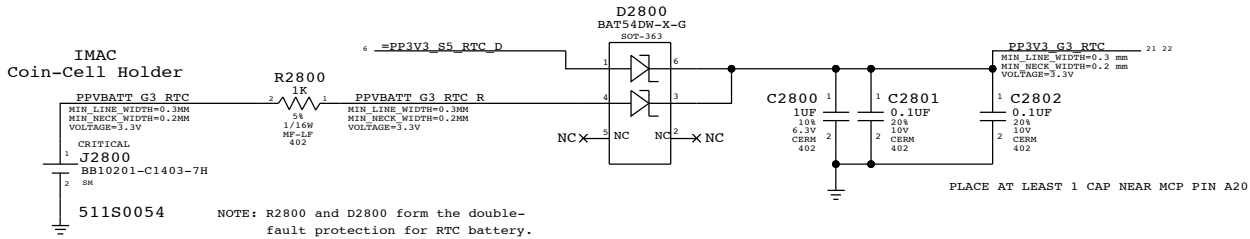
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116S0004	1	RES,0,5%,402	C2641		MXM

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MCP Graphics Support			
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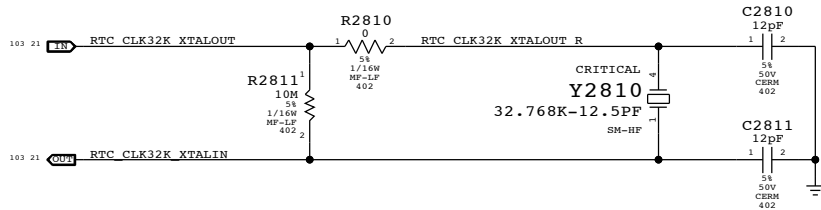
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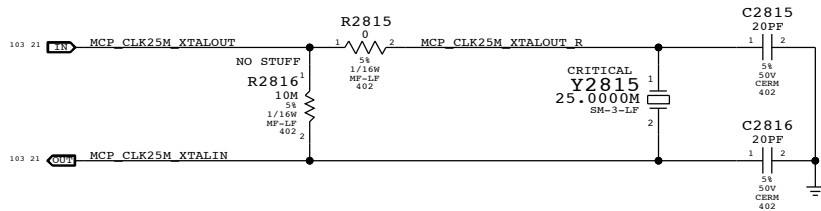
RTC Power Sources



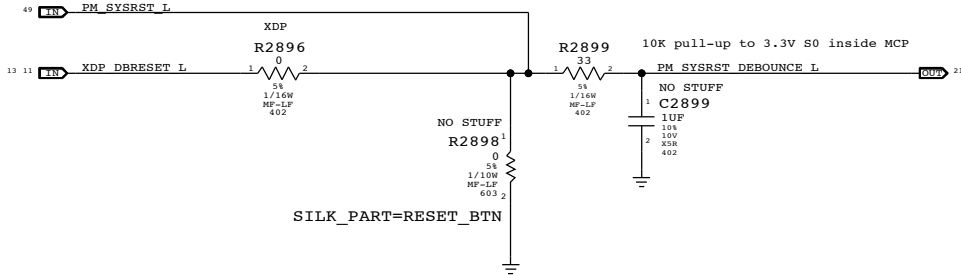
RTC Crystal



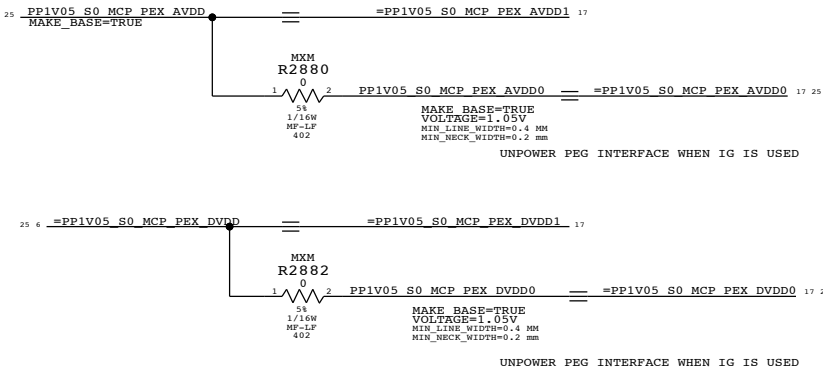
MCP 25MHz Crystal



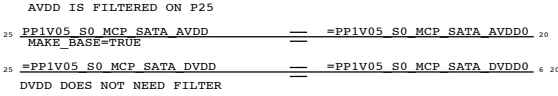
Reset Button



PEG POWER ALIAS/OPTION TO GND UNUSED POWER PIN



SATA ALIAS/GROUNDING UNUSED DVDD1 AND AVDD1



SYNC MASTER=MASTER		SYNC DATE=N/A			
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SB Misc					
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28 OF 110		<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>			

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

VREFMRGN
PRODUCTION

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

MEM A VREF DQ

MEM A VREF CA

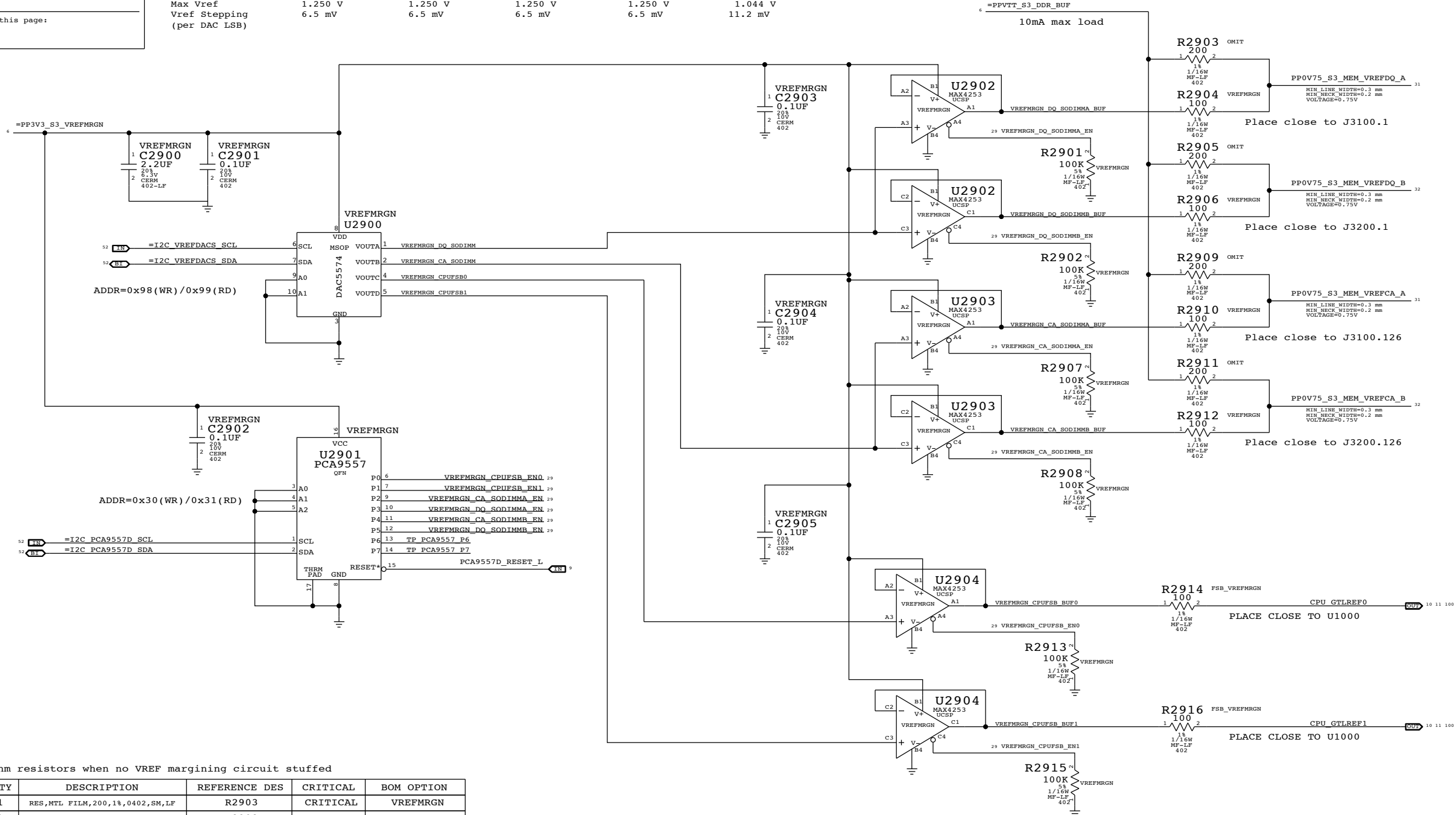
MEM B VREF DQ

MEM B VREF CA

CPU FSB VREF

	A	B	A	B	C
Min DAC code	0x00	0x00	0x00	0x00	0x00
Max DAC code	0x87	0x87	0x87	0x87	0x55
Max sink I	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-0.91 mA
Max source I	5 mA	5 mA	5 mA	5 mA	0.52 mA
Nominal Vref	0.75 V	0.75 V	0.75 V	0.75 V	0.70 V
Min Vref	0.375 V	0.375 V	0.375 V	0.375 V	0.091 V
Max Vref	1.250 V	1.250 V	1.250 V	1.250 V	1.044 V
Vref Stepping (per DAC LSB)	6.5 mV	6.5 mV	6.5 mV	6.5 mV	11.2 mV

SO-DIMM A and SO-DIMM B Vref settings should be margined separately
(i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2903	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2905	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2909	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	PRODUCTION
114S0149	1	RES,MTL FILM,200,1%,0402,SM,LF	R2911	CRITICAL	VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	PRODUCTION

SYNC MASTER=MASTER SYNC DATE=MASTER

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FSB/DDR3 Vref Margining

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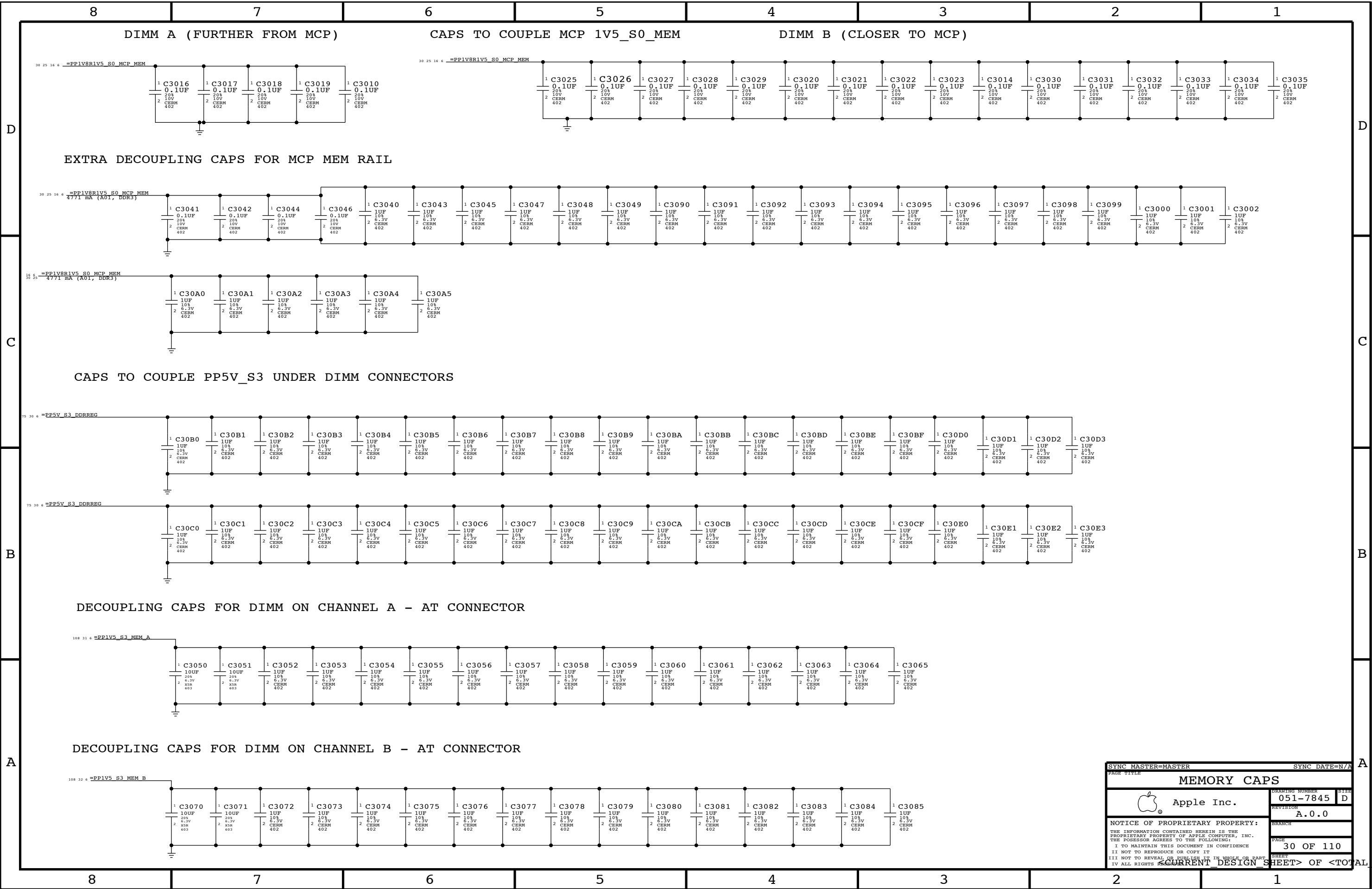
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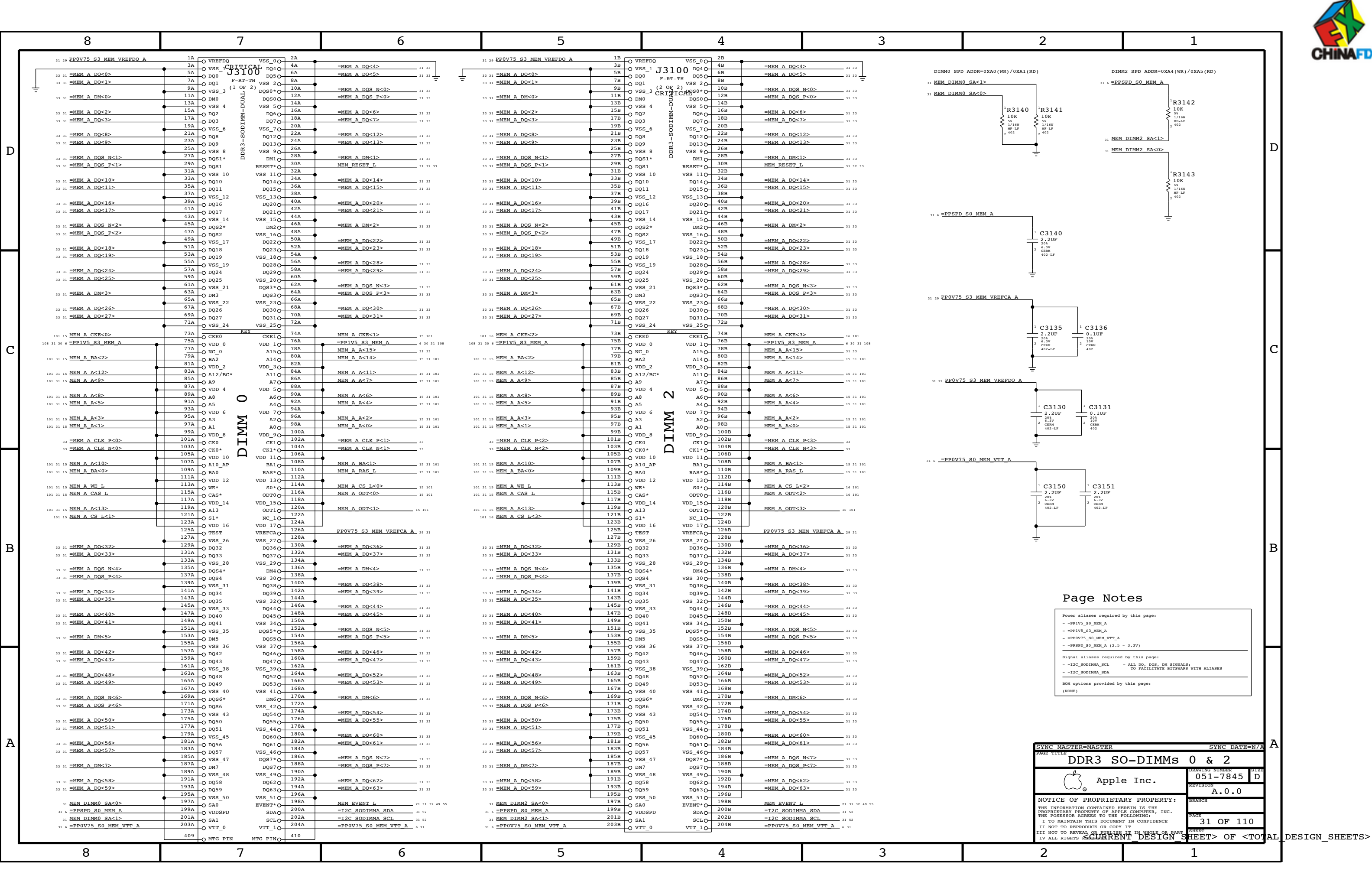
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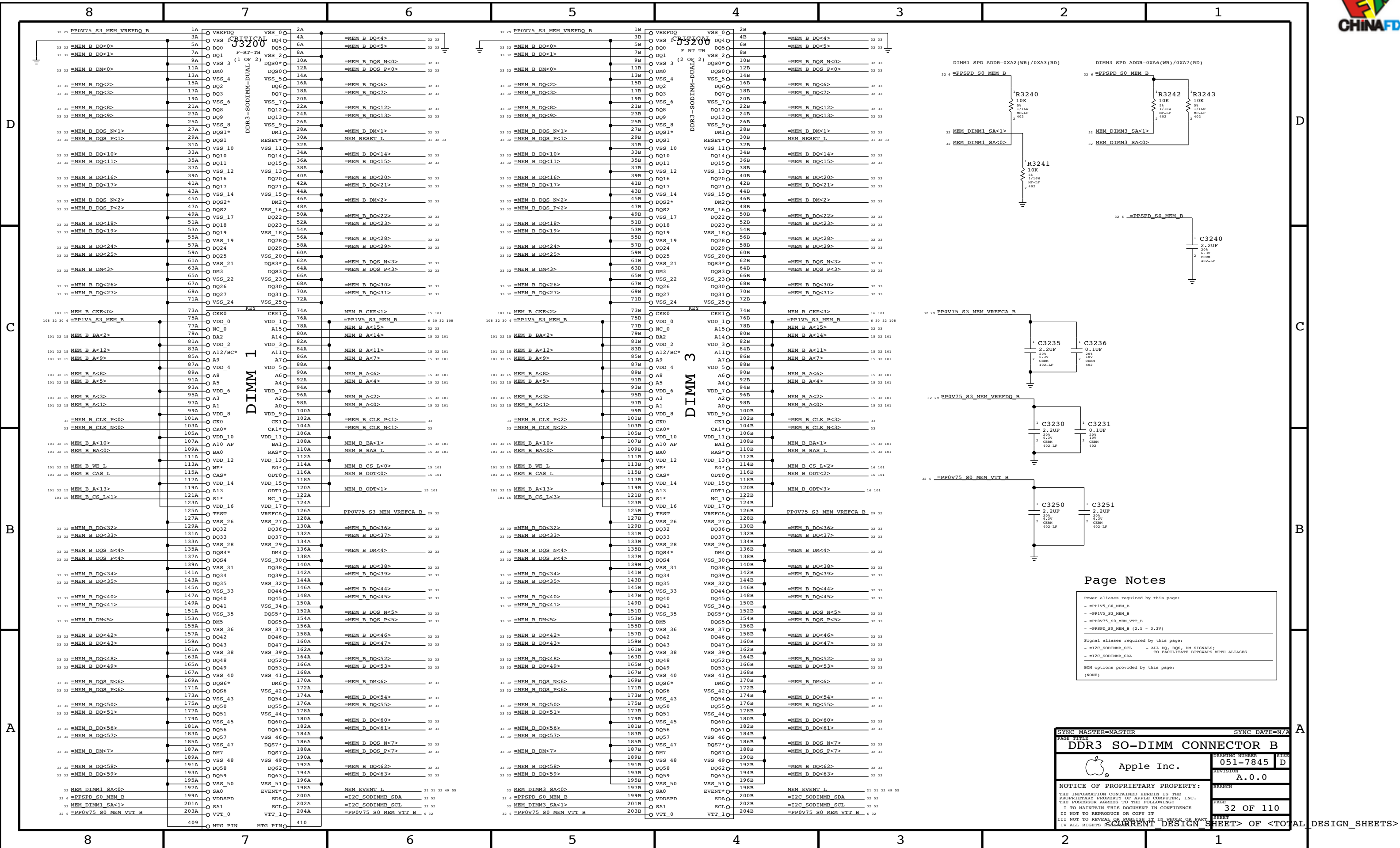
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

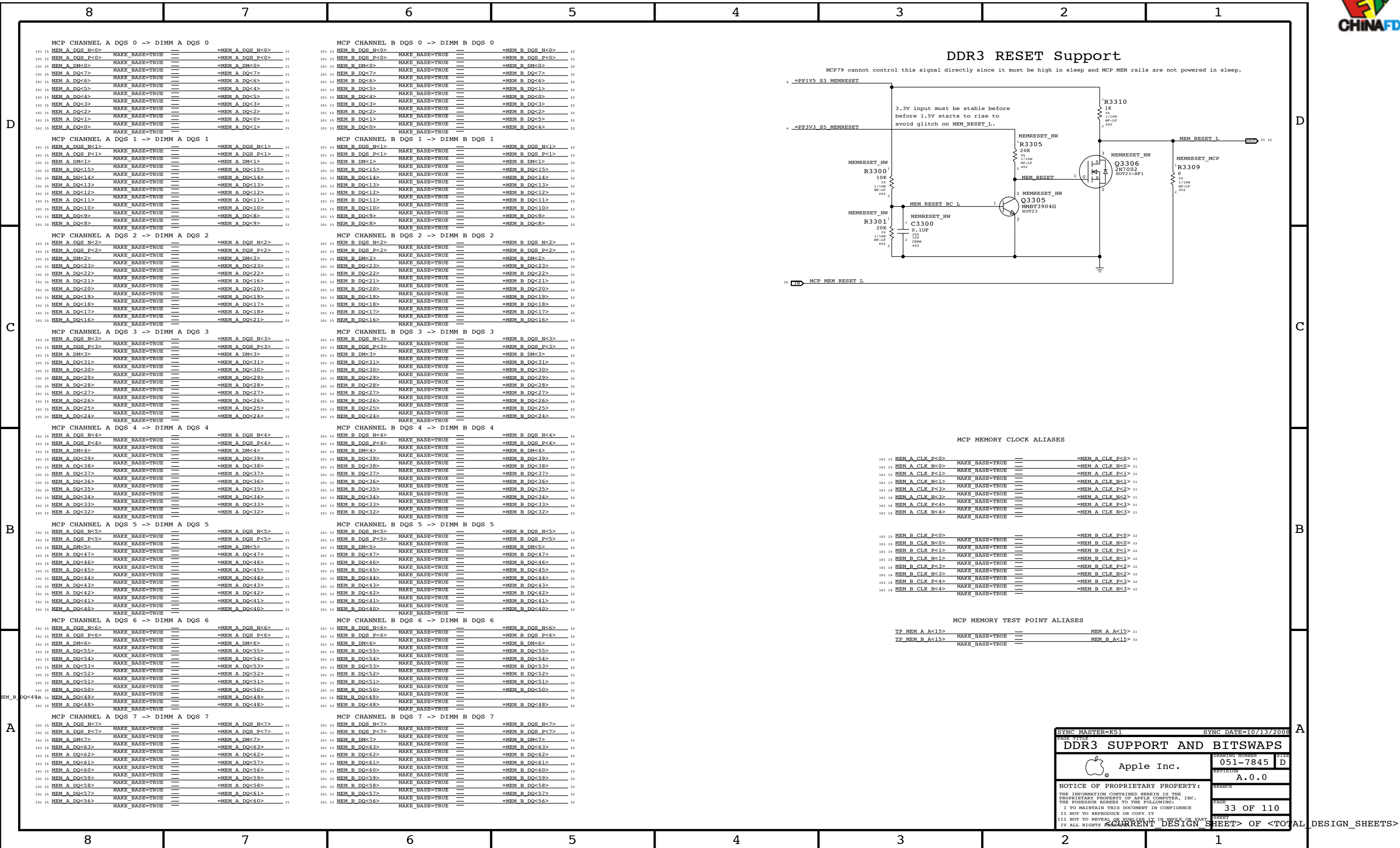
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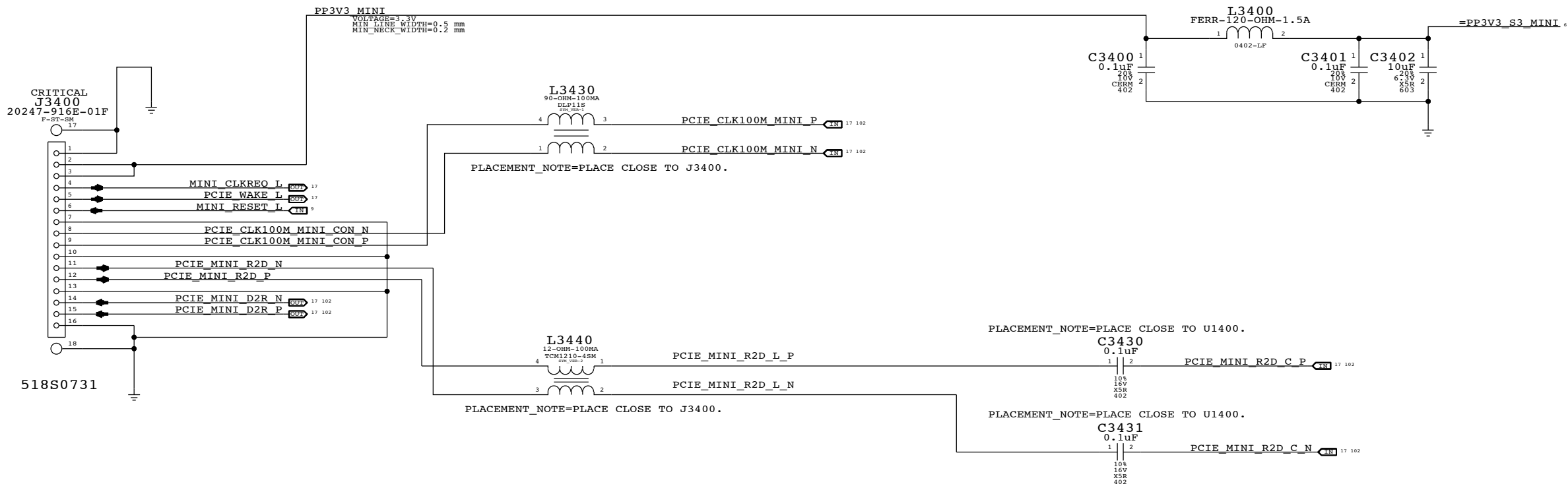
- =I2C_SODIMMB_SCL - ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSTREAMS WITH ALIASES
- =I2C_SODIMMB_SDA

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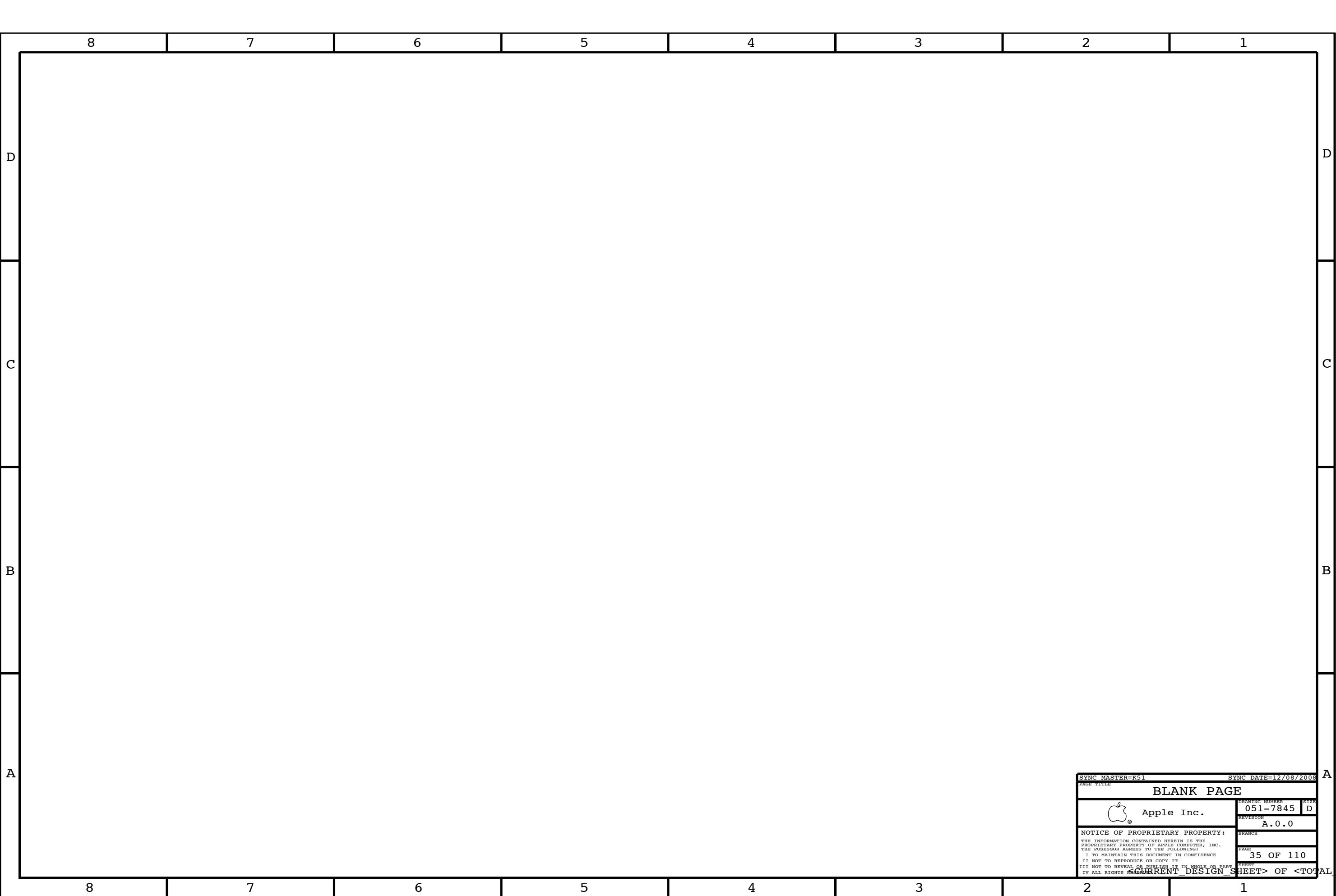
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PCI-E Wireless Connector			
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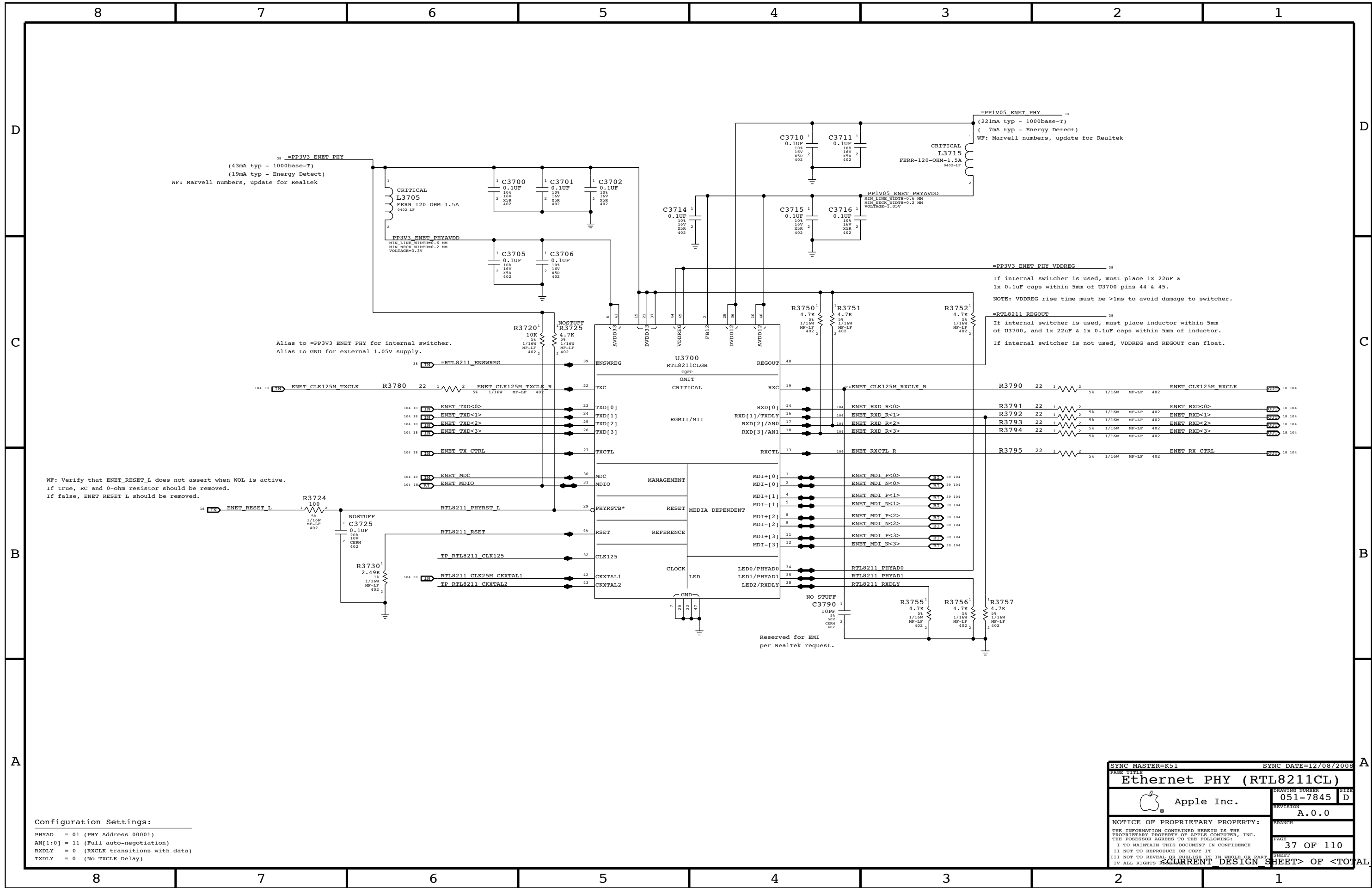
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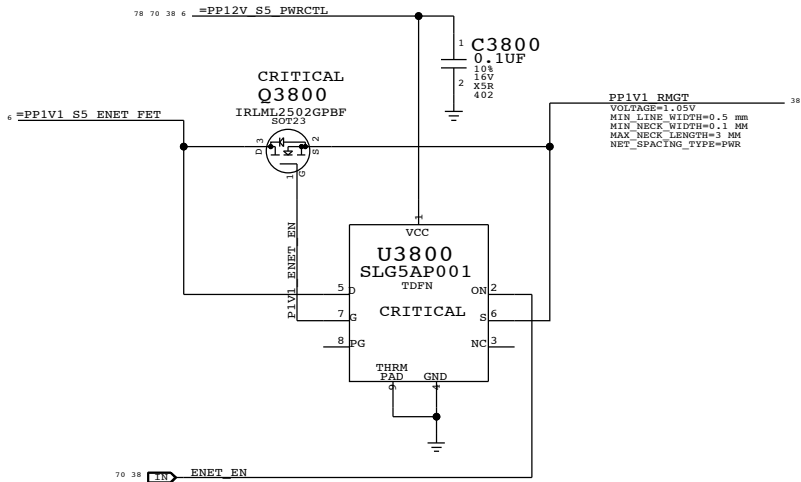
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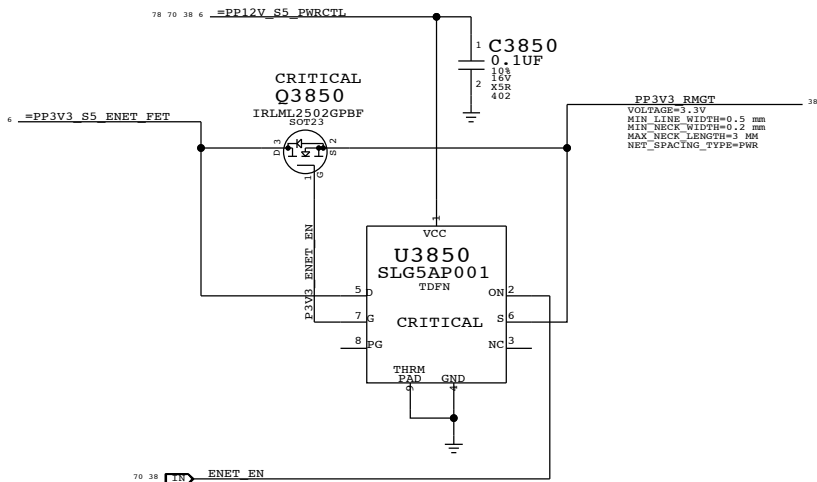
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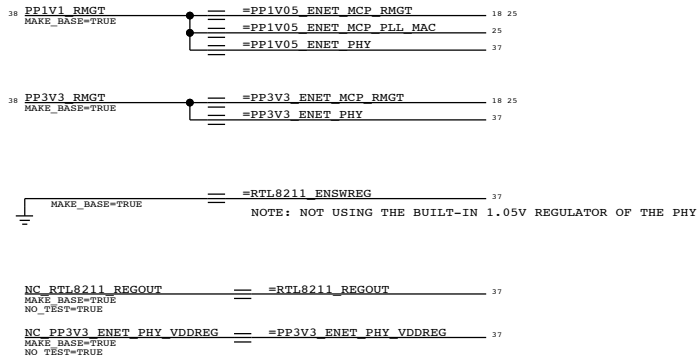
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3.3V ENET FET

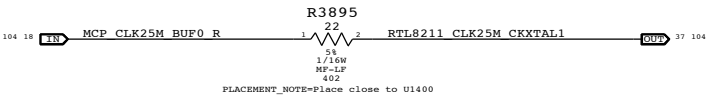


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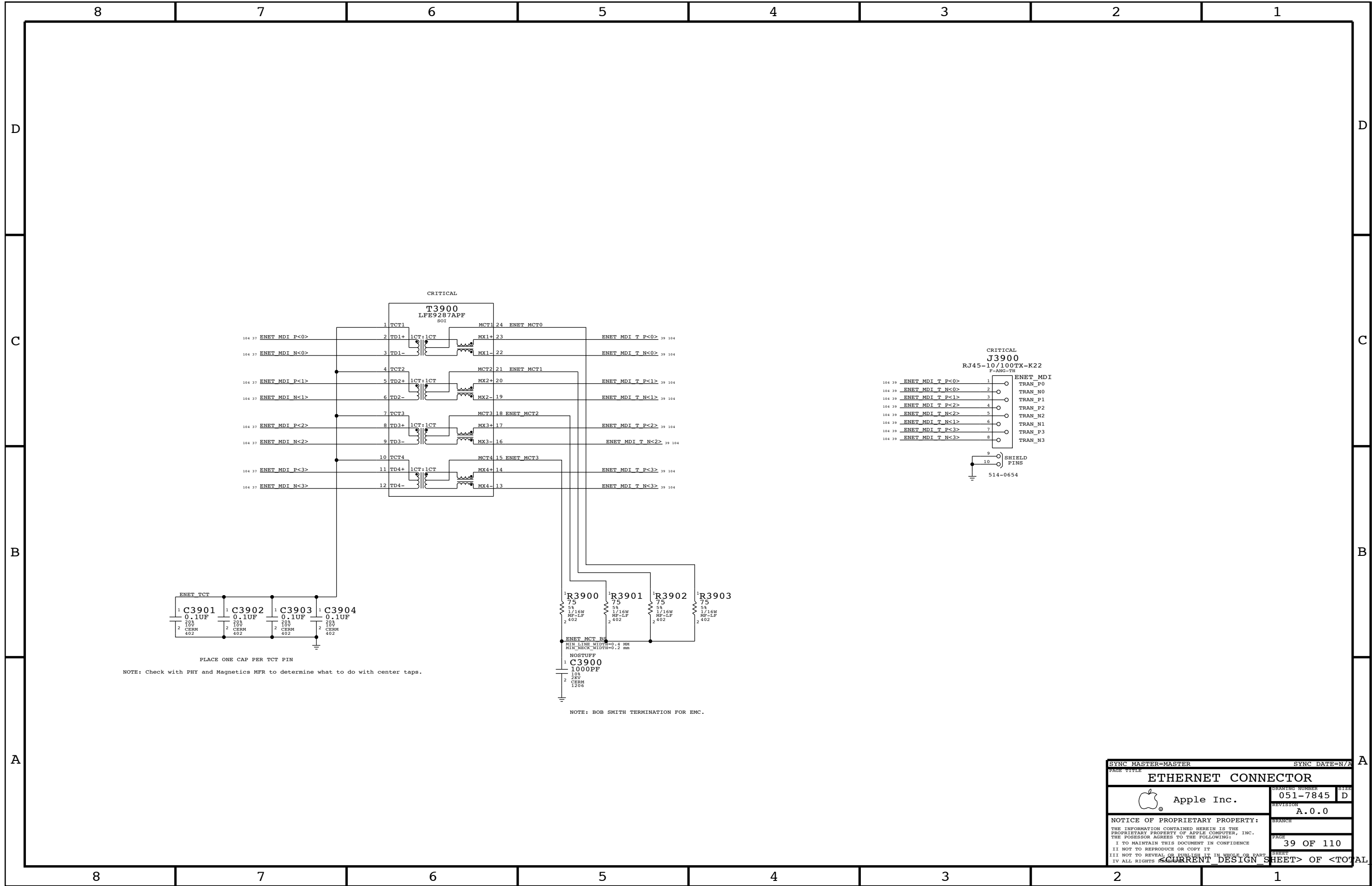


RTL8211 25MHz Clock

NOTE: MCP79 can provide 25MHz clock, but clock runs whenever RMGT rails are powered. Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.



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Ethernet Support			
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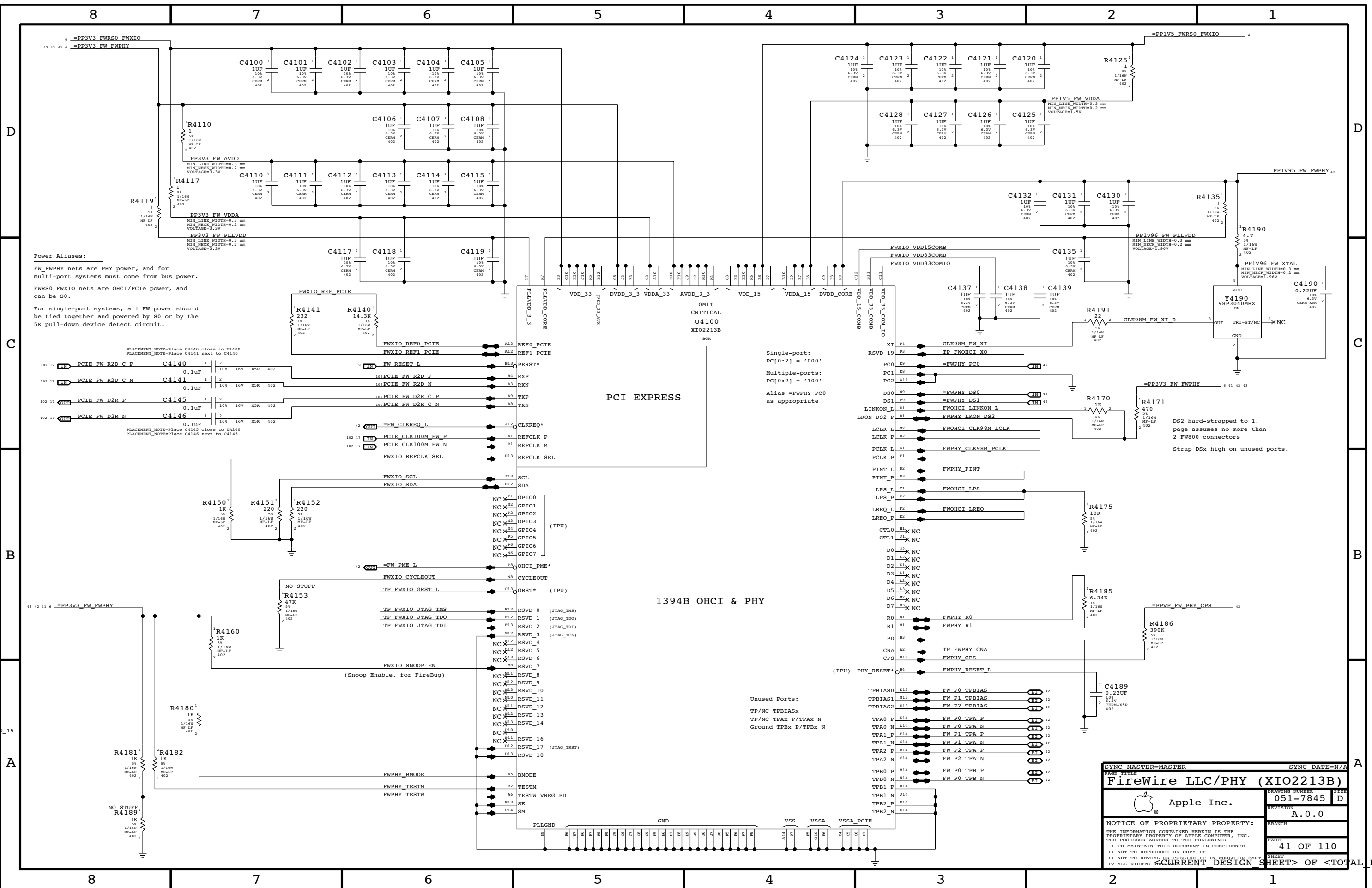
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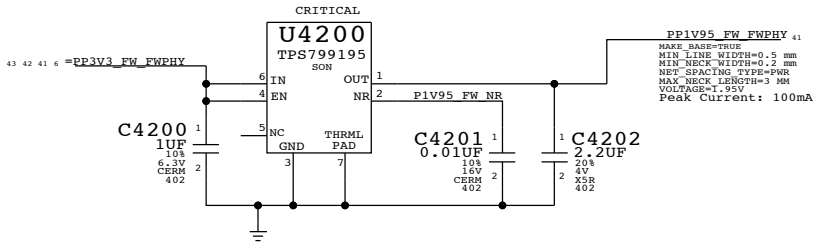
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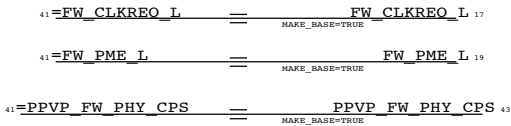
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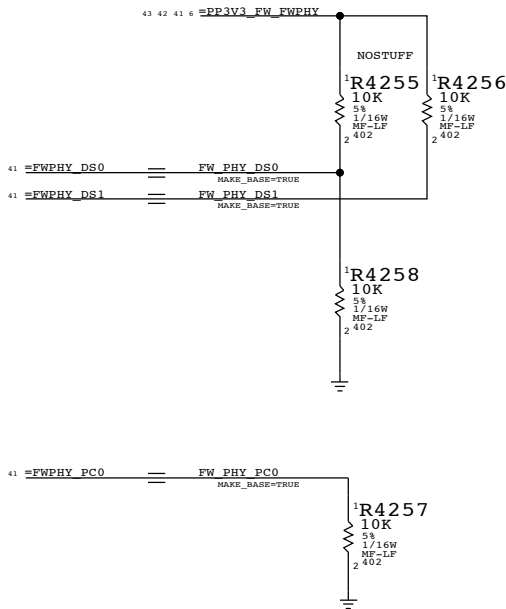
1394 PHY 1.95V SUPPLY



FireWire Aliases For Connectivity



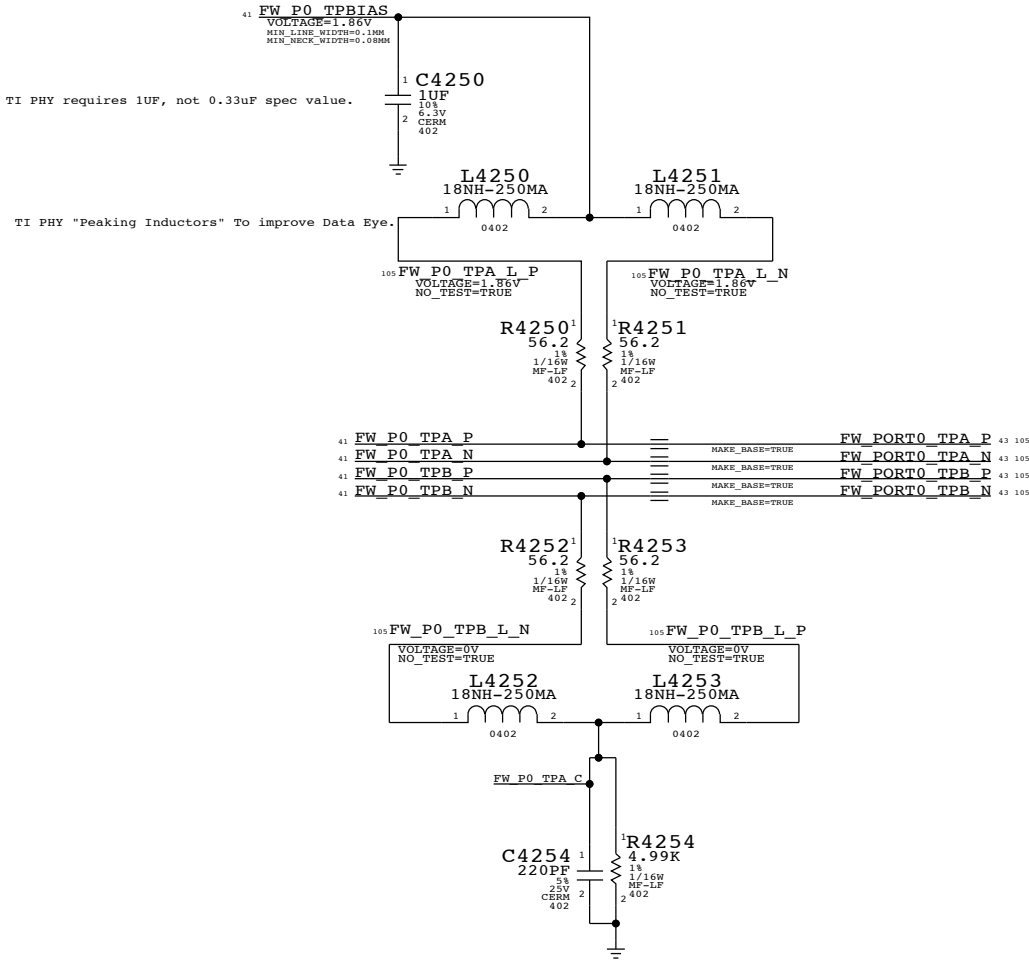
1394 PHY STRAPPING OPTIONS



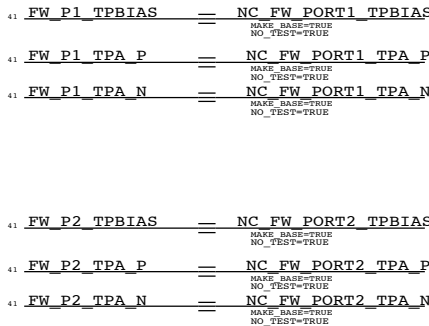
THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

iMacs are now one port only and have Power Code "000"

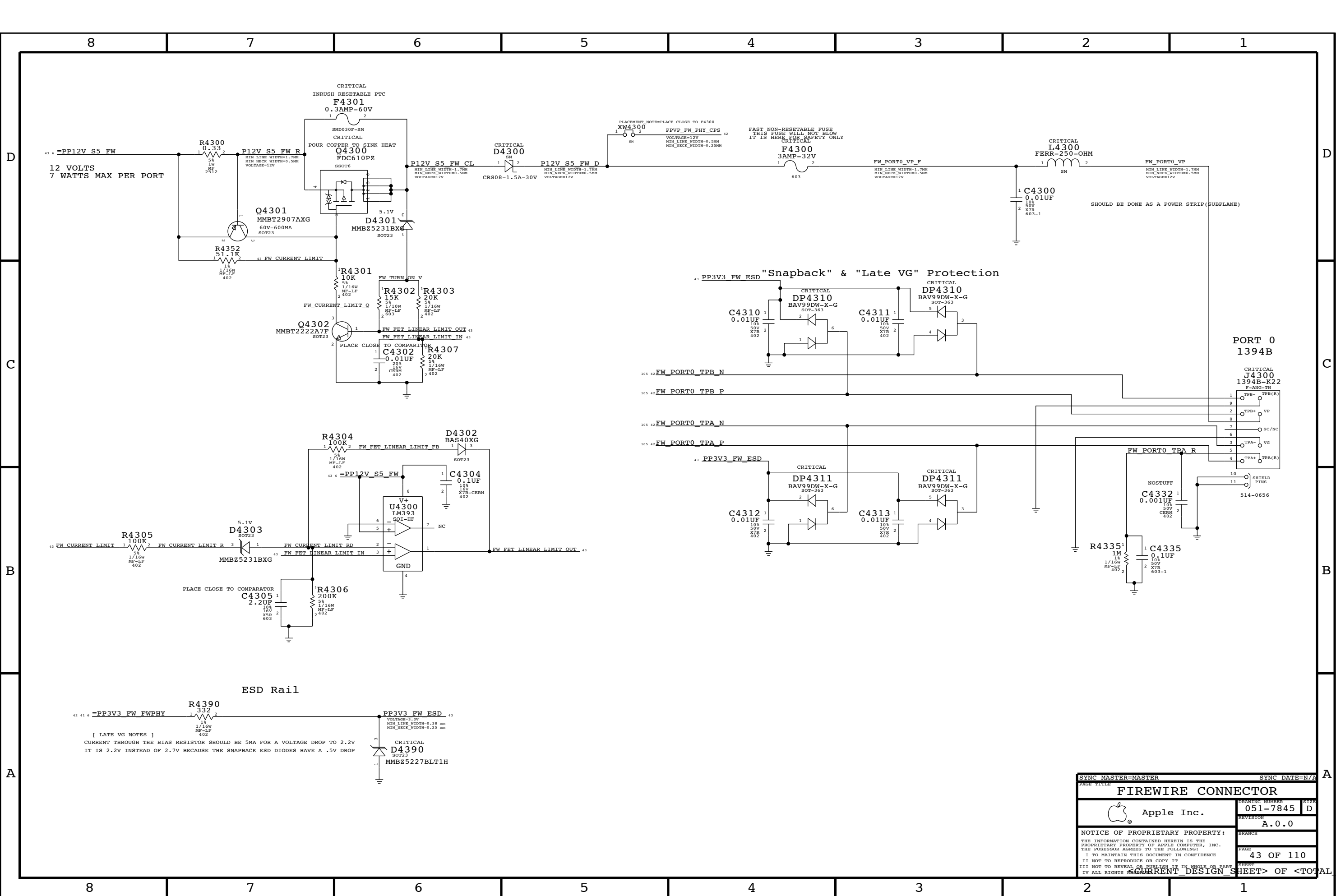
Termination
Place close to FireWire PHY



2ND & 3RD TPA/TPB PAIR UNUSED



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
FW: 1394B MISC			
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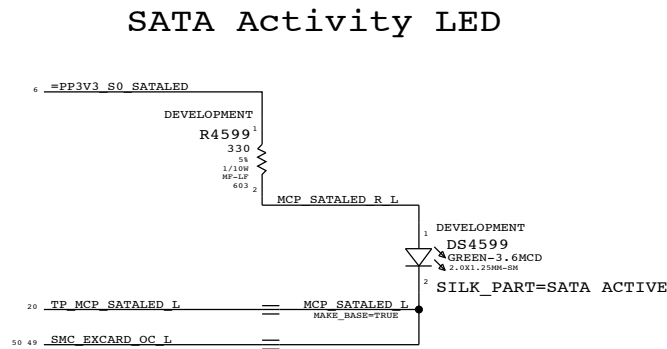
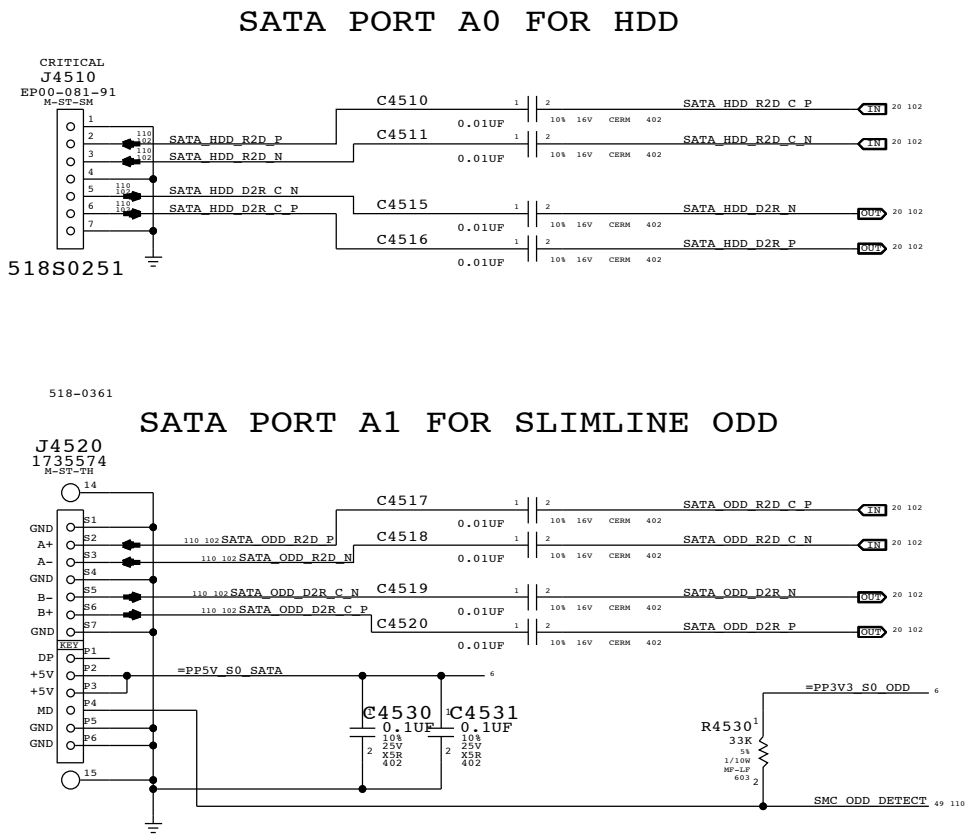
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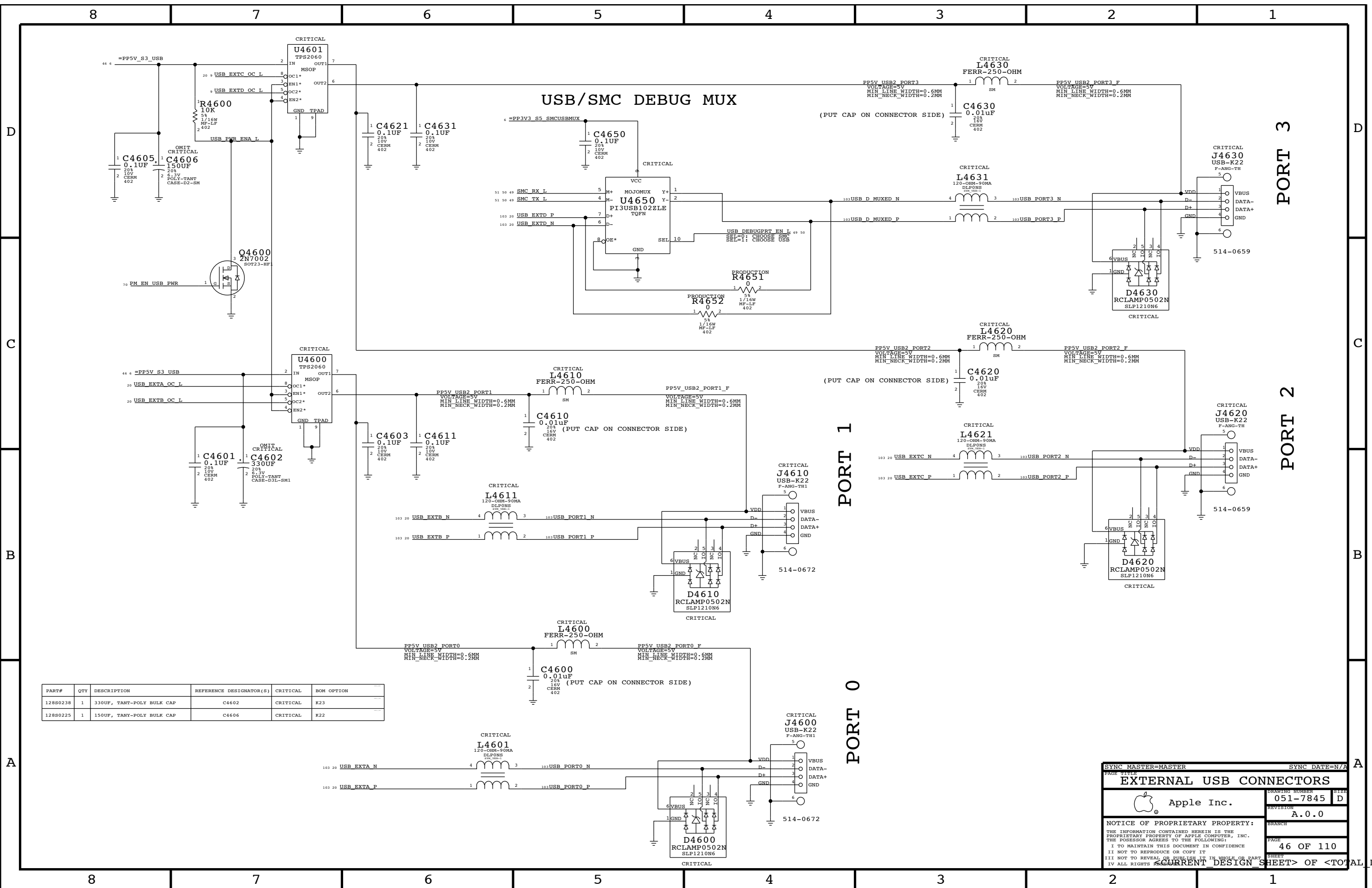
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<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>





PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
128S0238	1	330UF, TANT-POLY BULK CAP	C4602	CRITICAL	K23
128S0225	1	150UF, TANT-POLY BULK CAP	C4606	CRITICAL	K22

SYNC MASTER=MASTER

SYNC DATE=N/A

EXTERNAL USB CONNECTORS

Apple Inc.

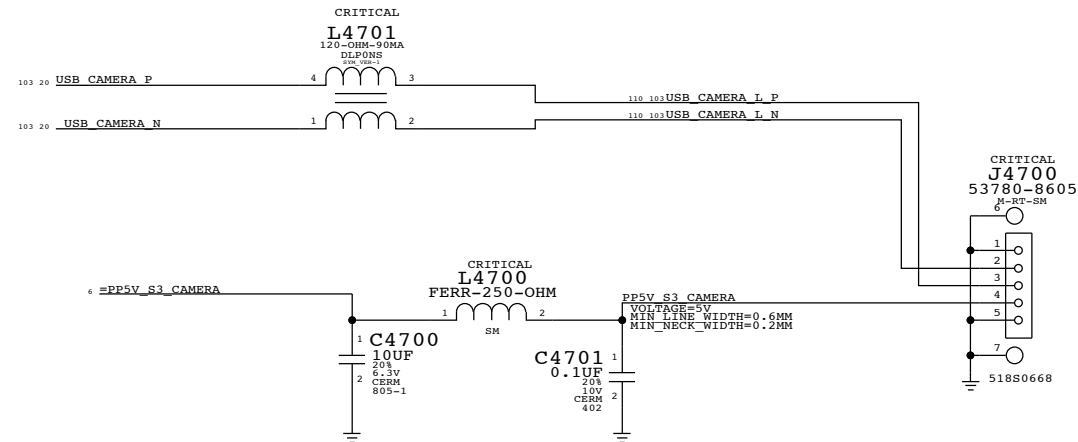
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SHEET

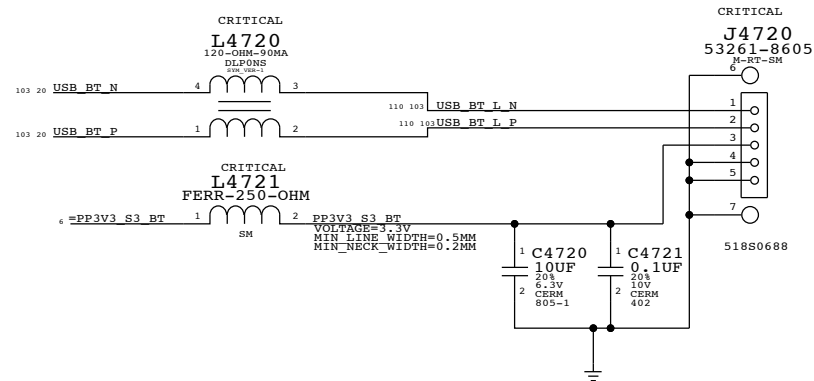
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CAMERA CONNECTOR & FILTER

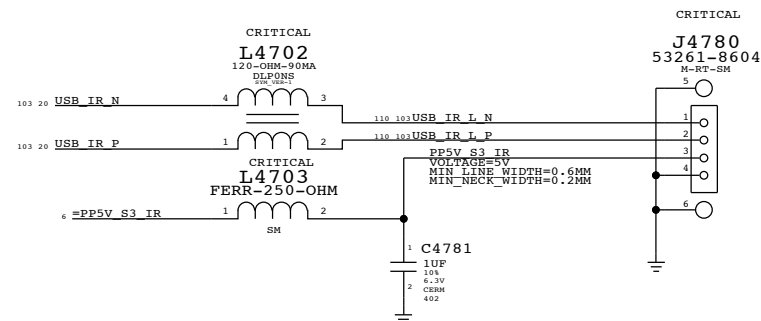


LAYOUT NOTE:
PLACE C4700, C4701 & L4700
NEAR J4700 PINS 4 AND 5 IN THE
ORDER LISTED, AND NOT ON
BOTH SIDES OF THE PIN.

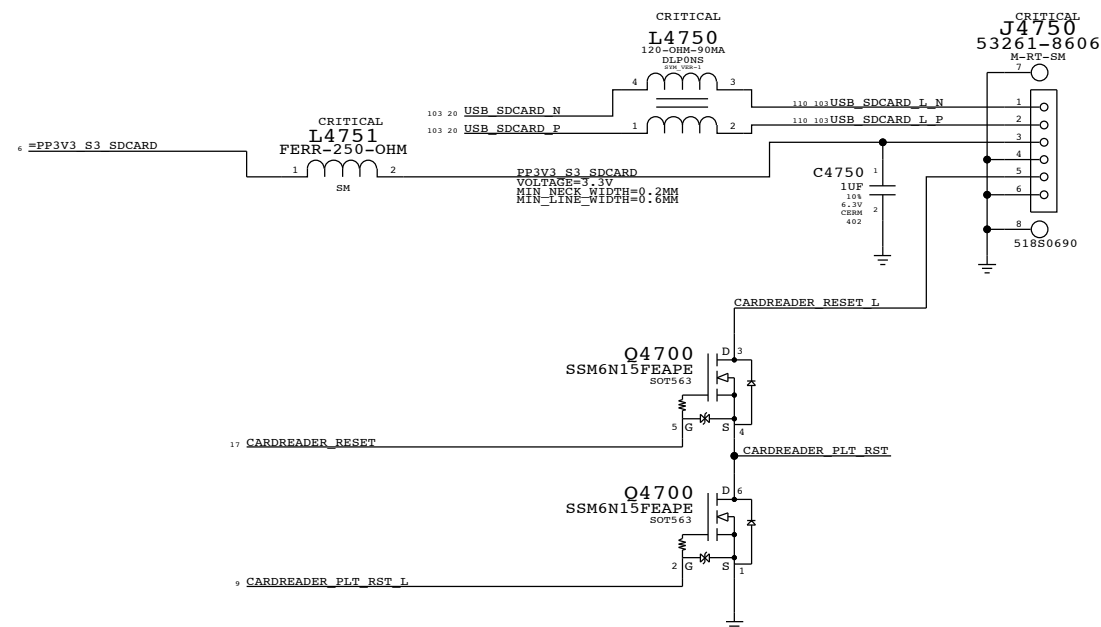
K37L (BLUETOOTH) CONNECTOR



IR RECEIVER CONNECTOR



SD Card Reader Board Connector

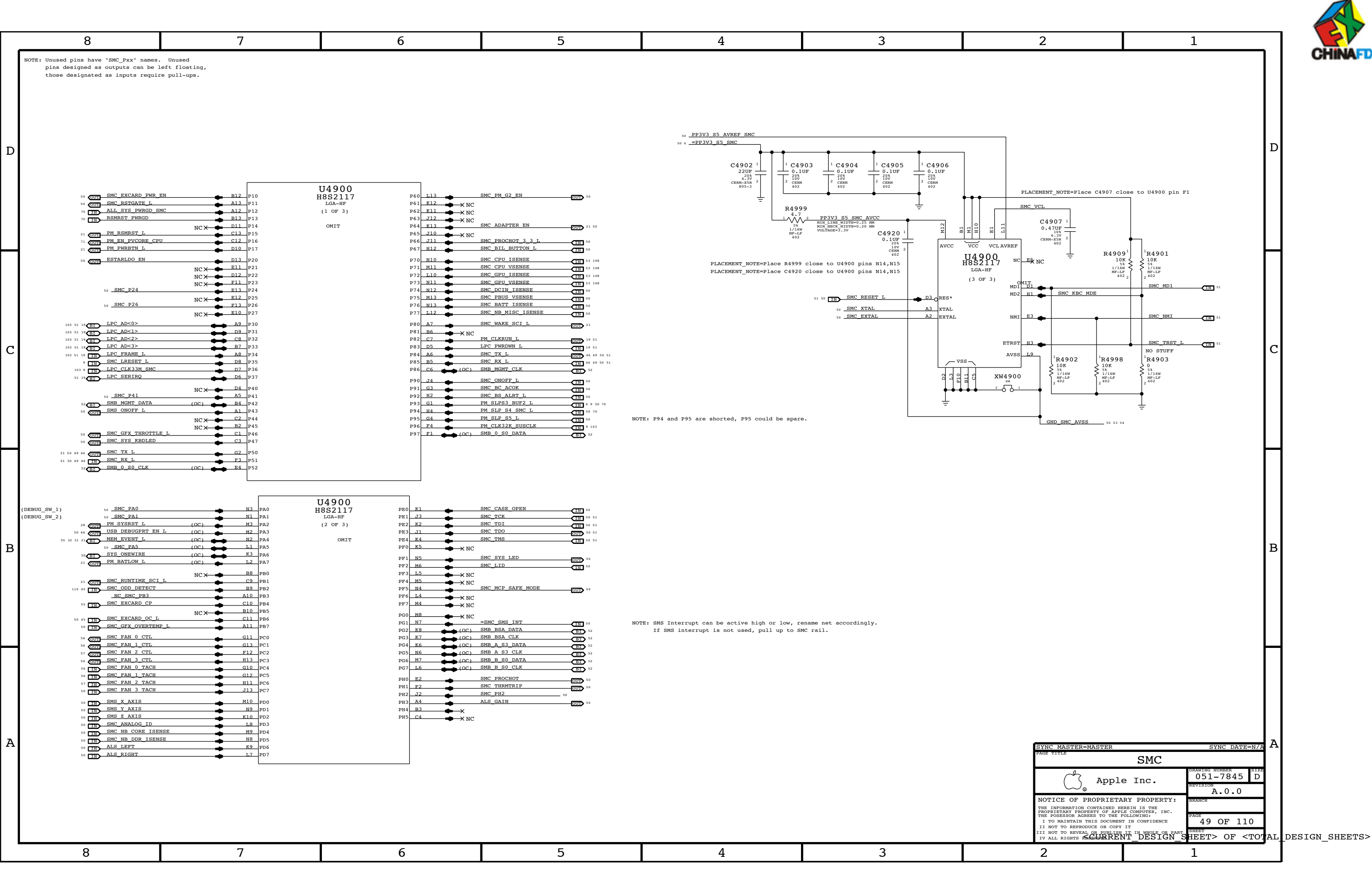


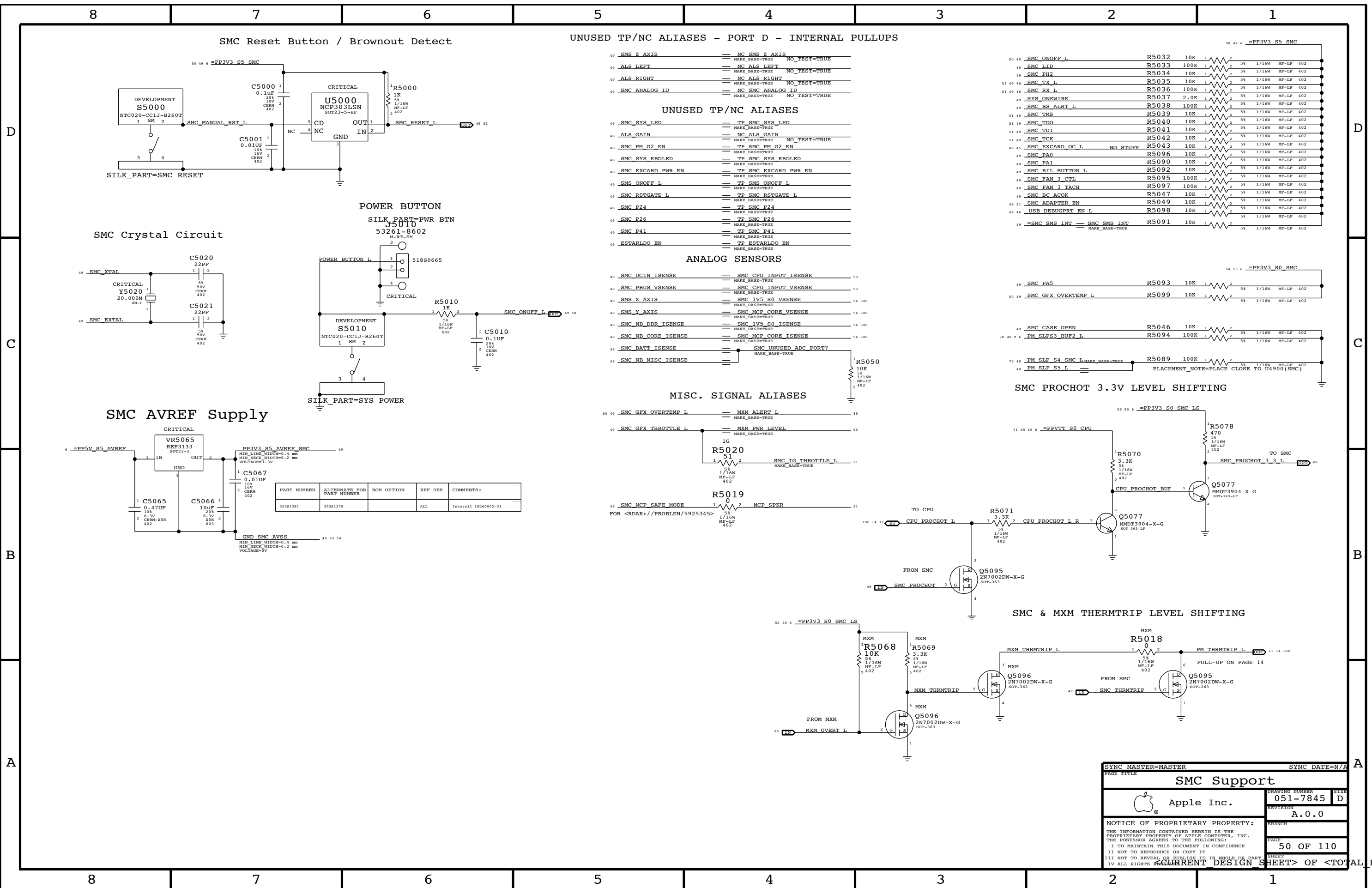


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SYNC MASTER=MASTER

SYNC DATE=N/A

PAGE TITLE

SMC Support

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DRAWING NUMBER
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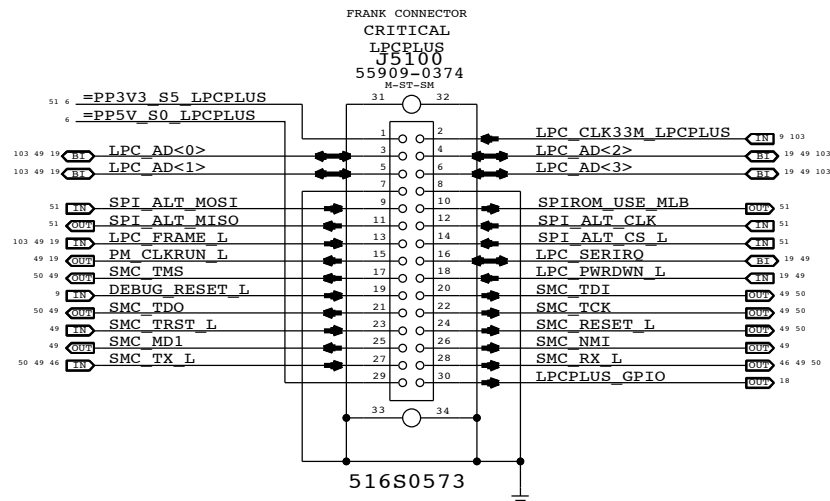
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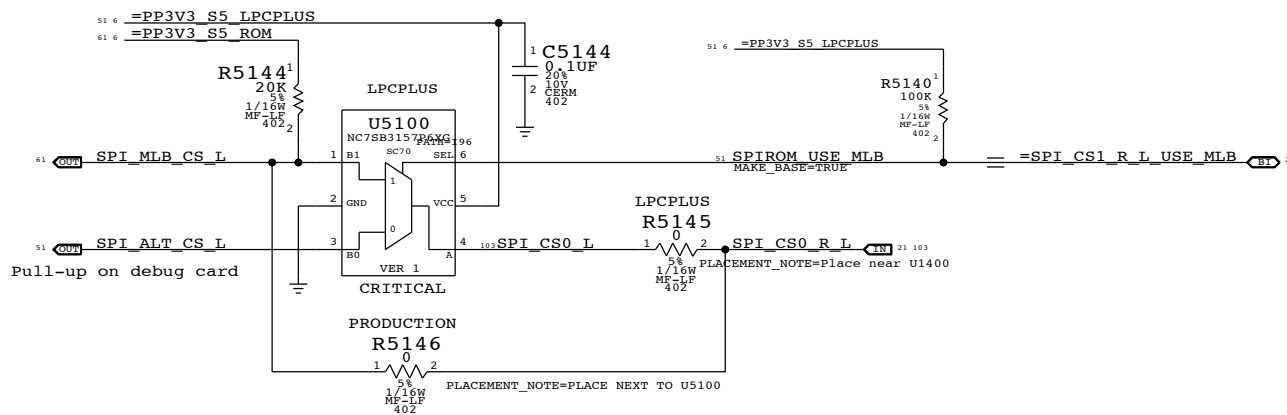
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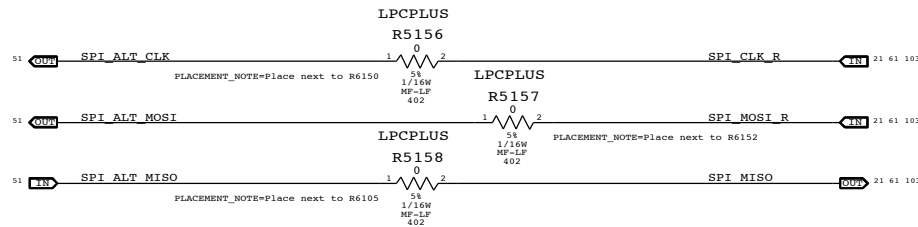
LPC+SPI Connector



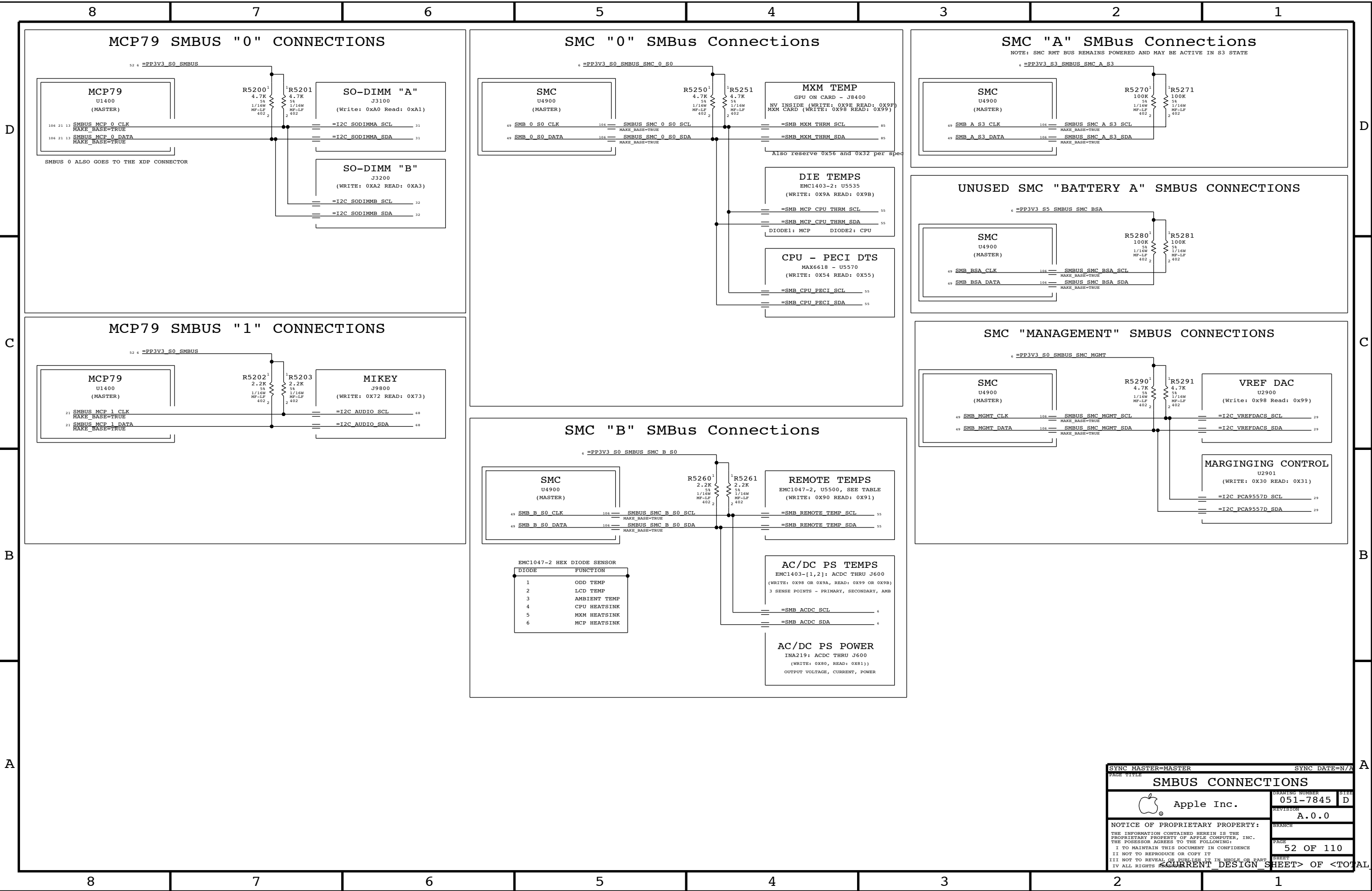
Alternate SPI ROM Support

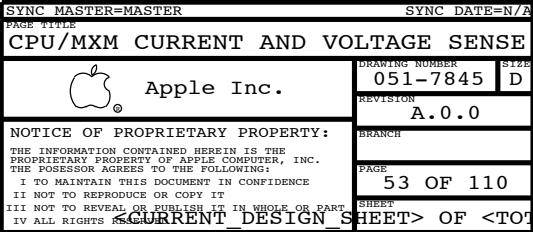


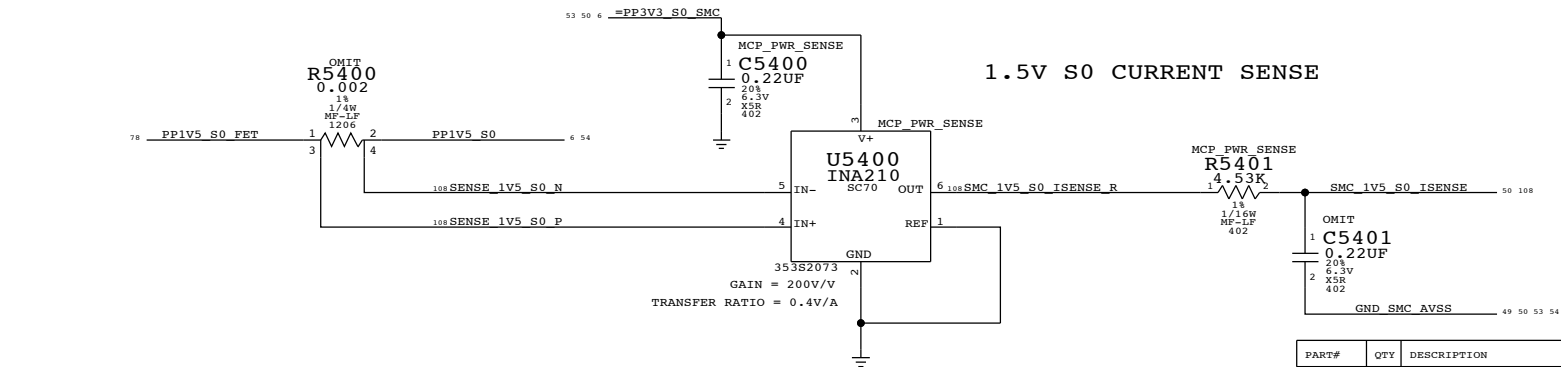
SPI Bus Series Resistance Option



SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE LPC+SPI Debug Connector			
DRAWING NUMBER 051-7845		SIZE D	
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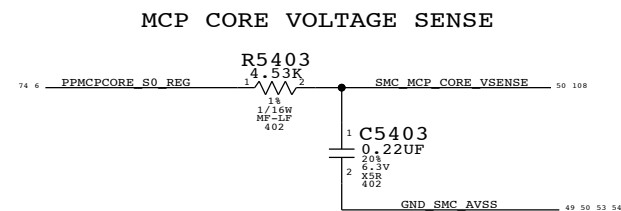
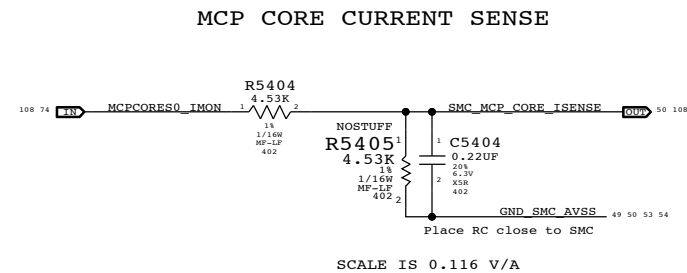
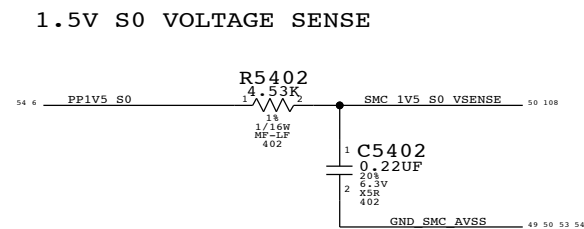






PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5400	CRITICAL	MCP_PWR_SENSE
101S0414	1	RES,0 OHM,1206,20MILLIOHM MAX	R5400	CRITICAL	PRODUCTION

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
132S0080	1	CAP, 0.22UF, 0402	C5401		MCP_FWR_SENSE
116S0004	1	RES, 0 OHM, 0402	C5401		PRODUCTION

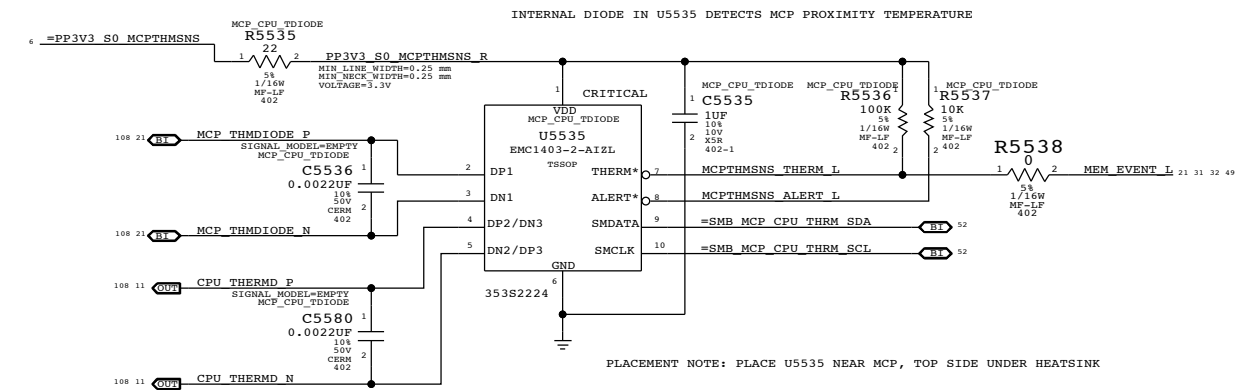
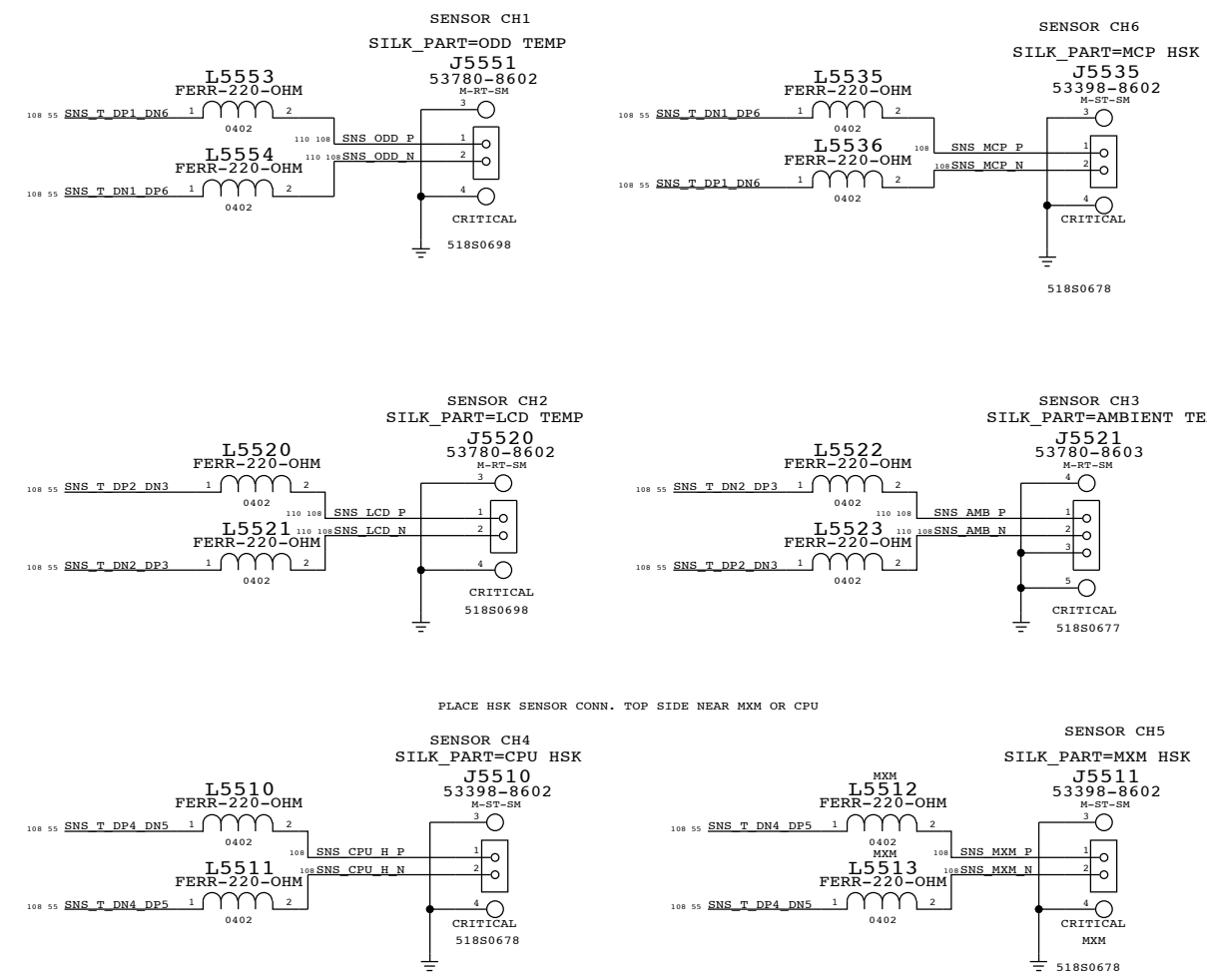


SYNC MASTER-K51		SYNC DATE=12/08/2008	
PAGE 111111			
MCP CURRENT AND VOLTAGE SENSE			
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MCP CURRENT DESIGN SHEET		TOTAL DESIGN SHEETS>	

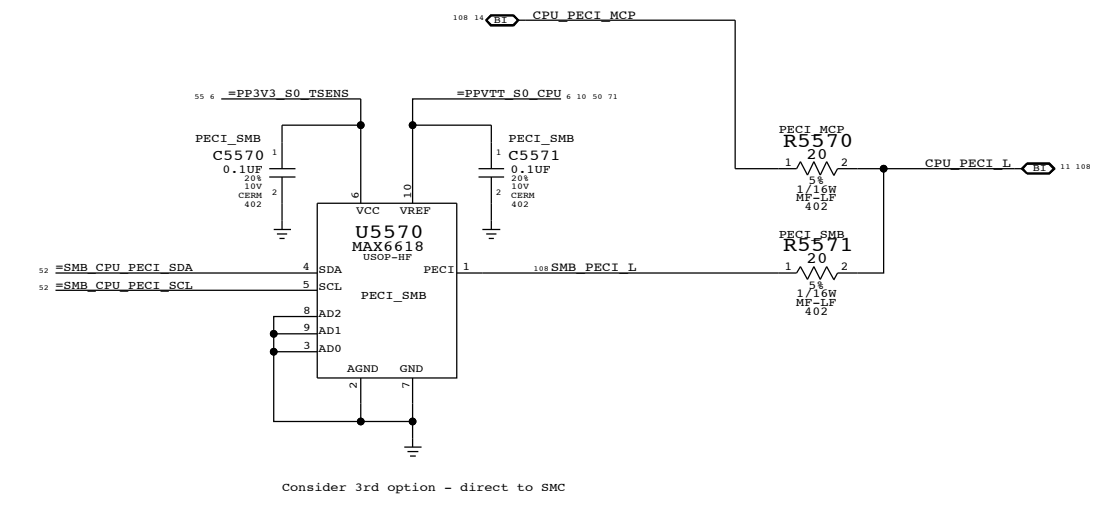


REMOTE THERMAL SENSORS HEATSINKS, AMBIENT, PANEL AND ODD

MCP & CPU T-Diode Thermal Sensor

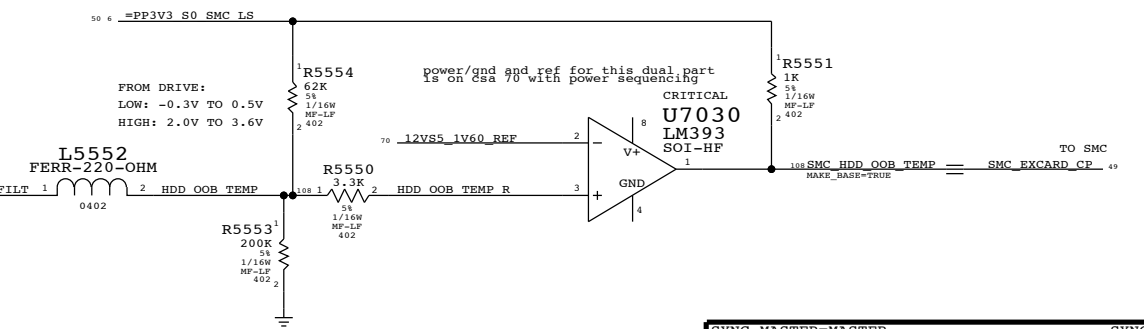
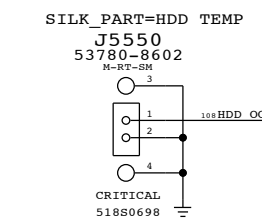
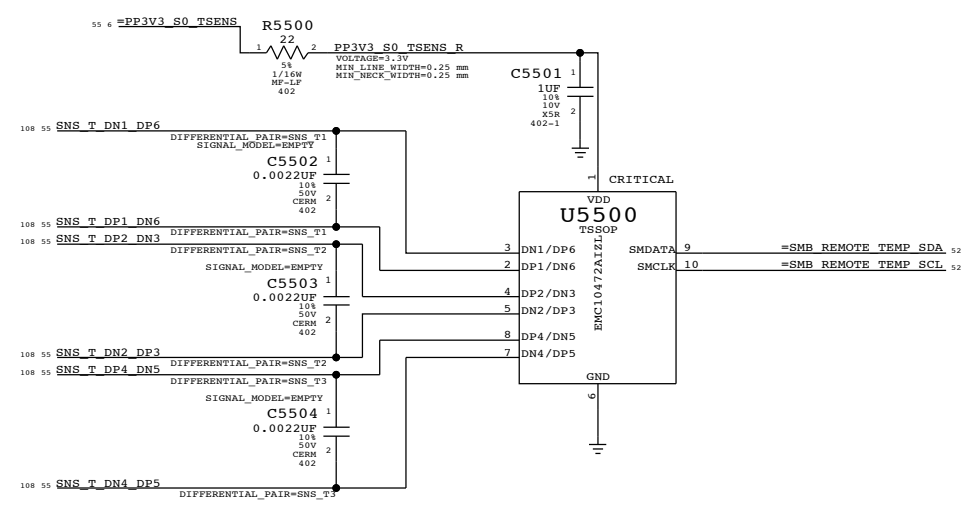


CPU PECCI DTS OPTIONS



REMOTE THERMAL SENSORS (HEATSINKS AND ODD)

HDD OUT OF BAND TEMPERATURE SENSING LEVEL SHIFTING

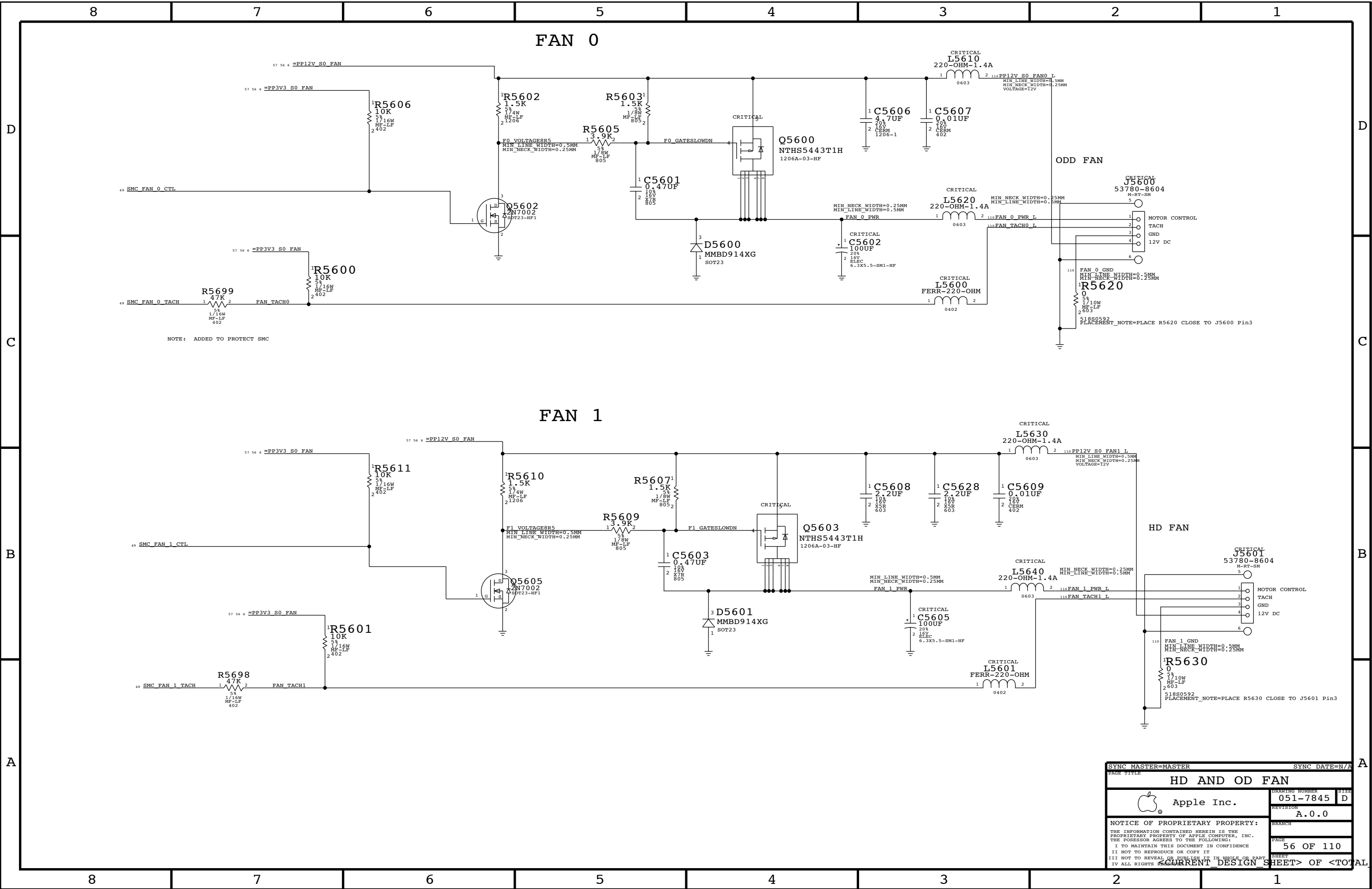


Drive active = valid signal and protocol
Drive asleep = HDD drives and protocols
Drive disconnected = pulled high

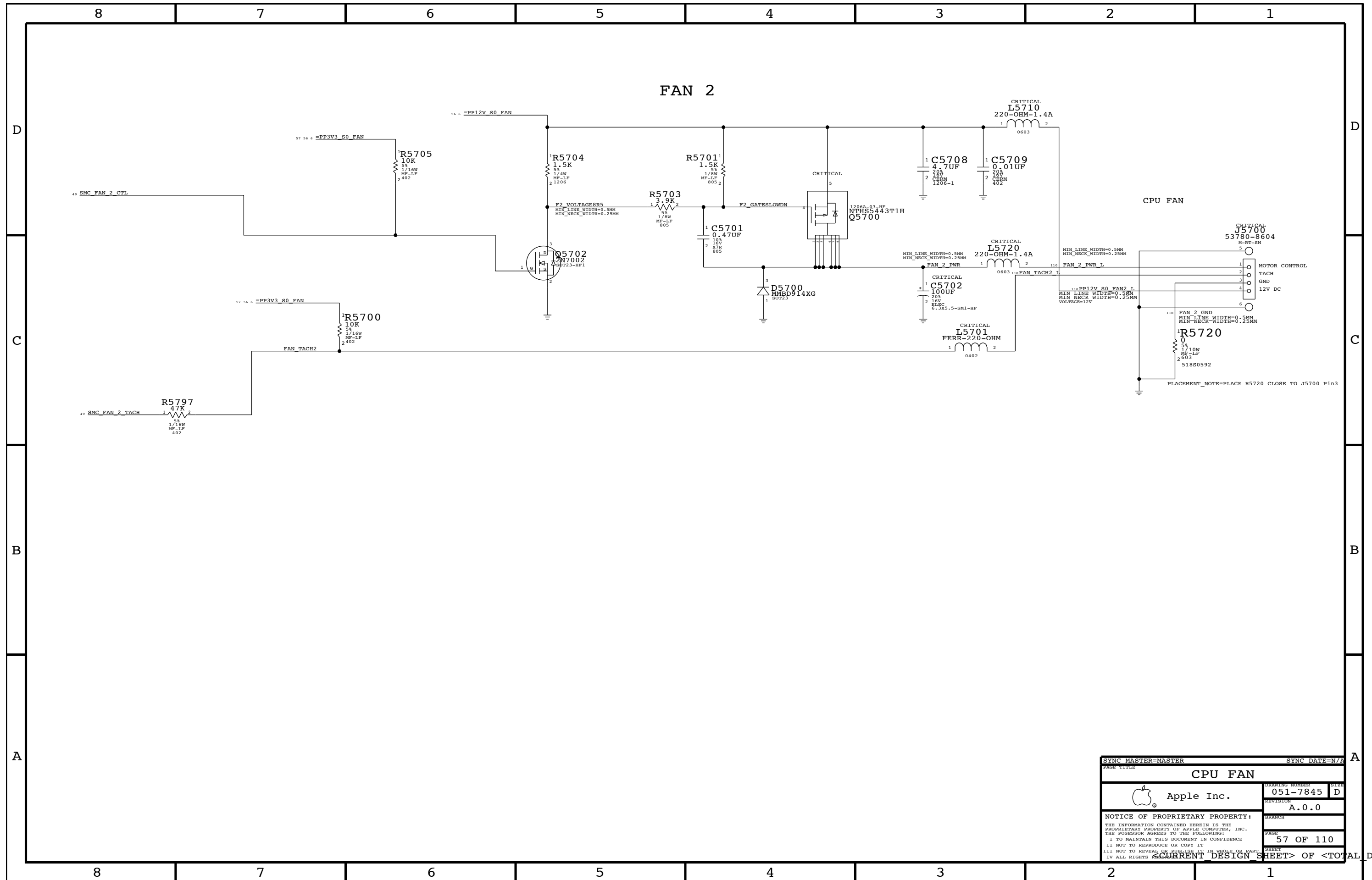
Cannot pull low because some drives use this bit to determine 1.5 Gbps vs. 3.0 Gbps SATA

Must pull high to 2.5V for compatibility with all drives

SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
Thermal Sensors		DRAWING NUMBER	
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SYNC MASTER=MASTER		SYNC DATE=N/A			
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HD AND OD FAN					
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SYNC DATE=12/08/2008

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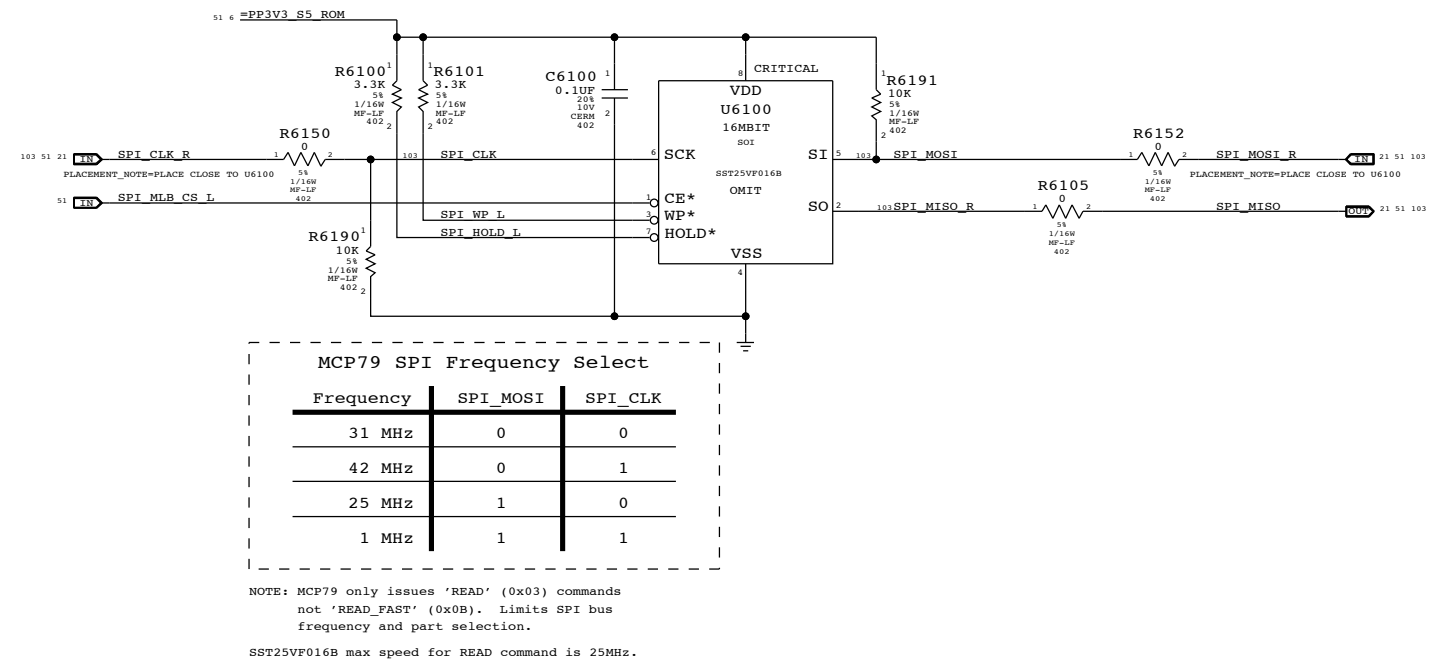


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8 7 6 5 4 3 2 1

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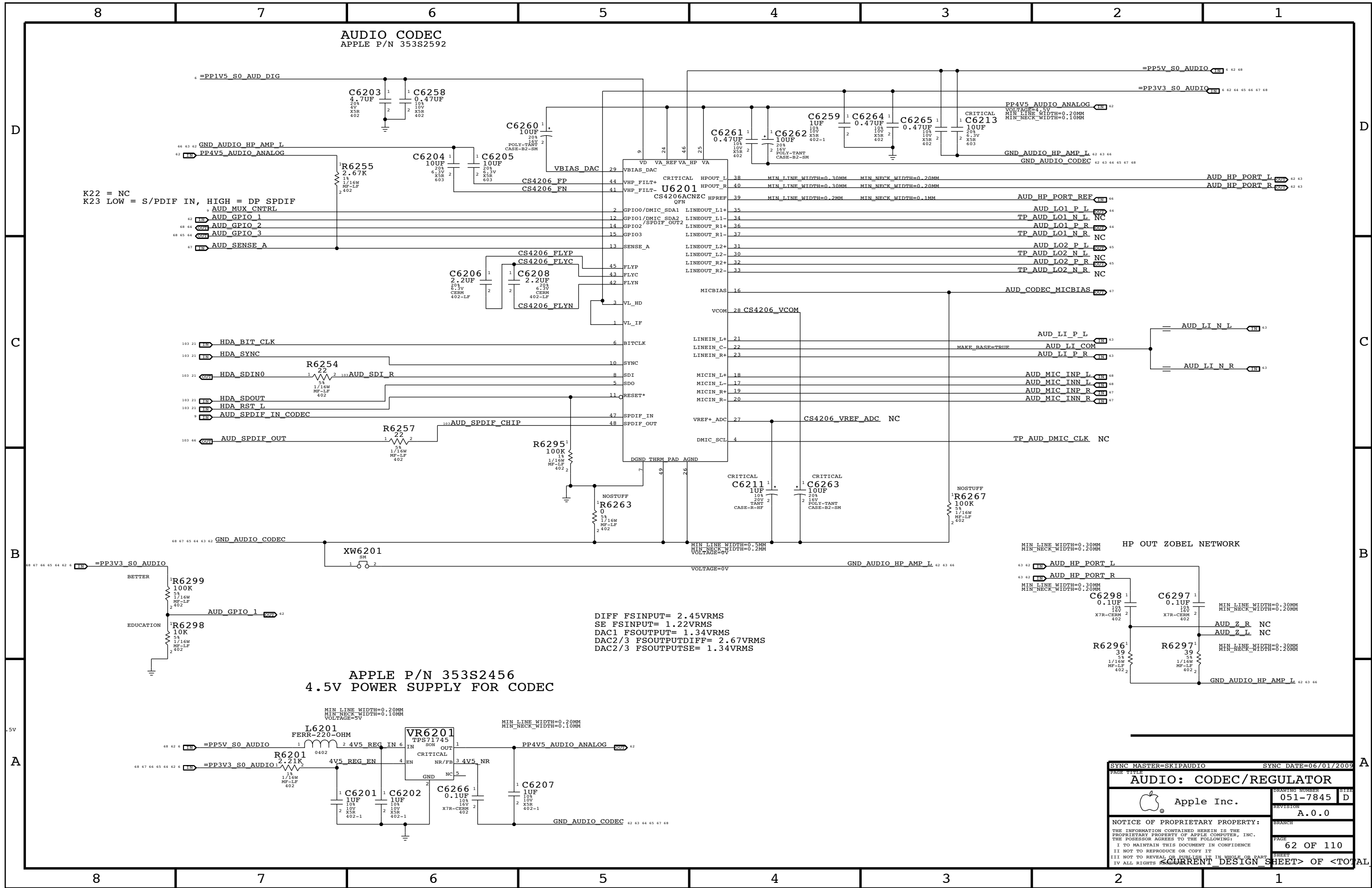
Frequency	SPI_MOSI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: MCP79 only issues 'READ' (0x03) commands
not 'READ_FAST' (0x0B). Limits SPI bus
frequency and part selection.

SST25VF016B max speed for READ command is 25MHz.

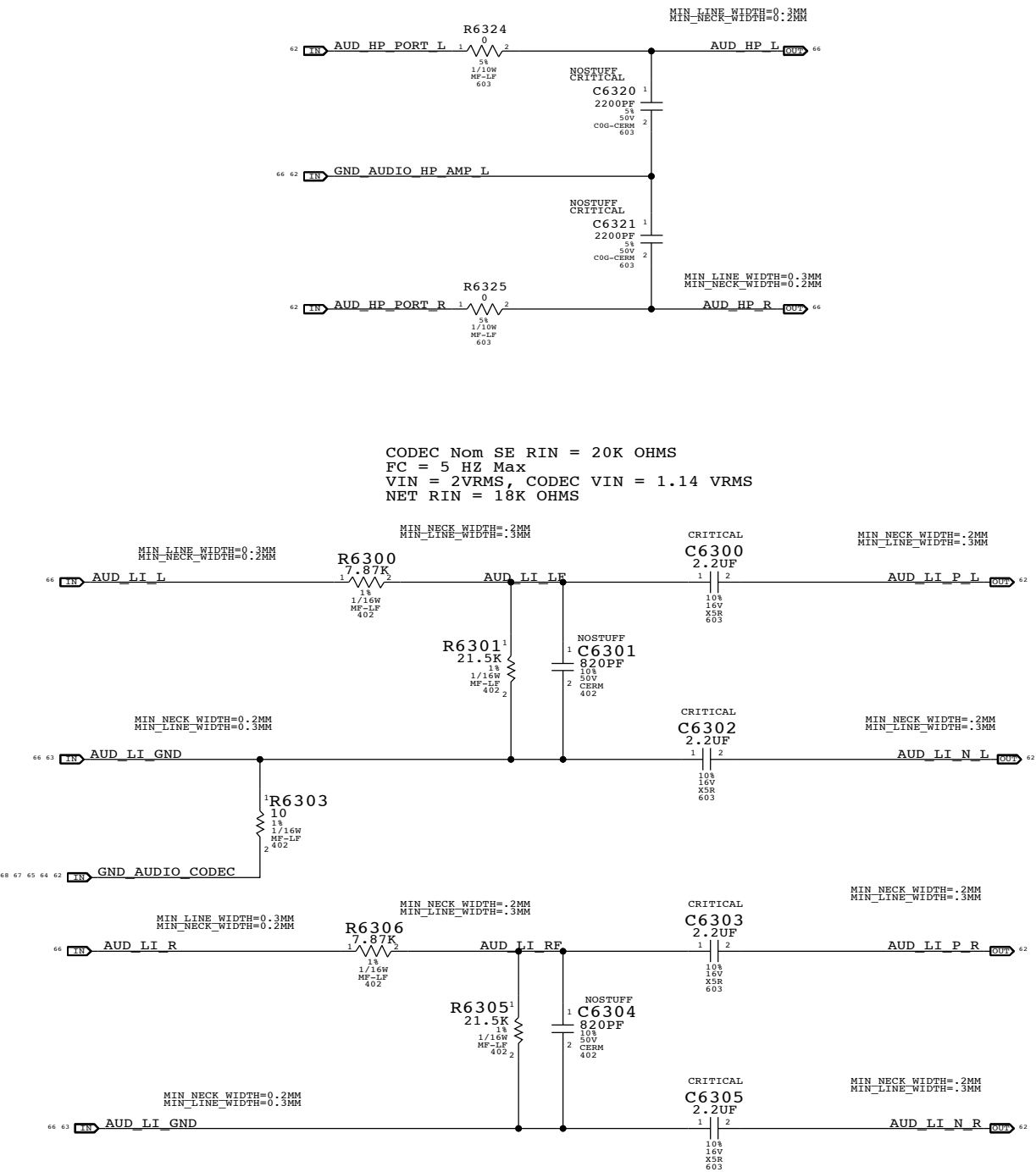
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SPI ROM			
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PAGE TITLE		SYNC DATE=06/01/2009	
AUDIO: CODEC/REGULATOR		DRAWING NUMBER	
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1ST ORDER DAC FILTER PLACEHOLDER



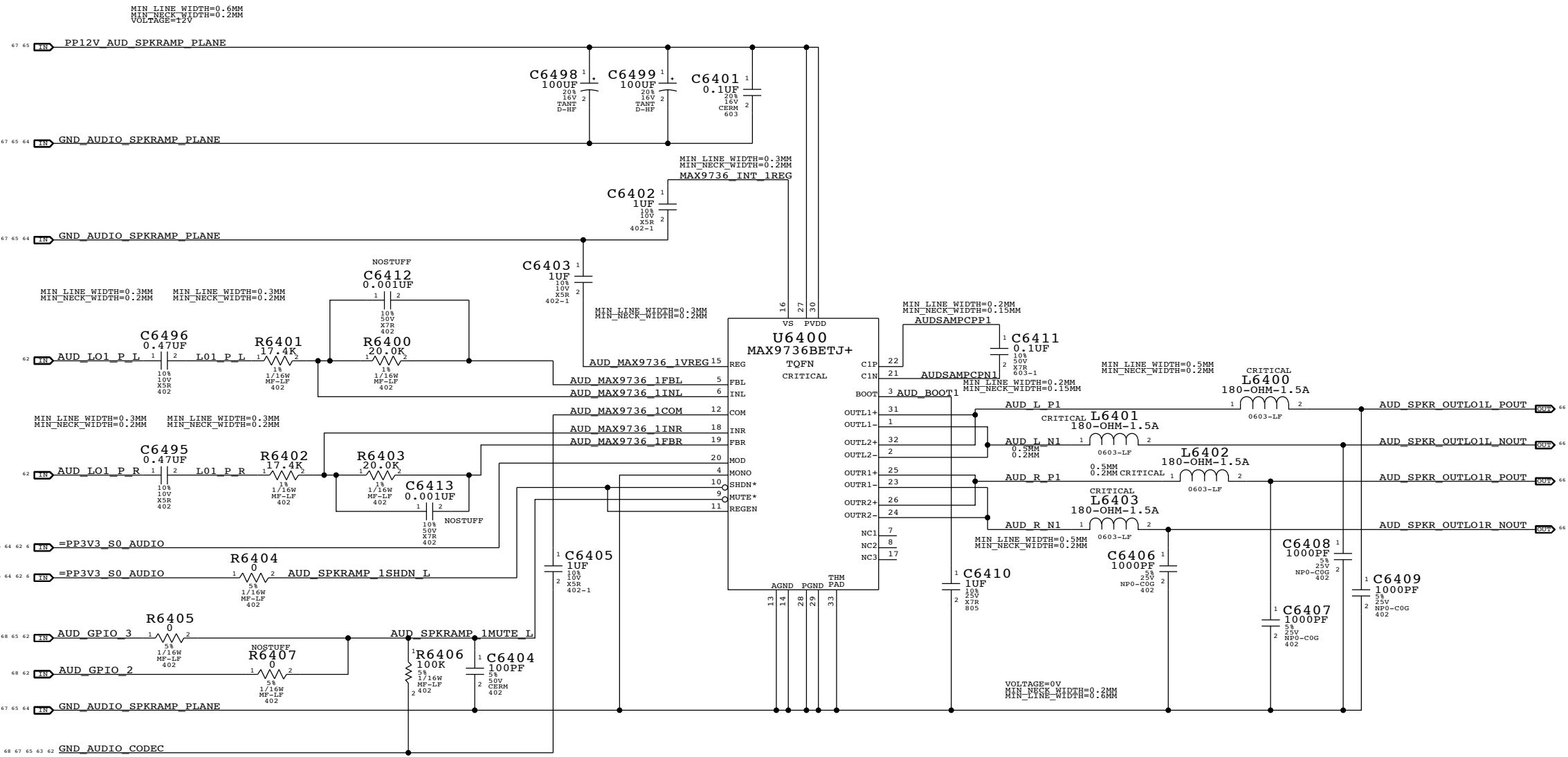
CODEC Nom SE RIN = 20K OHMS
FC = 5 HZ Max
VIN = 2VRMS, CODEC VIN = 1.14 VRMS
NET RIN = 18K OHMS

SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: FILTER/BUFFER			
	Apple Inc.		DRAWING NUMBER
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SHEET			
SHEET> OF <TOTAL>			

TWEETER SPEAKER AMPLIFIER

MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
 CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
 AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
 FC = 19.5 HZ
 POUT = 6.76 W INTO 8 OHMS @ 1% THD+N



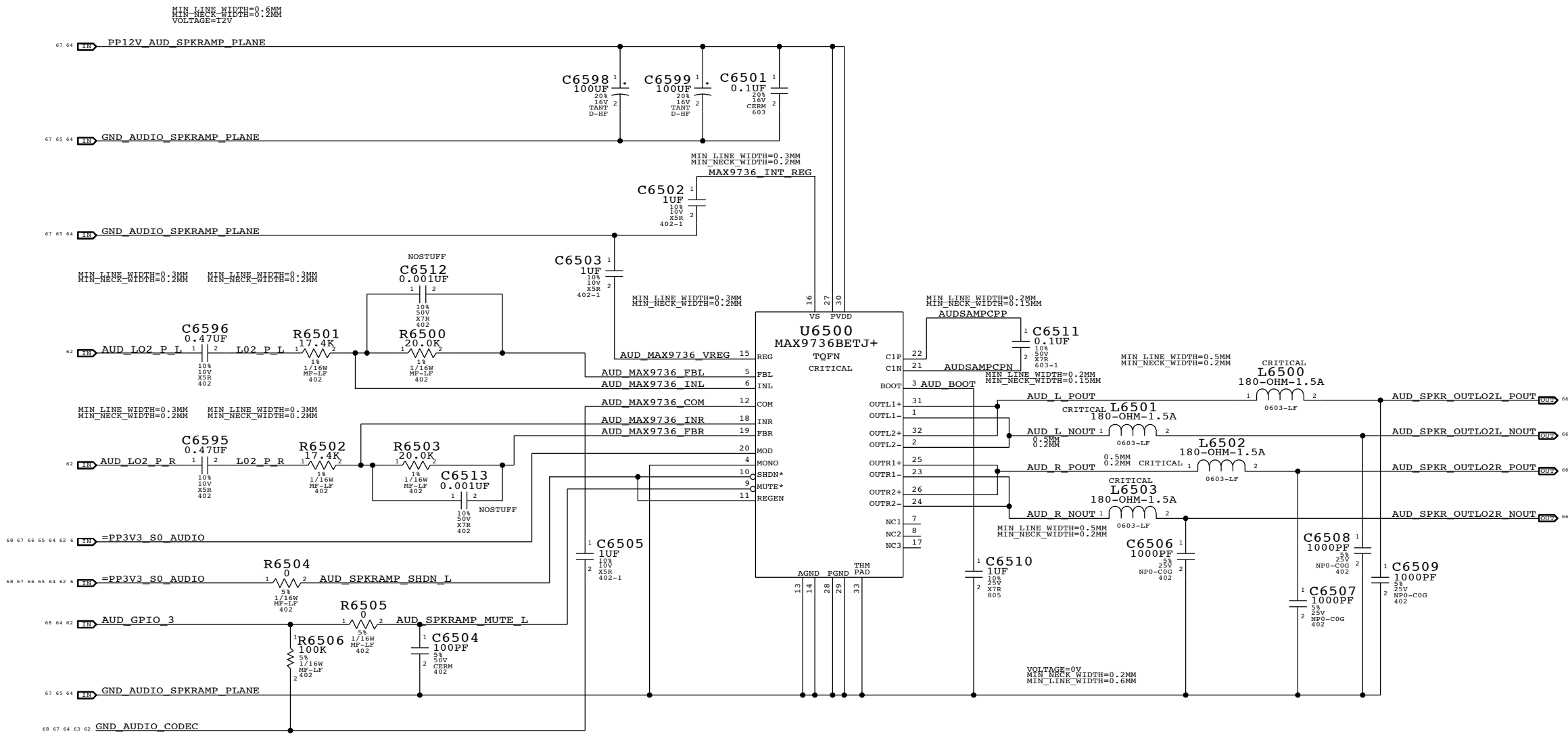
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PAGE TITLE			
AUDIO: SPEAKER AMP			
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		REVISION	A.0.0
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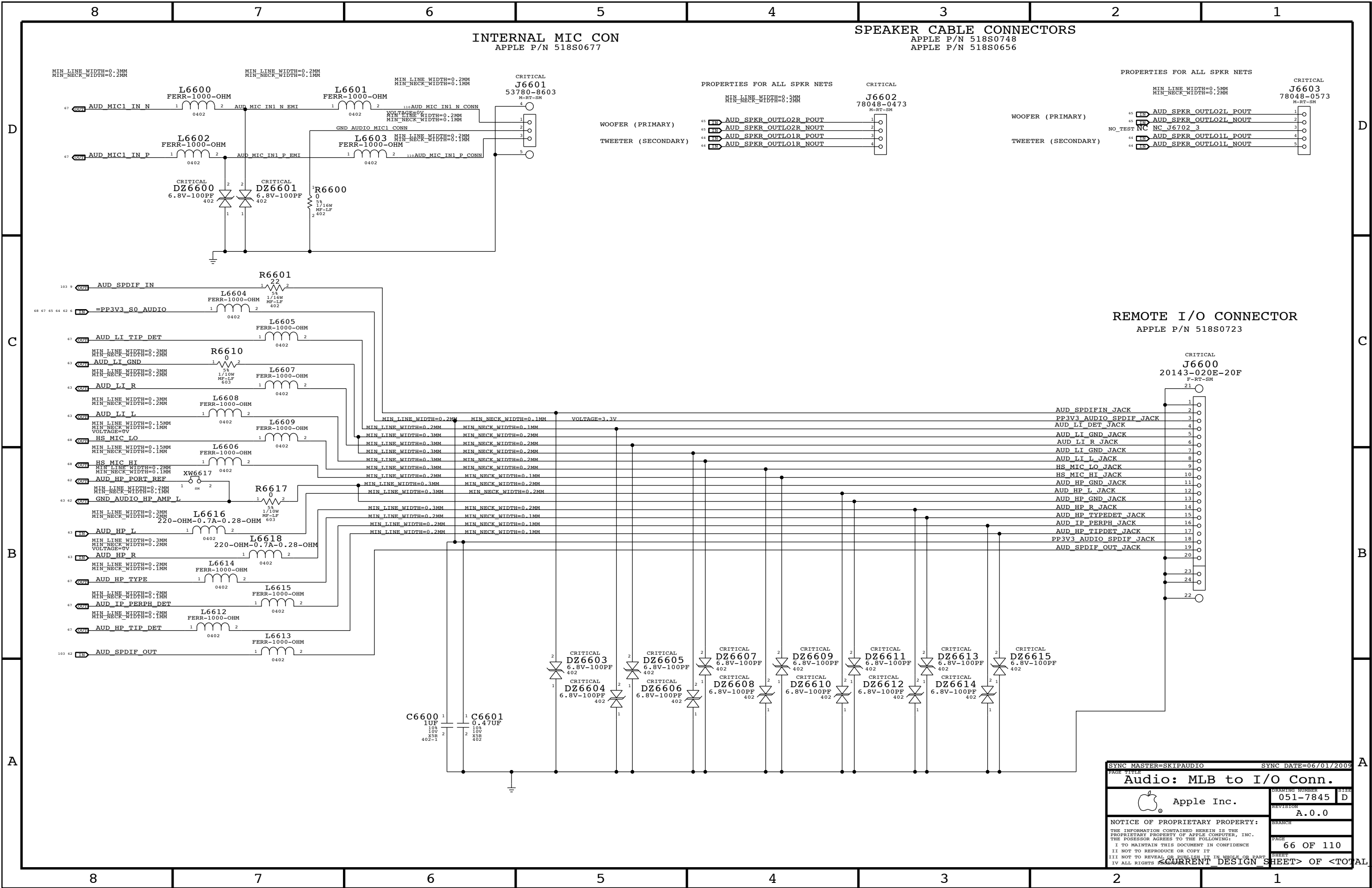
WOOFER SPEAKER AMPLIFIER

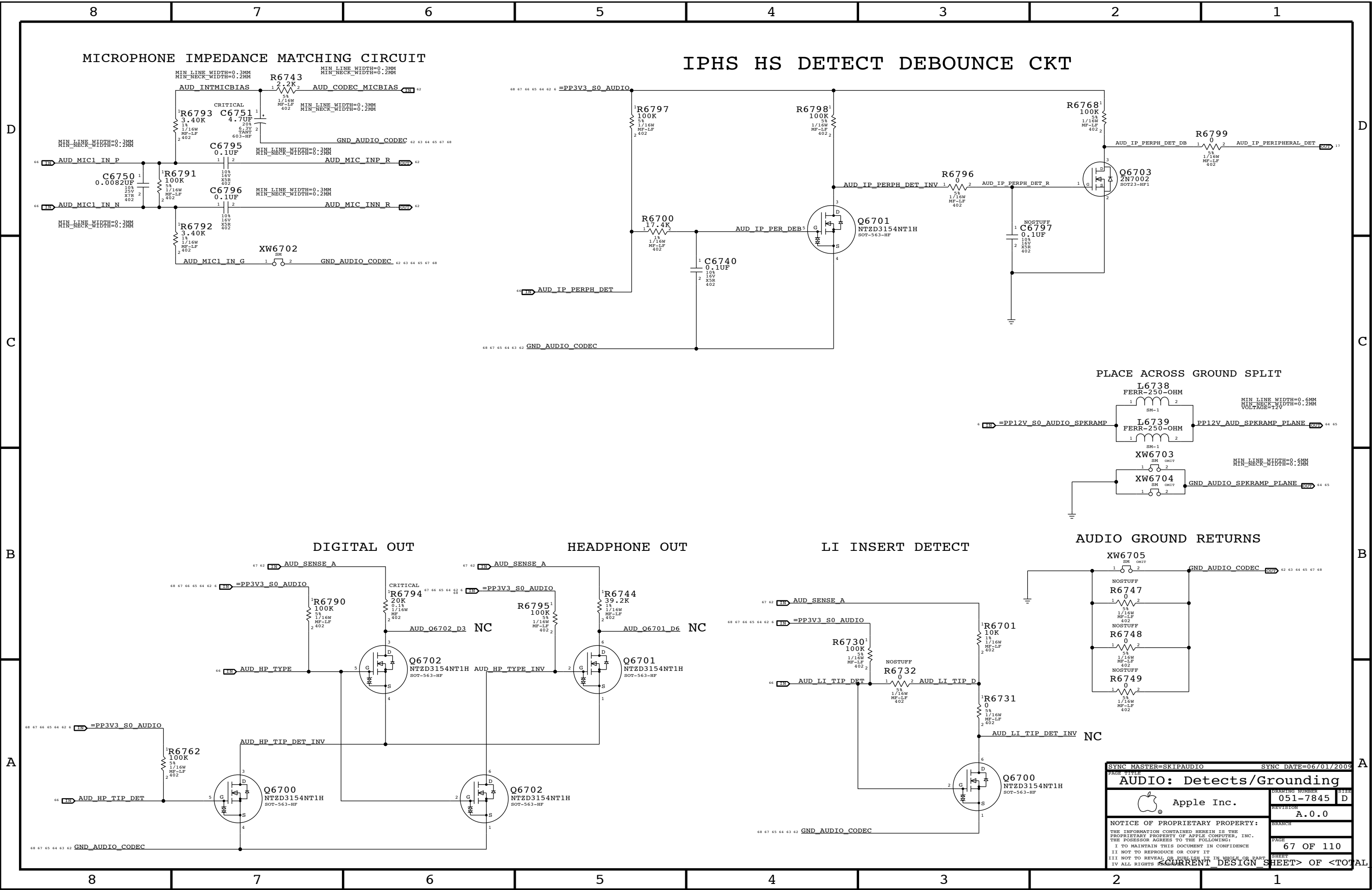
MAX9736B APN:353S2042

GAIN = -4.8(20K/17.4K) TURN ON TIME: 110MS
CODEC OUT = 1.335VRMS TURN ON DELAY: 150MS
AMP VOUT = 7.355VRMS RIN = 17.4 OHMS
POUT = 6.76 W INTO 8 OHMS @ 1% THD+N FC = 19.5 HZ



SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: SPEAKER AMP			
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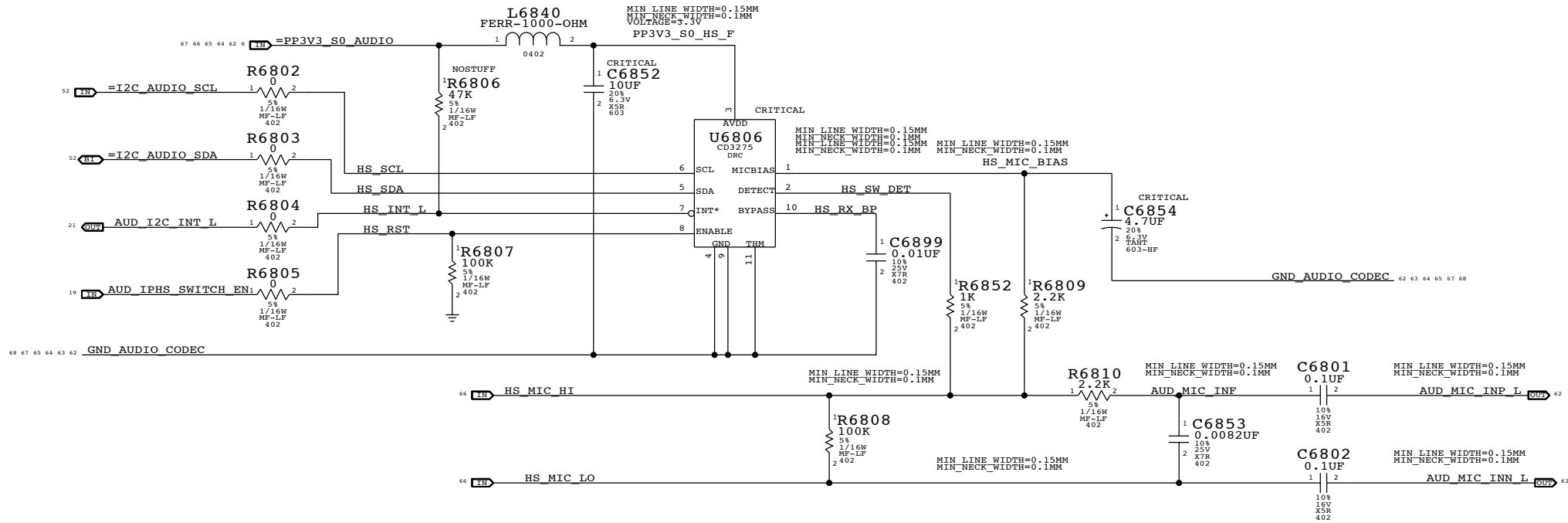


SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
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AUDIO: Detects/Grounding			
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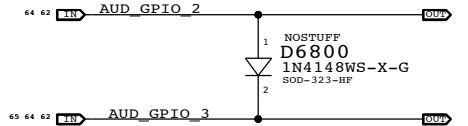
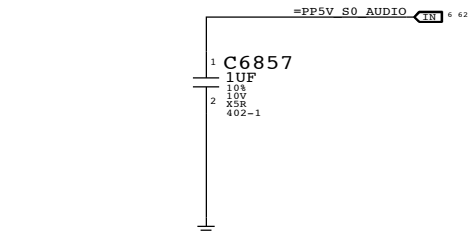
FUNCTION	PIN	CONVERTER	VOLUME	ENABLE/ CNTRL TYPE	DETECT/ INTERRUPT
PRIMARY	0X0B	0X04	0X04	GPIO 3	N/A
SECONDARY	0X0A	0X03	0X03	GPIO 3	N/A
HEADPHONES	0X09	0X02	0X02	N/A	0X09 (A)
LINE INPUT	0X0C	0X05	0X05	N/A	LINE IN
BUILT-IN MICROPHONE	0X0D (13,B,RIGHT)	0X06	0X06	MICBIAS 80%	N/A
HEADSET MICROPHONE	0X0D (13,V22,B,LEFT)	0X06	0X06	MIKEY	MIKEY
SPDIF OUT	0X10	0X08	N/A	N/A	0X0C (B)
SPDIF IN	0X0F	0X07	N/A	N/A	N/A
MIKEY	N/A	N/A	N/A	MCP GPIO_38	MCP GPIO_5

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256



FLP = 8.82 KHZ
FHP = 80 HZ



SYNC MASTER=SKIPAUDIO		SYNC DATE=06/01/2009	
PAGE TITLE			
AUDIO: Mikey			
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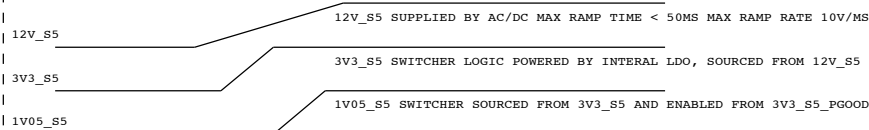
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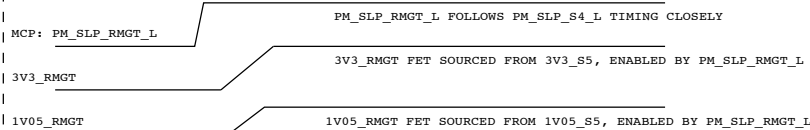
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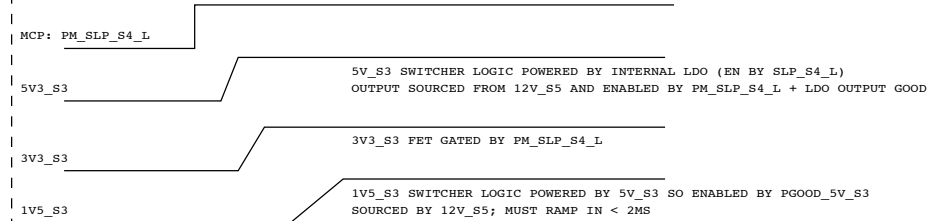
S5 POWER RAIL SEQUENCING



RMGT POWER RAIL SEQUENCING

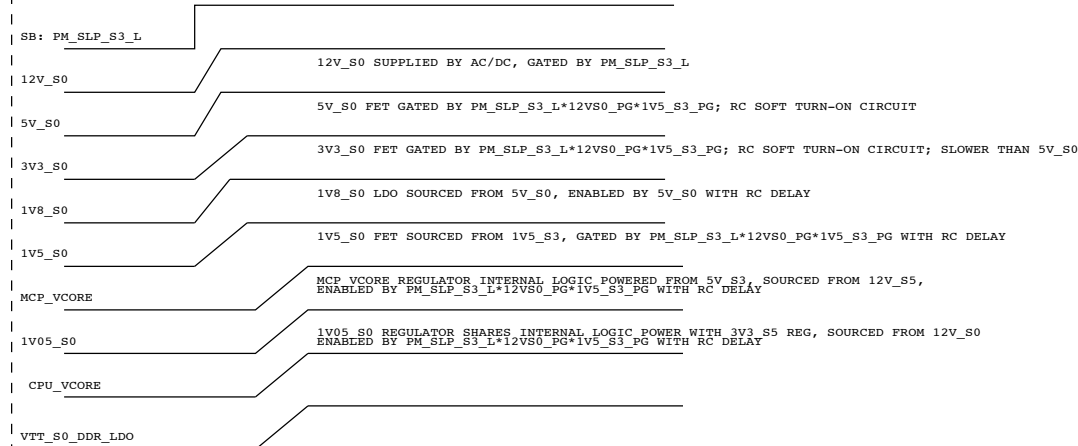


S3 POWER RAIL SEQUENCING

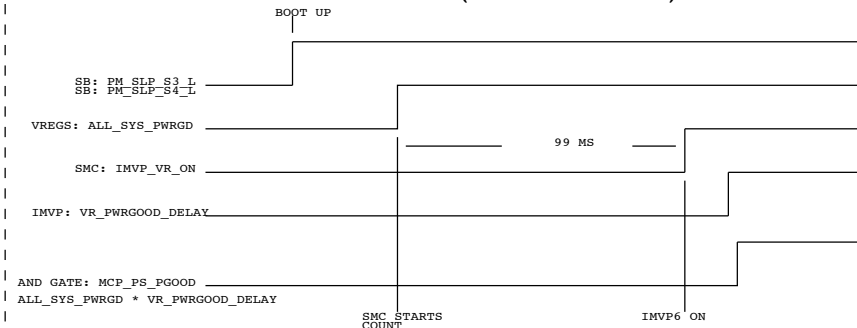


NOTE: NO SEQUENCING REQUIREMENTS FOR THESE 3 RAILS

S0 POWER RAIL SEQUENCING



STARTUP (BOOT OR WAKE) TIMING



State	Manageability	SMC_PM_G2_ENABLE	PM_S4_STATE_L	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_M_L
Run (S0/M0)	N/A	1	1	1	1	1
Sleep (S3/M1)	On	1	1	0	1	1
Soft-Off (S5/M1)	On	1	0	0	1	1
Sleep (S3/M-Off)	Off	1	1	0	1	0
Soft-Off (S5/M-Off)	Off	1	0	0	0	0
Battery Off (G3Hot)	N/A	0	0	0	0	0

SHUT DOWN (SHUTDOWN OR SLEEP) TIMING

POWER RAILS ON DURING THIS TIME



SYNC MASTER=K51 SYNC DATE=12/08/2008

PAGE TITLE

POWER SEQUENCING BLOCK DIAGRAM

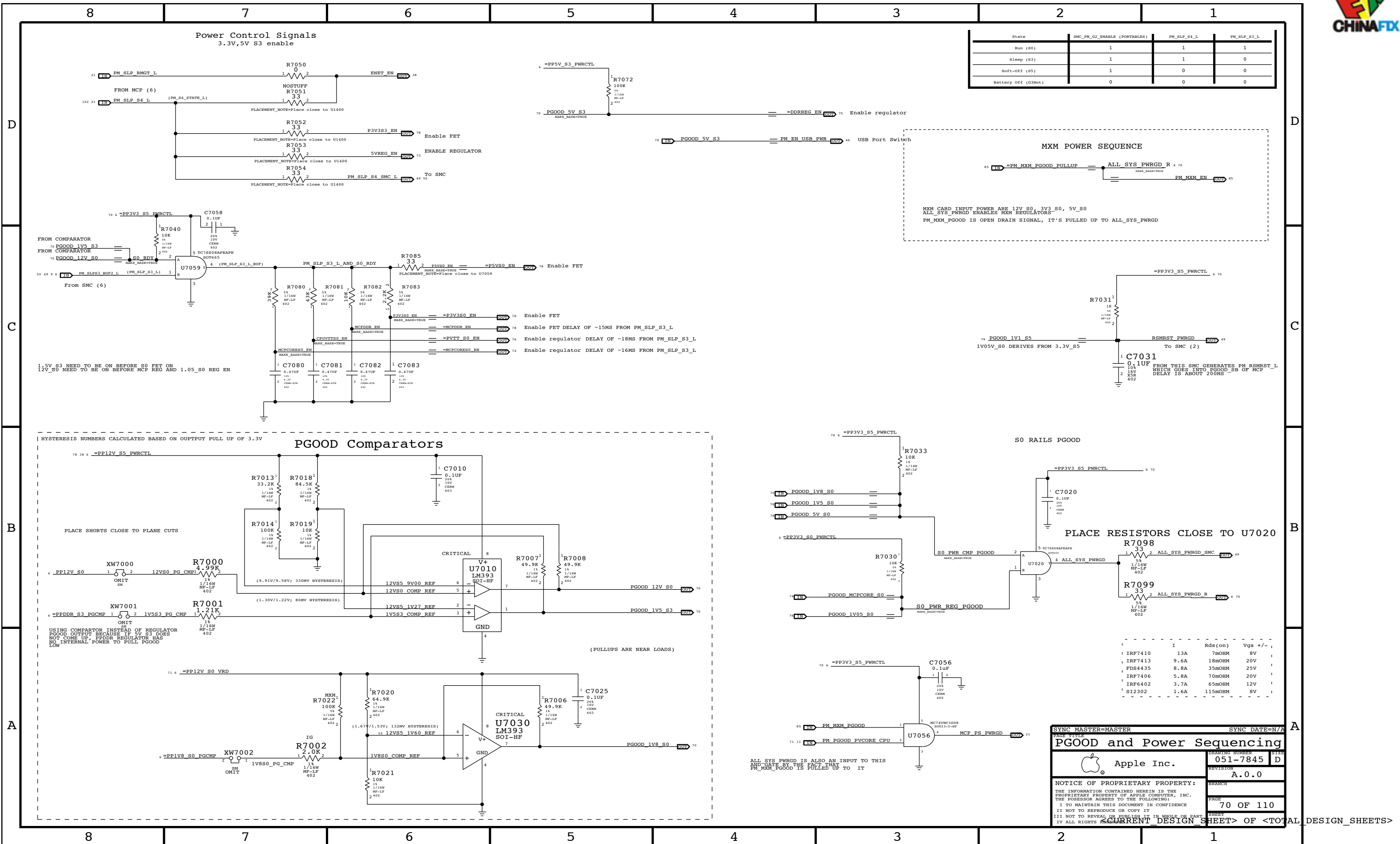
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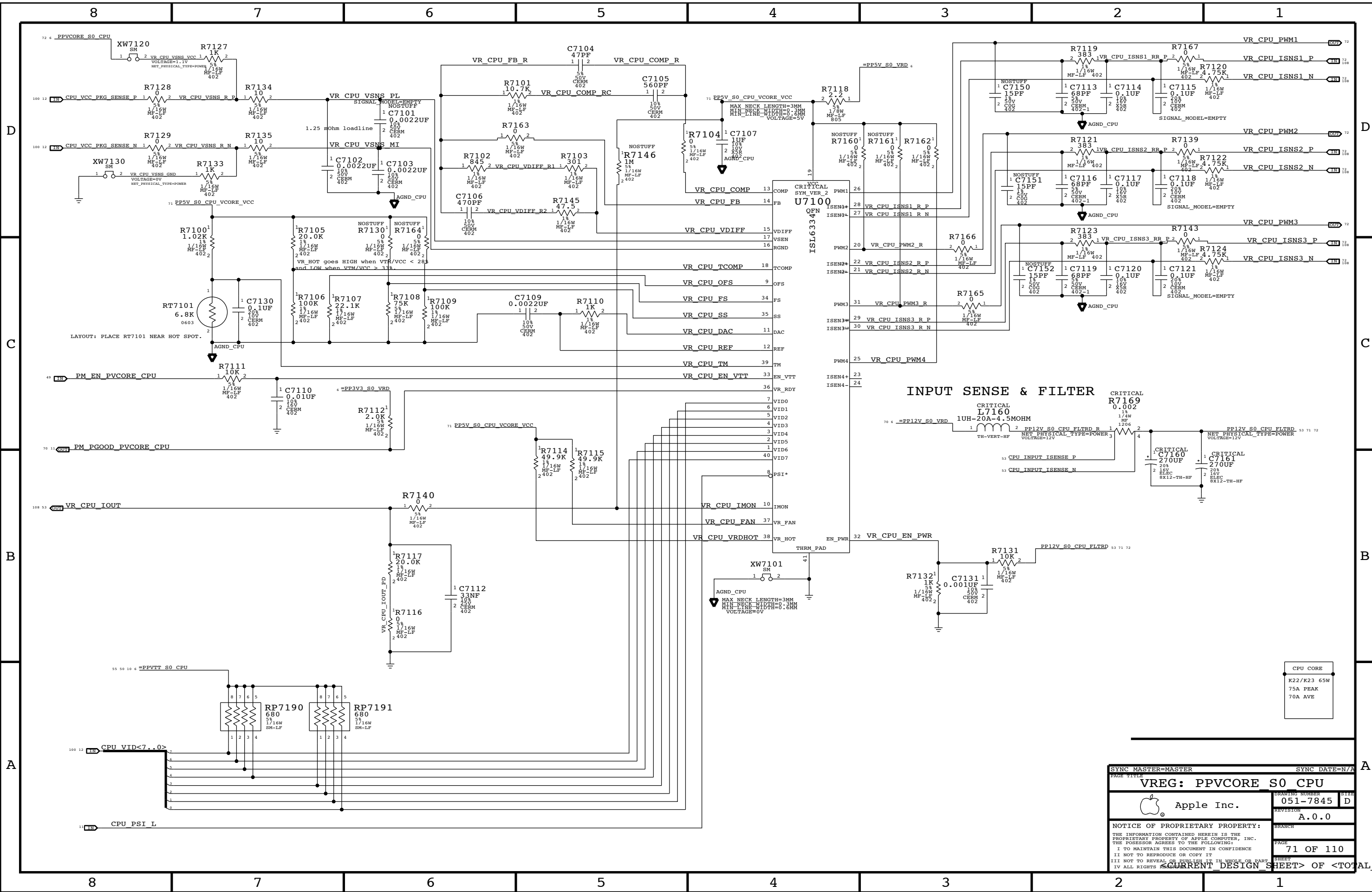
051-7845 D

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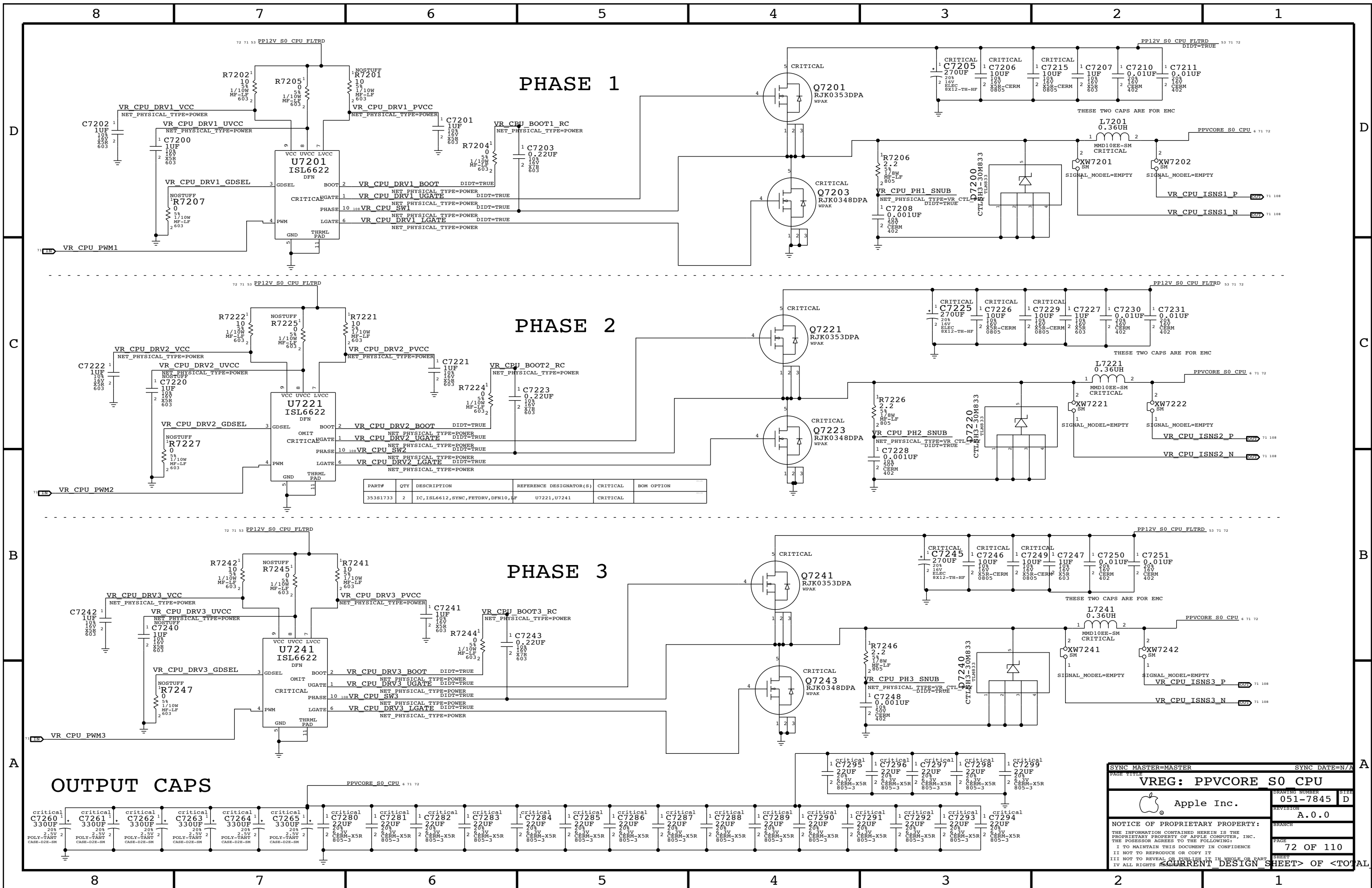
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SYNC MASTER=MASTER		SYNC DATE=N/A	
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VREG: PPVCORE S0 CPU		051-7845	
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VREG: PPVCORE S0 CPU

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051-7845 D

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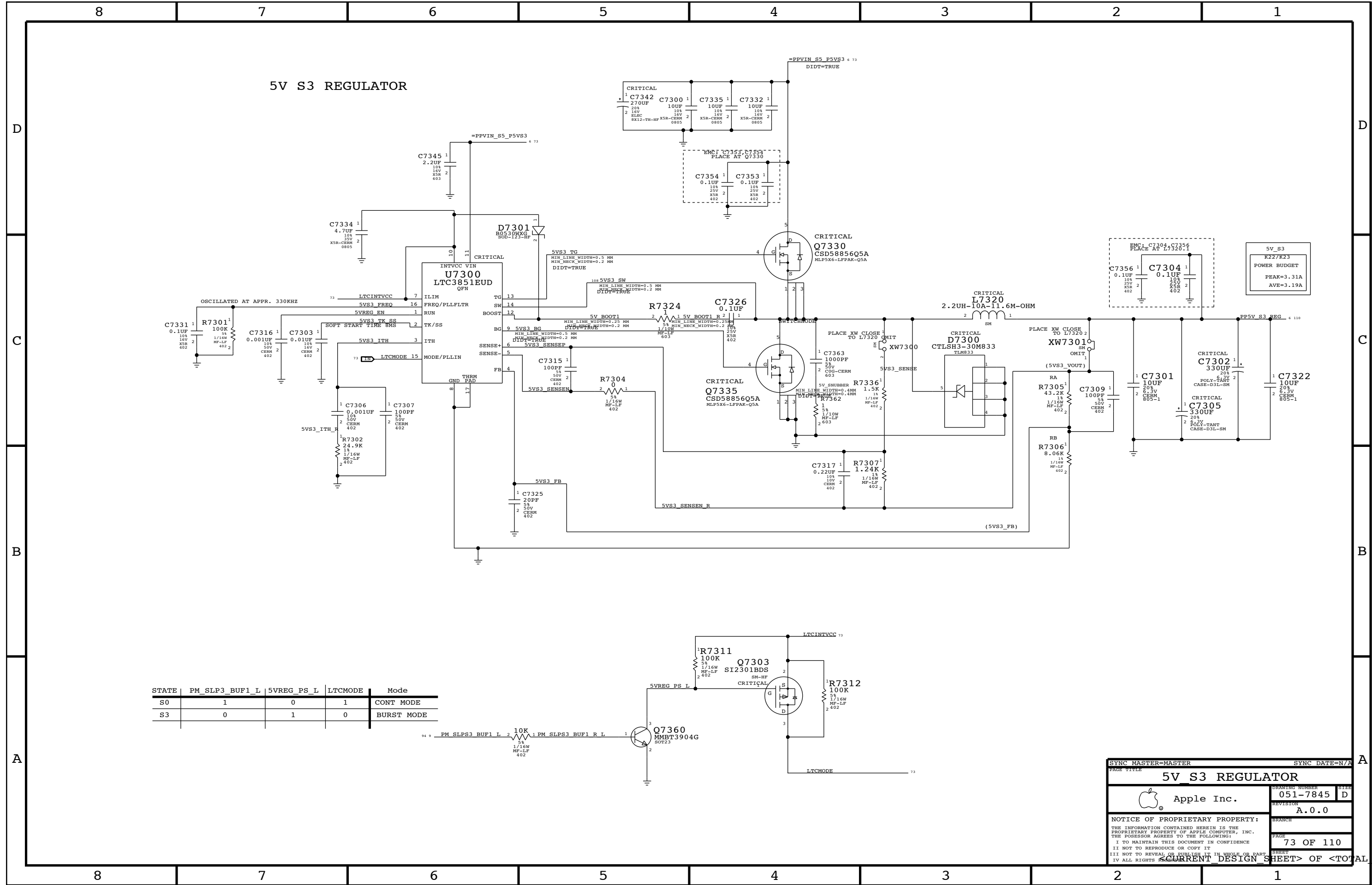
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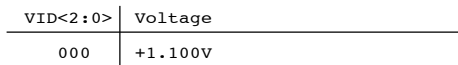
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SYNC MASTER=MASTER		SYNC DATE=N/A	
PAGE TITLE			
MCP CORE REGULATOR			
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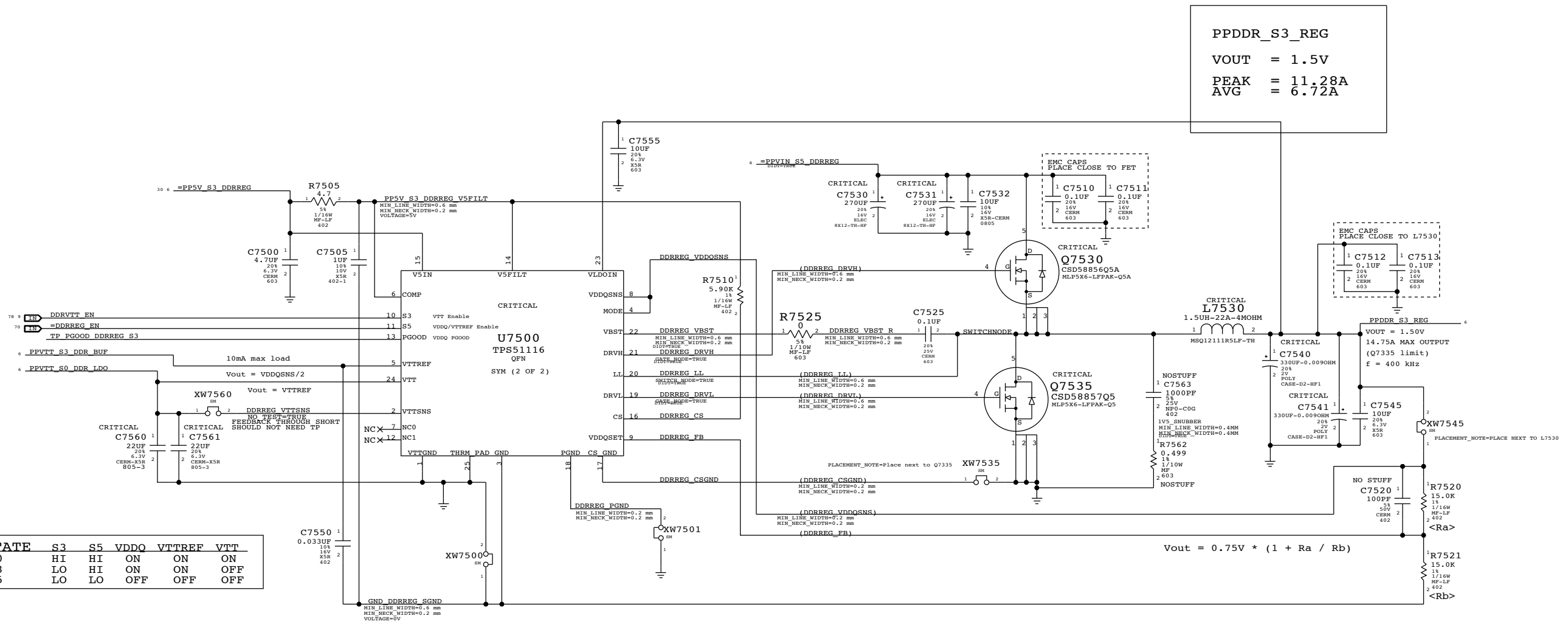
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74 OF 110		

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1.5 V DDR SUPPLY



PPDDR_S3_REG
VOUT = 1.5V
PEAK = 11.28A
AVG = 6.72A

STATE	S3	S5	VDDQ	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

SYNC MASTER=MASTER

SYNC DATE=N/A

1.5V DDR SUPPLY

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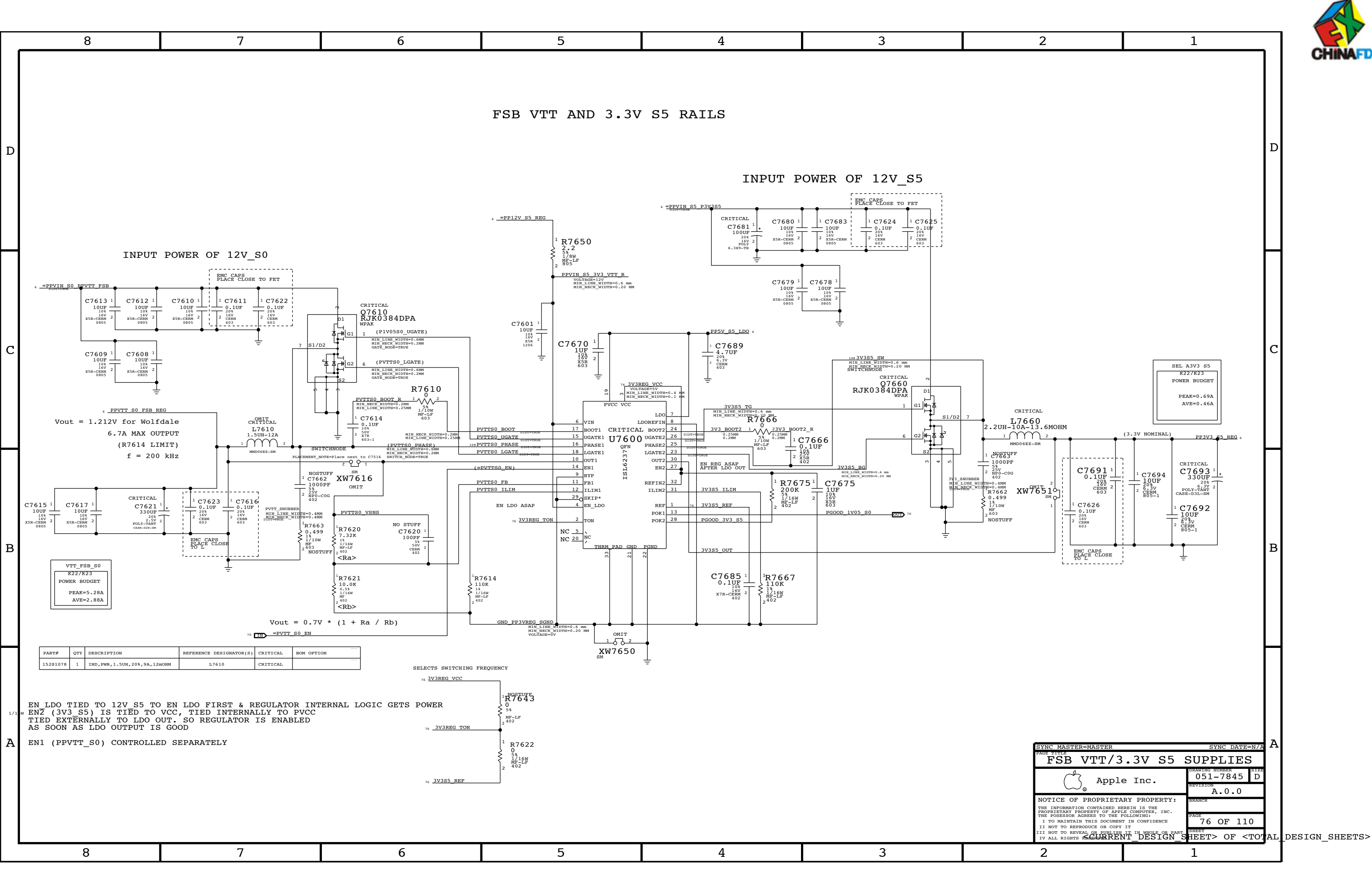
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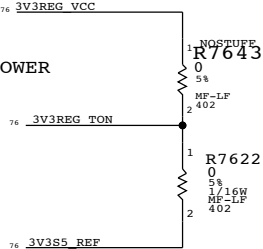


```

LDO TIED TO 12V S5 TO EN LDO FIRST & REGULATOR INTERNAL LOGIC GETS POWER
V2 (3V3 S5) IS TIED TO VCC, TIED INTERNALLY TO PVCC
TIED EXTERNALLY TO LDO OUT. SO REGULATOR IS ENABLED
AS SOON AS LDO OUTPUT IS GOOD

```

SELECTS SWITCHING FREQUENCY



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	8	7	6	5	4	3	2	1
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SYNC MASTER=K51

SYNC DATE=12/08/2008

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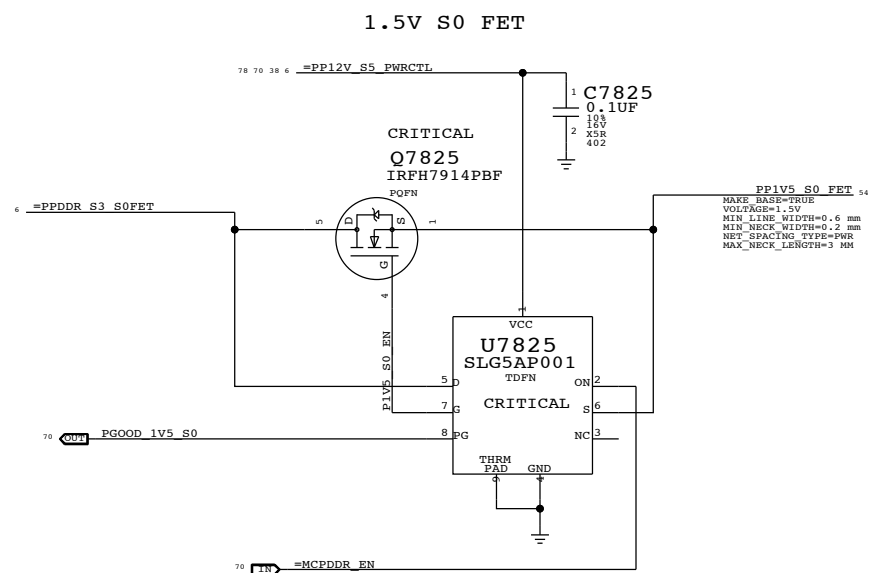
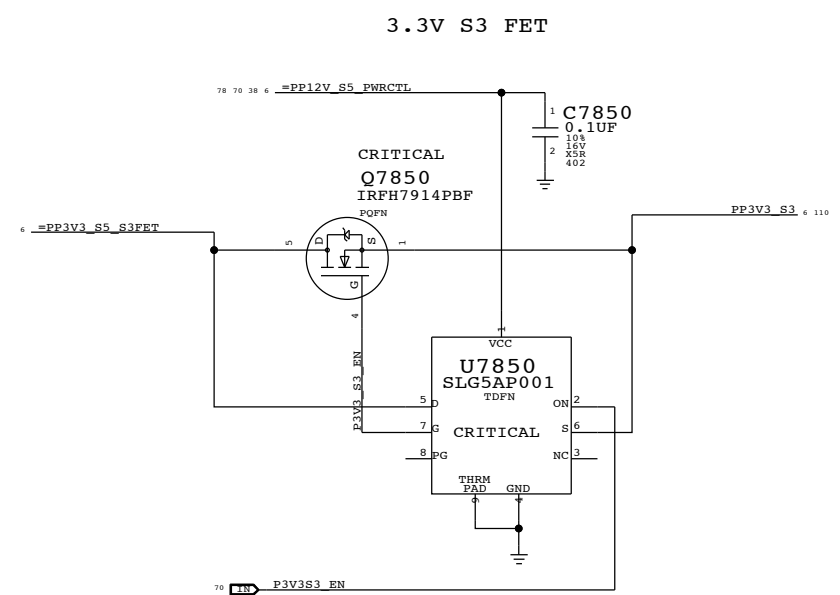
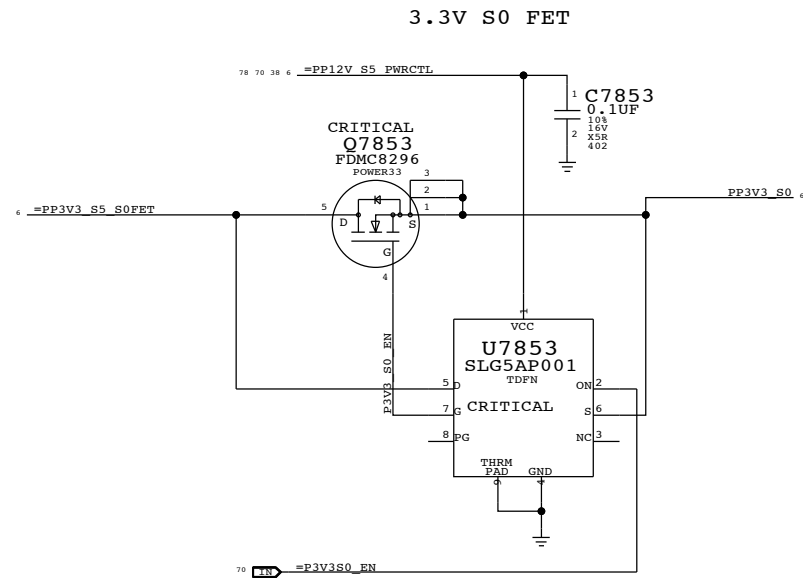
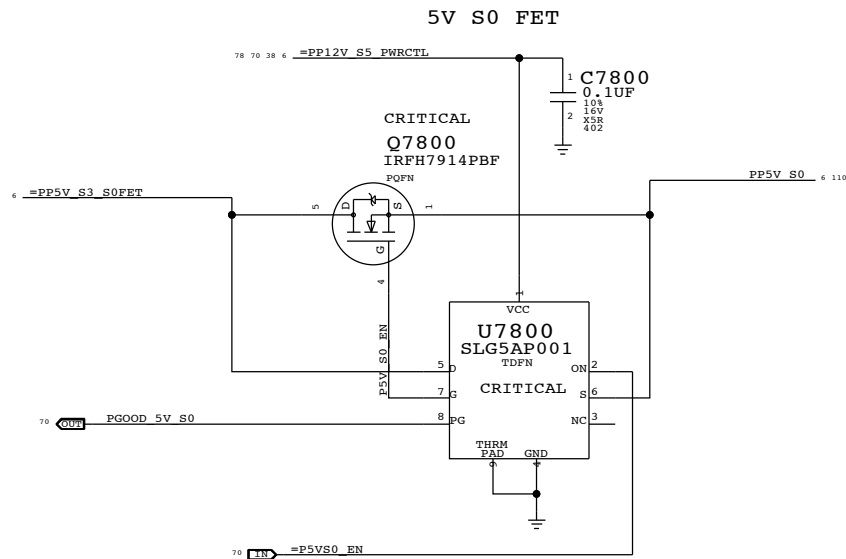
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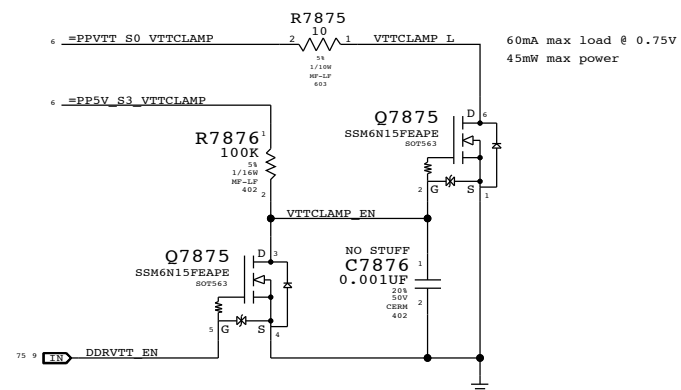
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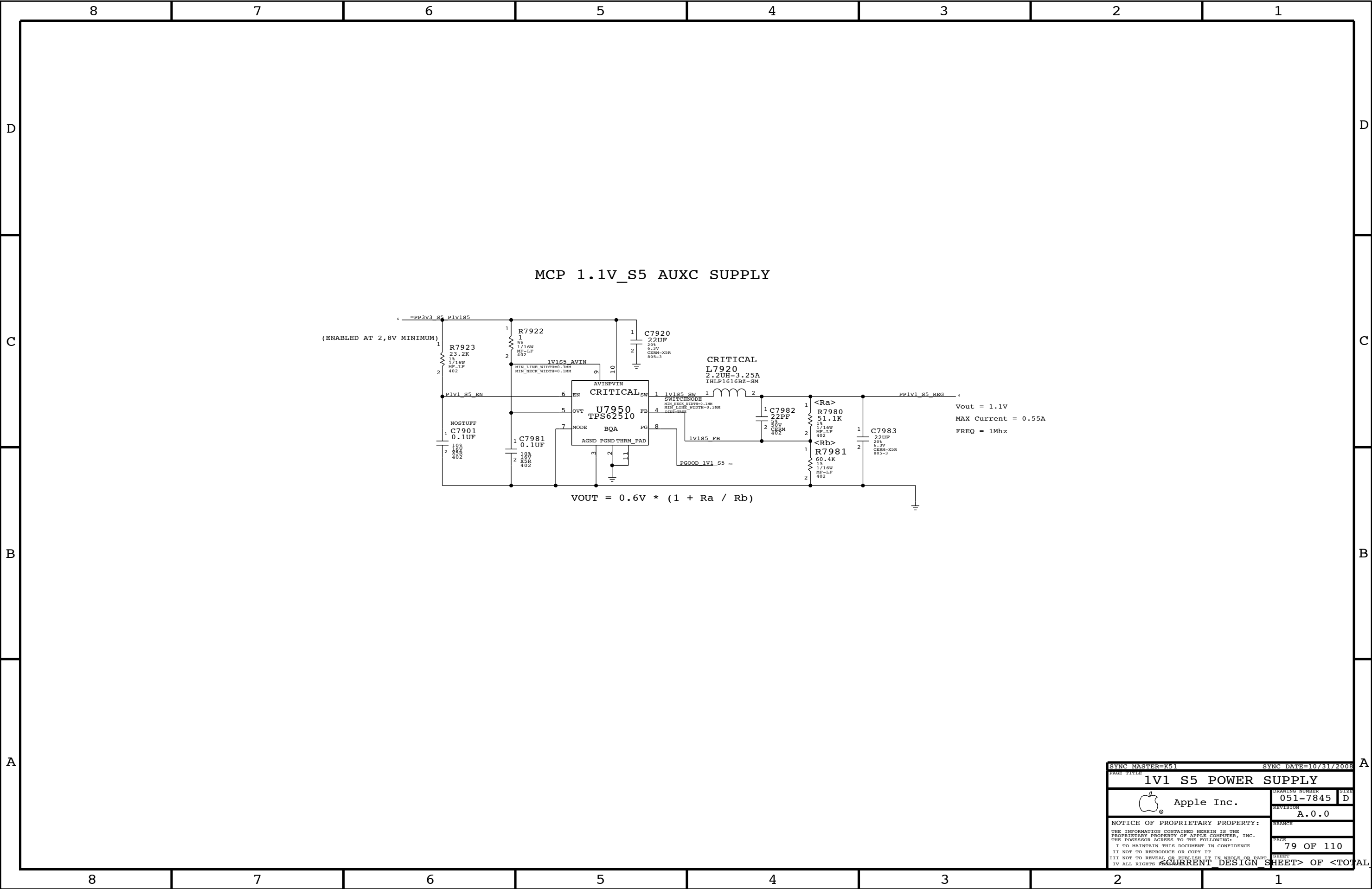


MCP79 DDRVTT FET

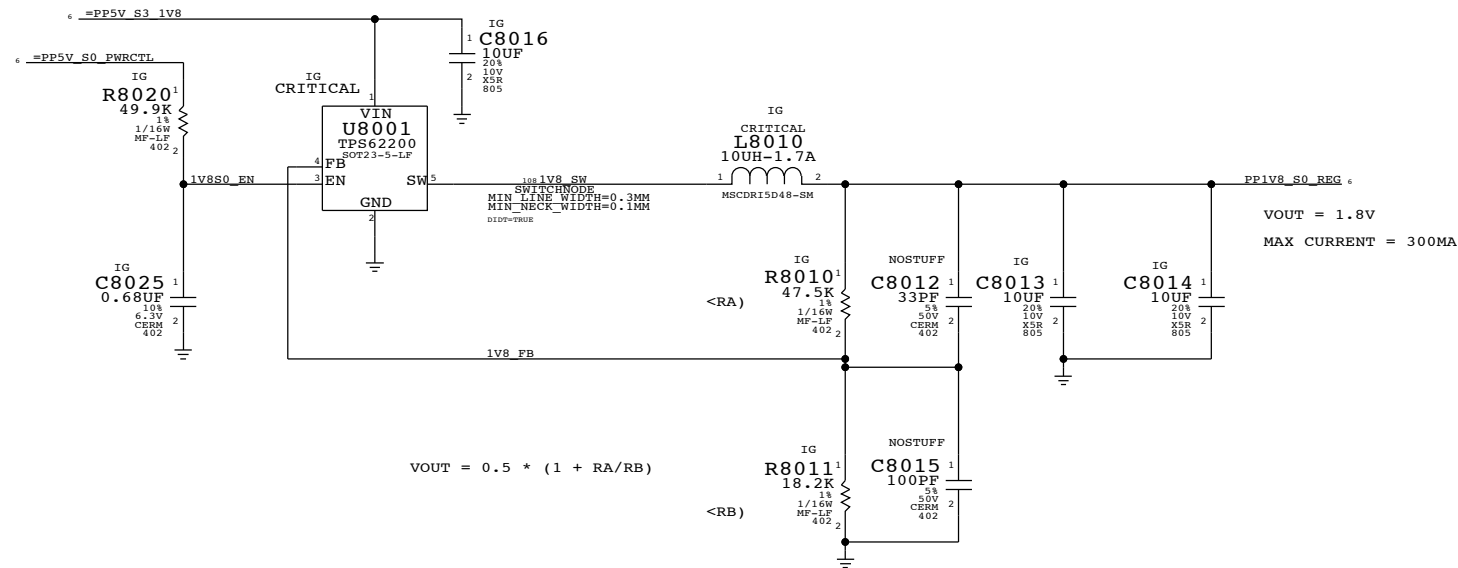
MC7P9 DDR PAD LEAKAGE IS HIGH ENOUGH THAT
NVIDIA RECOMMENDS UNPOWERING DURING SLEEP.
IN ORDER TO SUPPORT UNPOWERING RAIL, HARDWARE
MUST GUARANTEE MEM_CKE SIGNALS ARE LOW
BEFORE RAIL IS TURNED OFF, AND REMAINS LOW
UNTIL AFTER RAIL TURNS BACK ON OR DIMMS
WILL EXIT SELF-REFRESH PREMATURELY.
MEM_VTT_EN OUTPUT FROM MC7P9 USED TO ENABLE CLAMP
ON VTT RAIL, WHICH PULLS ALL CKE SIGNALS
LOW THROUGH VTT TERMINATION RESISTORS.



SYNC MASTER=MASTER		SYNC DATE=N/A	
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S3 & S0 FETs			
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MCP ONLY 1.8V_S0 POWER SUPPLY



SYNC MASTER=MASTER		SYNC DATE=N/A	
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1V8 POWER SUPPLY			
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SYNC MASTER=K51

SYNC DATE=12/08/2008

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SYNC DATE=12/08/2008

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SYNC MASTER=K51

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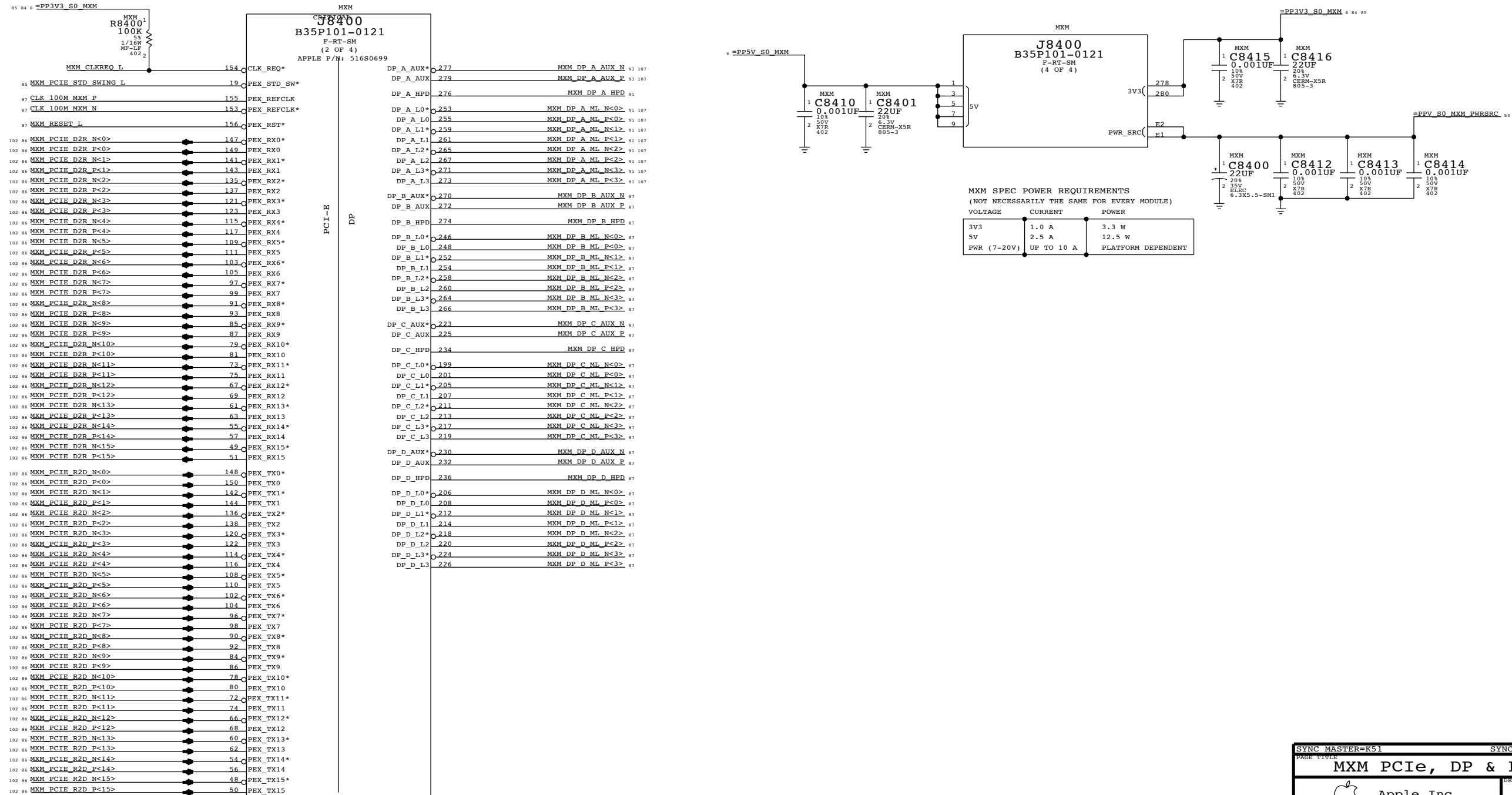


Power aliases required by this page:

```
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC
```

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
- MXM



MXM SPEC POWER REQUIREMENTS (NOT NECESSARILY THE SAME FOR EVERY MODULE)		
VOLTAGE	CURRENT	POWER
3V3	2.0 A	3.3 W
5V	1.5 A	12.5 W
PWR (7-20V)	UP TO 10 A	PLATFORM DEPEND

SYNCH MASTER=K51		SYNCH DATE=10/31/2008	
MXM PCIE, DP & Power			
 Apple Inc.		DRAWING NUMBER	SIZE
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SECURE CURRENT DESIGN SHEET		BRANCH	
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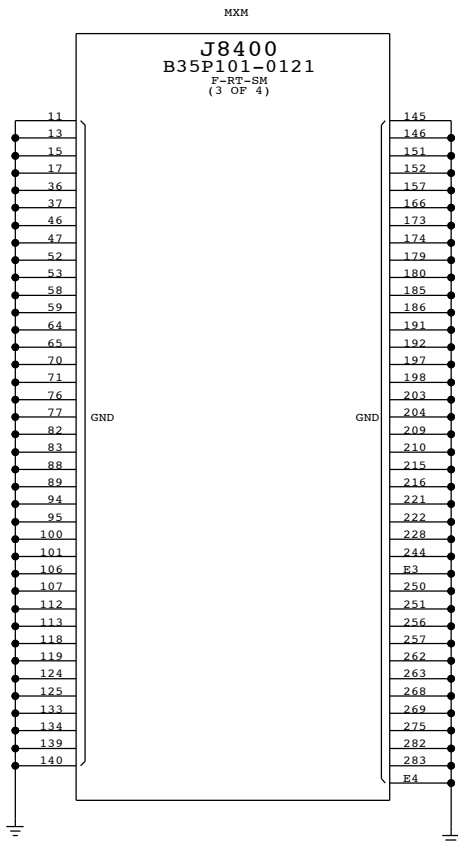
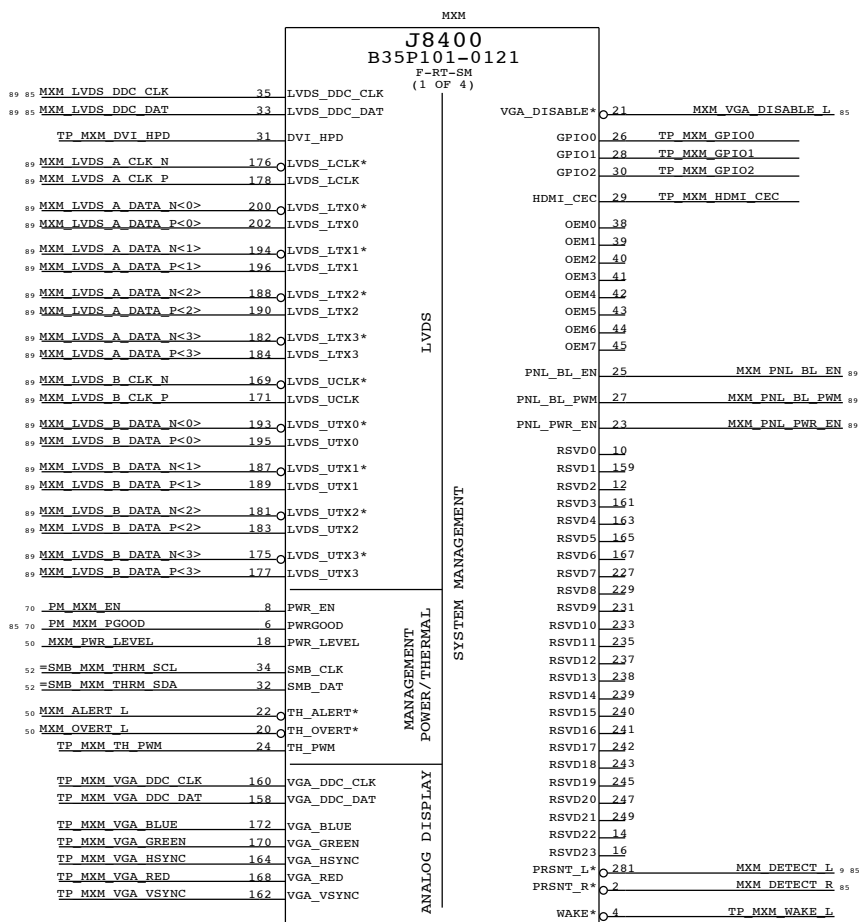
- =PP3V3_S0_MXM

Signal aliases required by this page:

- =SMB_MXM_THRM_DATA - =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

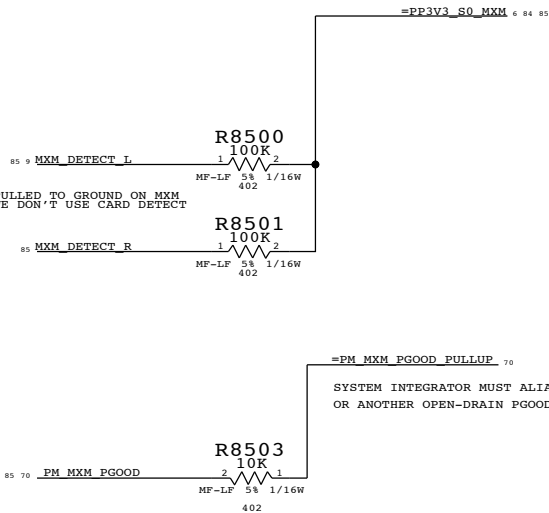
BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



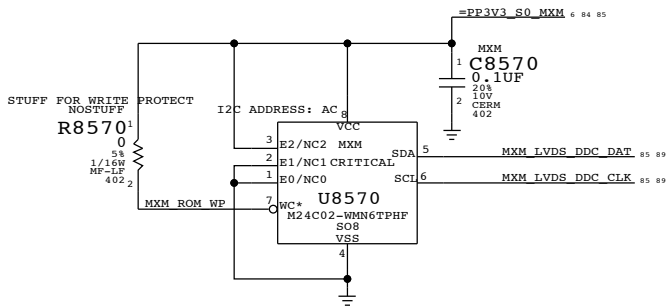
FLOAT = NORMAL VGA MODE
GND = SECONDARY DISPLAY CARD

FLOAT = LOW SWING
GND = HIGH SWING



MXM SYSTEM INFORMATION ROM

PLACE CLOSE TO J7800



SYNC MASTER=K51 SYNC DATE=10/31/2008

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MXM I/O

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MXM TX CAPS

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MXM RX CAPS

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102 84	MXM PCIE D2R P<9>	MXM C8644 0.1UF 1	2 10% 16V X5R 402	PEG D2R P<6>	9 102
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102 84	MXM PCIE D2R P<2>	MXM C8658 0.1UF 1	2 10% 16V X5R 402	PEG D2R N<13>	9 102
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SYNC MASTER=MASTER

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MXM PCIE CAPS

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Page Notes

Power aliases required by this page:
- =PP5V_DP_AUX

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

MCP CONNECTIONS

64	CLK_100M_MXM_P	==	GPU_CLK100M_PCIE_P	9 102
			MAKE_BASE=TRUE	
64	CLK_100M_MXM_N	==	GPU_CLK100M_PCIE_N	9 102
			MAKE_BASE=TRUE	
64	MXM_RESET_L	==	PEG_RESET_L	9 90
			MAKE_BASE=TRUE	

UNUSED DP INTERFACES

04	<u>MXM_DP_C_ML_N<0..3></u>	==	NC_MXM_DP_C_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_C_ML_P<0..3></u>	==	NC_MXM_DP_C_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_C_AUX_N</u>	==	NC_MXM_DP_C_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_C_AUX_P</u>	==	NC_MXM_DP_C_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_C_HPD</u>	==	TP_MXM_DP_C_HPD	MAKE_BASE=TRUE
04	<u>MXM_DP_B_ML_N<0..3></u>	==	NC_MXM_DP_B_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_B_ML_P<0..3></u>	==	NC_MXM_DP_B_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_B_AUX_N</u>	==	NC_MXM_DP_B_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_B_AUX_P</u>	==	NC_MXM_DP_B_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_B_HPD</u>	==	TP_MXM_DP_B_HPD	MAKE_BASE=TRUE
04	<u>MXM_DP_D_ML_N<0..3></u>	==	NC_MXM_DP_D_ML_N<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_D_ML_P<0..3></u>	==	NC_MXM_DP_D_ML_P<0..3>	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_D_AUX_N</u>	==	NC_MXM_DP_D_AUX_N	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_D_AUX_P</u>	==	NC_MXM_DP_D_AUX_P	MAKE_BASE=TRUE NO_TEST=TRUE
04	<u>MXM_DP_D_HPD</u>	==	TP_MXM_DP_D_HPD	MAKE_BASE=TRUE

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MXM ALIASES					
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SYNC DATE=10/01/2008

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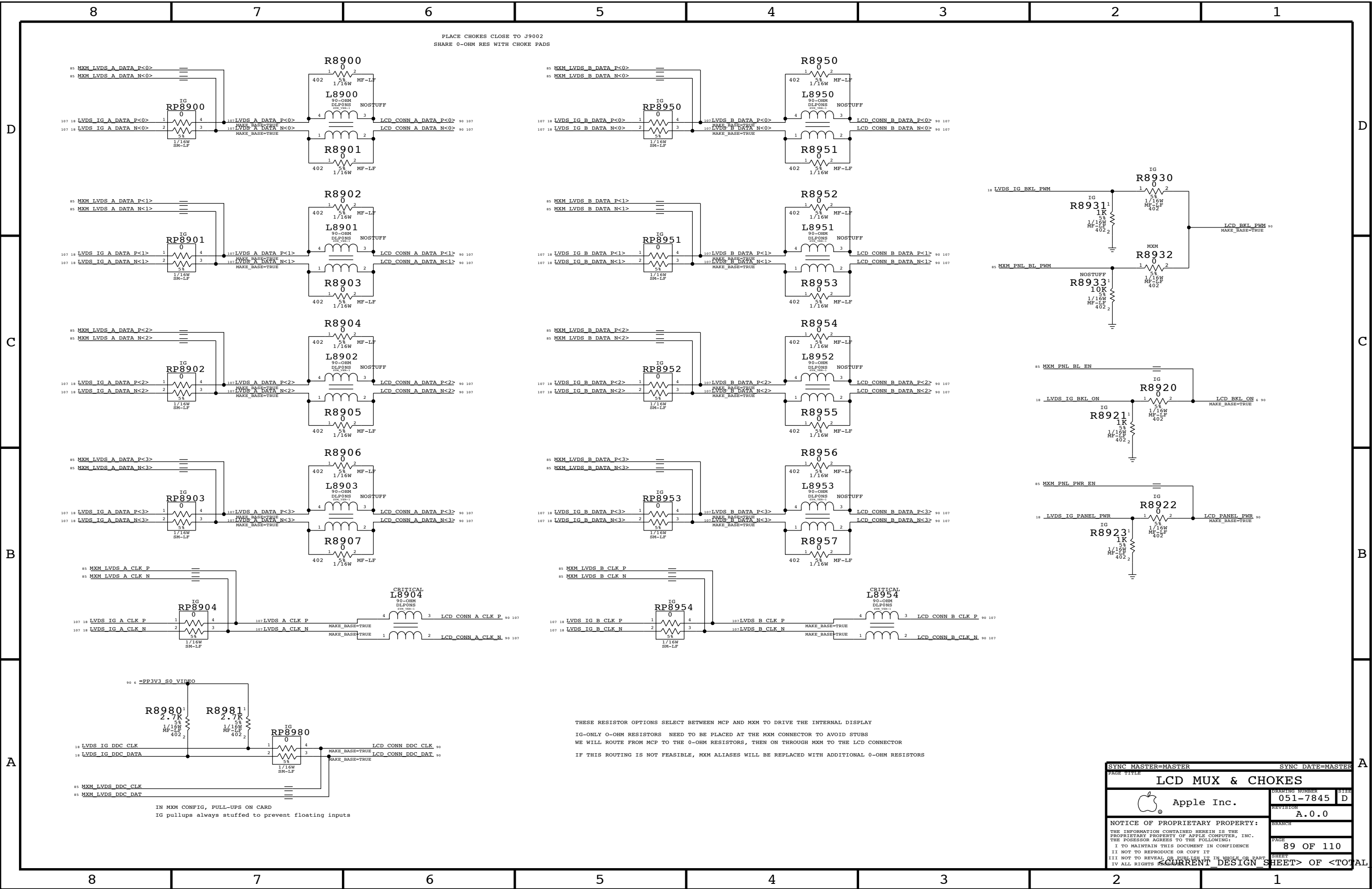
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LCD MUX & CHOKES			
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Page Notes

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- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:

(NONE)

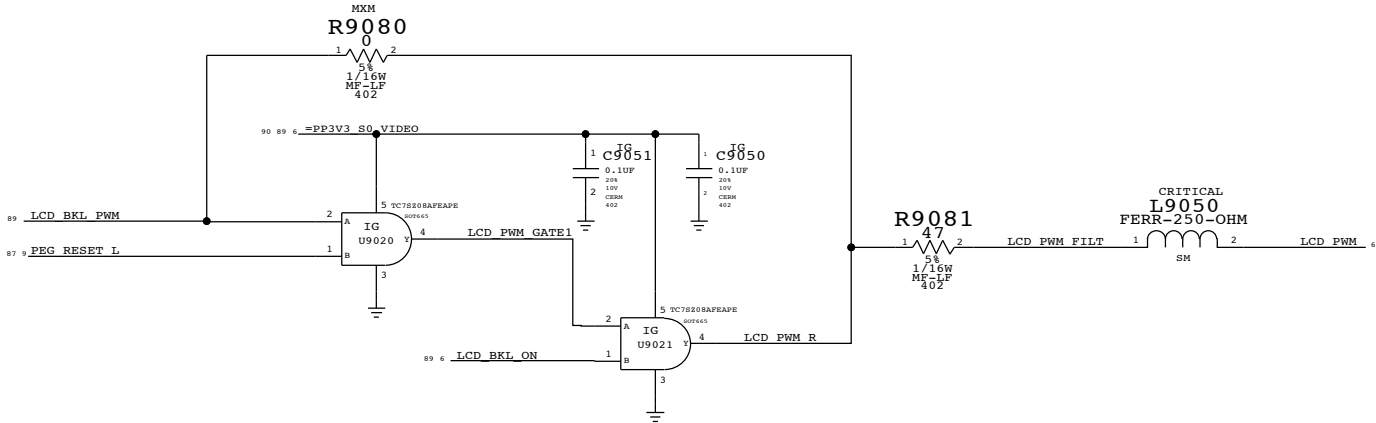
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IG, MXM

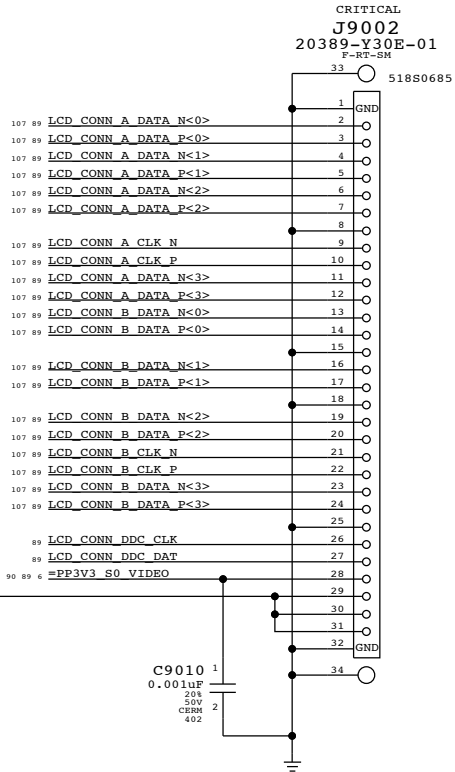
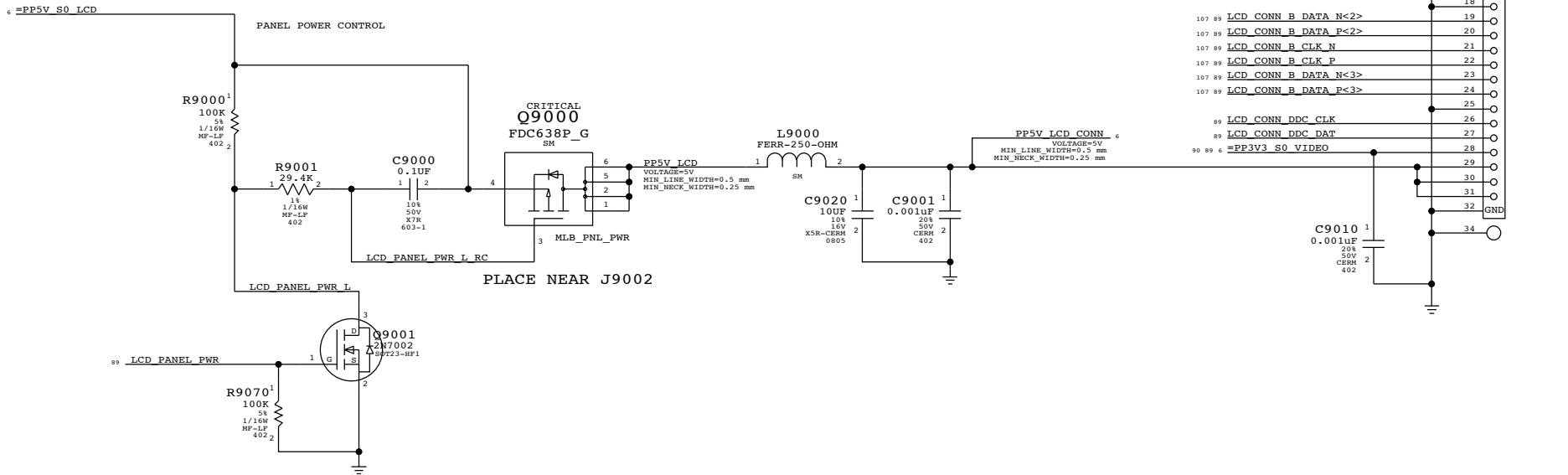
BACKLIGHT CONTROL SUPPORT

THIS AND GATE CIRCUIT PROVIDES BACKLIGHT GLITCH PREVENTION WHEN MCP GLITCHES GPIOs ON POWERUP
IT MAY BE BYPASSED IF THE PWM SOURCE IS THE MXM

IF NOT BYPASSED, THIS CAN BE USED TO FORCE THE USE OF THE BACKLIGHT ENABLE SIGNAL EVEN IF THE INVERTER DOES NOT TAKE THIS AS AN INPUT



INTERNAL LCD INTERFACE



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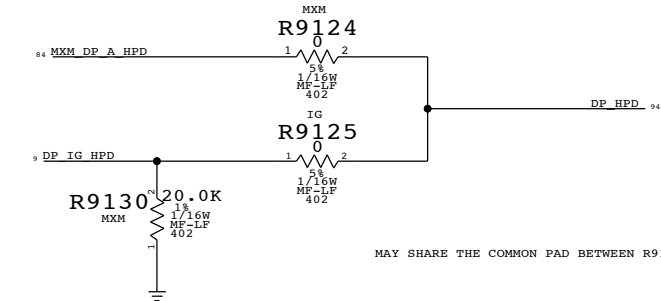
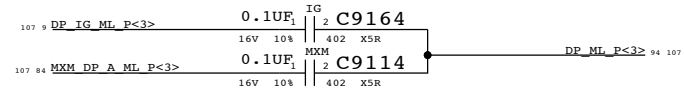
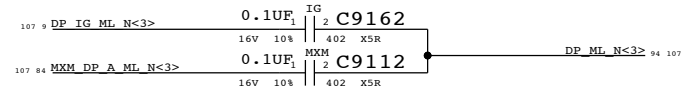
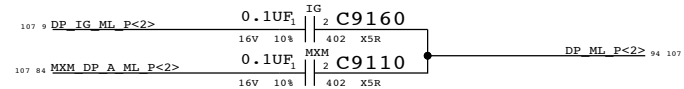
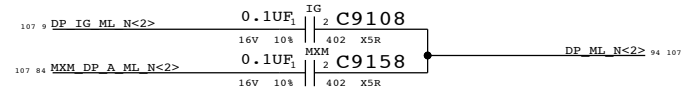
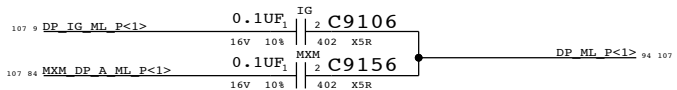
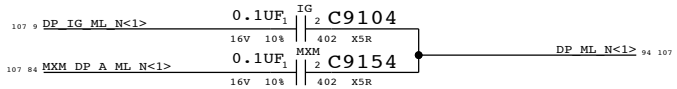
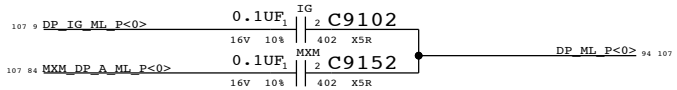
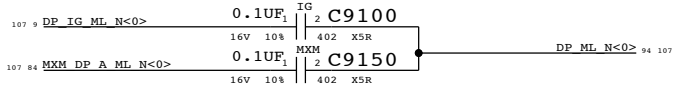
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K50 NOTE: PLACE THESE CAPACITORS ATLEAST 1INCH AWAY FROM DP CONNECTOR
DCOX: PLACE AT MXM CONNECTOR IF THERE IS ROOM



SYNC MASTER=MASTER		SYNC DATE=N/A	
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DP MUX SUPPORT			
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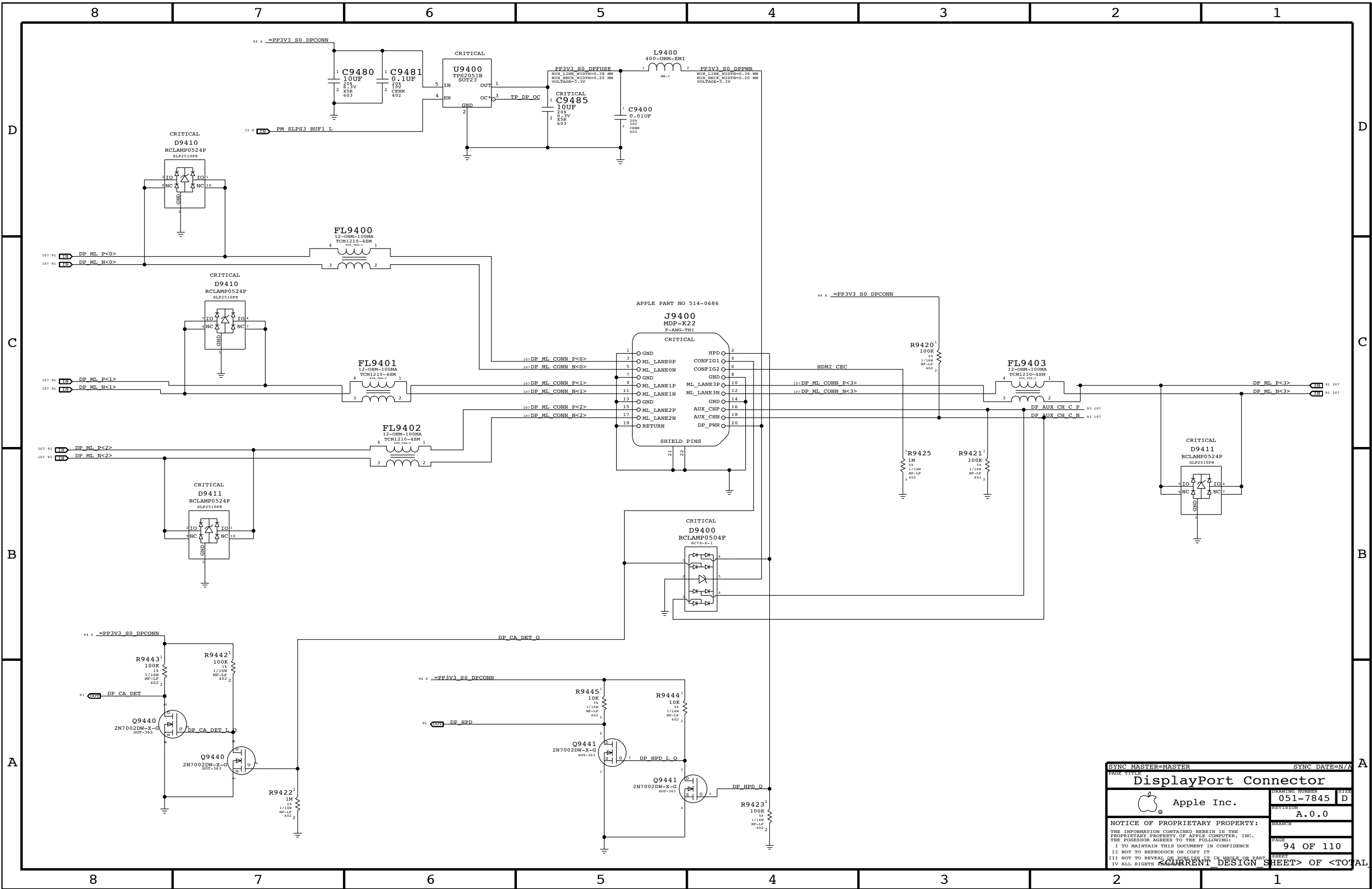
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FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?	FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?	FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?	FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	*	=2x_DIELECTRIC	?	FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?	FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 2x/1x/Async FSB signals with impedance requirements are 50-ohm single-ended.
All 4x FSB signals with impedance requirements are 42-ohm single-ended.

FSB 4x signals / groups shown in signal table on right.
Signals within each 4x group should be matched within 5 ps of strobe.
DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 90 ps. (Tighter than MCP79)
Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.
DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2x signals / groups shown in signal table on right.
Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 300 ps.
Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1x signals shown in signal table on right.
Signals within each 1x group should be matched to CPU clock, +/-1000 mils.
Design Guide recommends each strobe/signal group is routed on the same layer.
Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.175 MM	0.175 MM

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_SMIL	*	0.2 MM	?				
CPU_COMP	*	0.6 MM	?				
CPU_GTLREF	*	0.6 MM	?				
CPU_ITP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	0.6 MM	?				

SR DG recommends at least 25 mils, >50 mils preferred

MOST CPU SIGNALS WITH IMPEDANCE REQUIREMENTS ARE 50-OHM SINGLE-ENDED.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2
SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_FSB_COMP	*	0.2 MM	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_FSB_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?	CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
Group 0	FSB_42S	FSB_DATA	FSB D L<15..0> 10 14
	FSB_42S	FSB_DATA	FSB DINV L<0> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L P<0> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L N<0> 10 14
Group 1	FSB_42S	FSB_DATA	FSB D L<31..16> 10 14
	FSB_42S	FSB_DATA	FSB DINV L<1> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L P<1> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L N<1> 10 14
Group 2	FSB_42S	FSB_DATA	FSB D L<47..32> 10 14
	FSB_42S	FSB_DATA	FSB DINV L<2> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L P<2> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L N<2> 10 14
Group 3	FSB_42S	FSB_DATA	FSB D L<63..48> 10 14
	FSB_42S	FSB_DATA	FSB DINV L<3> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L P<3> 10 14
	FSB_DSTB_42S	FSB_DSTB	FSB DSTB L N<3> 10 14
Group 0	FSB_50S	FSB_ADDR	FSB A L<16..3> 10 14
	FSB_50S	FSB_ADDR	FSB REQ L<4..0> 10 14
	FSB_50S	FSB_ADSTB	FSB ADSTB L<0> 10 14
Group 1	FSB_50S	FSB_ADDR	FSB A L<35..17> 10 14
	FSB_50S	FSB_ADSTB	FSB ADSTB L<1> 10 14
FSB 1X Signals	FSB_50S	FSB_1X	FSB ADS L 10 14
	FSB_50S	FSB_1X	FSB BREQ0 L 10 14
	FSB_50S	FSB_1X	FSB BREQ1 L 14
	FSB_50S	FSB_1X	FSB BNR L 10 14
	FSB_50S	FSB_1X	FSB BPRI L 10 14
	FSB_50S	FSB_1X	FSB DBSY L 10 14
	FSB_50S	FSB_1X	FSB DEFER L 10 14
	FSB_50S	FSB_1X	FSB DRDY L 10 14
	FSB_50S	FSB_1X	FSB HIT L 10 14
	FSB_50S	FSB_1X	FSB HITM L 10 14
	FSB_50S	FSB_1X	FSB LOCK L 10 14
	FSB_50S	FSB_1X	FSB CPURST L 10 13 14
	FSB_50S	FSB_1X	FSB RS L<2..0> 10 14
	FSB_50S	FSB_1X	FSB TRDY L 10 14
	CPU_50S	CPU_ACTL	CPU A20M L 10 14
	CPU_50S	CPU_ACTL	CPU BSEL<2..0> 11 14
FSB 2X Signals	CPU_50S	CPU_BN1L	CPU FERR L 10 14
	CPU_50S	CPU_ACTL	CPU IGNNE L 10 14
	CPU_50S	CPU_ACTL	CPU INIT L 10 14
	CPU_50S	CPU_ACTL	CPU INTR 10 14
	CPU_50S	CPU_ACTL	CPU NMI 10 14
	CPU_50S	CPU_ACTL	CPU PROCHOT L 11 14 50
	CPU_50S	CPU_ACTL	CPU PWRGD 11 13 14
	CPU_50S	CPU_ACTL	CPU SMI L 10 14
	CPU_50S	CPU_ACTL	CPU STRCLK L 10 14
	CPU_50S	CPU_BN1L	PM THRMTRIP L 11 14 50
	CPU_50S	CPU_ACTL	FSB CPUSLP L 11 14
	CPU_50S	CPU_ACTL	CPU DPSLP L 11 14
	CPU_50S	CPU_ACTL	CPU DPRSTP L 11 14
	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP VDD 14
	MCP_50S	MCP_FSB_COMP	MCP BCLK VML COMP GND 14
	MCP_50S	MCP_FSB_COMP	MCP CPU COMP VCC 14
	MCP_50S	MCP_FSB_COMP	MCP CPU COMP GND 14
FSB 4X Signals	CLK_FSB_100D	CLK_FSB	FSB CLK CPU P 10 14
	CLK_FSB_100D	CLK_FSB	FSB CLK CPU N 10 14
	CLK_FSB_100D	CLK_FSB	FSB CLK ITP P 13 14
	CLK_FSB_100D	CLK_FSB	FSB CLK ITP N 13 14
	CLK_FSB_100D	CLK_FSB	FSB CLK MCP P 14
	CLK_FSB_100D	CLK_FSB	FSB CLK MCP N 14
	CPU_50S	CPU_ACTL	CPU IERR L 10
	CPU_50S	CPU_GTLREF	CPU GTLREF0 10 11 29
	CPU_50S	CPU_GTLREF	CPU GTLREF1 10 11 29
	CPU_27P4S	CPU_COMP	CPU COMP<8> 11
	CPU_27P4S	CPU_COMP	CPU COMP<3> 11
	CPU_27P4S	CPU_COMP	CPU COMP<2> 11
	CPU_27P4S	CPU_COMP	CPU COMP<1> 11
	CPU_27P4S	CPU_COMP	CPU COMP<0> 11
	CPU_50S	CPU_ITP	CPU XDP TDI 11 13
	CPU_50S	CPU_ITP	CPU XDP TDO 11 13
	CPU_50S	CPU_ITP	CPU XDP TMS 11 13
	CPU_50S	CPU_ITP	CPU XDP TCK 11 13
	CPU_50S	CPU_ITP	CPU XDP TRST L 11 13
	CPU_50S	CPU_ITP	CPU XDP BPM L<5..0> 11 13
	CPU_50S	CPU_ITP	CPU XDP BPMB<3..0> 11 13
	CPU_50S	CPU_ITP	XDP CPURST L 13
	CPU_50S	CPU_BN1L	CPU VID<7..0> 12 71
	CPU_27P4S	CPU_VCCSENSE	CPU VCC PKG SENSE P 12 71
	CPU_27P4S	CPU_VCCSENSE	CPU VCC PKG SENSE N 12 71
	CPU_27P4S	CPU_VCCSENSE	VR CPU VSNS R P 71
	CPU_27P4S	CPU_VCCSENSE	VR CPU VSNS R_N 71

SYNC MASTER=MASTER

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CPU/FSB Constraints

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_40S_VDD	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_70D_VDD	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	=3:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_2OTHER
MEM_CTRL	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_DATA	*	*	MEM_2OTHER
MEM_DQS	*	*	MEM_2OTHER

Need to support MEM *-style wildcards!

DDR2:

DQ signals should be matched within 20 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement.
All DQS pairs should be matched within 100 ps of clocks.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps.
A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 180 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
A/BA/cmd signals should be matched within 5 ps of CLK pairs.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.3

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MEM_COMP	*	Y	0.175 MM	0.175 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_MEM_COMP	*	0.2 MM	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.3.4

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE	
	PHYSICAL	SPACING
MEM_70D_VDD	MEM_CLK	MEM A CLK P<1..0>
MEM_70D_VDD	MEM_CLK	MEM A CLK N<1..0>
MEM_70D_VDD	MEM_CLK	MEM A CLK P<4..3>
MEM_70D_VDD	MEM_CLK	MEM A CLK N<4..3>
MEM_40S_VDD	MEM_CTRL	MEM A CKE<3..0>
MEM_40S_VDD	MEM_CTRL	MEM A CS L<3..0>
MEM_40S_VDD	MEM_CTRL	MEM A ODT<3..0>
MEM_40S_VDD	MEM_CMD	MEM A A<14..0>
MEM_40S_VDD	MEM_CMD	MEM A BA<2..0>
MEM_40S_VDD	MEM_CMD	MEM A RAS L
MEM_40S_VDD	MEM_CMD	MEM A CAS L
MEM_40S_VDD	MEM_CMD	MEM A WE L
MEM_40S	MEM_DATA	MEM A DQ<7..0>
MEM_40S	MEM_DATA	MEM A DM<0>
MEM_40S	MEM_DATA	MEM A DQ<15..8>
MEM_40S	MEM_DATA	MEM A DM<1>
MEM_40S	MEM_DATA	MEM A DQ<23..16>
MEM_40S	MEM_DATA	MEM A DM<2>
MEM_40S	MEM_DATA	MEM A DQ<31..24>
MEM_40S	MEM_DATA	MEM A DM<3>
MEM_40S	MEM_DATA	MEM A DQ<39..32>
MEM_40S	MEM_DATA	MEM A DM<4>
MEM_40S	MEM_DATA	MEM A DQ<47..40>
MEM_40S	MEM_DATA	MEM A DM<5>
MEM_40S	MEM_DATA	MEM A DQ<55..48>
MEM_40S	MEM_DATA	MEM A DM<6>
MEM_40S	MEM_DATA	MEM A DQ<63..56>
MEM_40S	MEM_DATA	MEM A DM<7>
MEM_70D	MEM_DQS	MEM A DQS P<0>
MEM_70D	MEM_DQS	MEM A DQS N<0>
MEM_70D	MEM_DQS	MEM A DQS P<1>
MEM_70D	MEM_DQS	MEM A DQS N<1>
MEM_70D	MEM_DQS	MEM A DQS P<2>
MEM_70D	MEM_DQS	MEM A DQS N<2>
MEM_70D	MEM_DQS	MEM A DQS P<3>
MEM_70D	MEM_DQS	MEM A DQS N<3>
MEM_70D	MEM_DQS	MEM A DQS P<4>
MEM_70D	MEM_DQS	MEM A DQS N<4>
MEM_70D	MEM_DQS	MEM A DQS P<5>
MEM_70D	MEM_DQS	MEM A DQS N<5>
MEM_70D	MEM_DQS	MEM A DQS P<6>
MEM_70D	MEM_DQS	MEM A DQS N<6>
MEM_70D	MEM_DQS	MEM A DQS P<7>
MEM_70D	MEM_DQS	MEM A DQS N<7>
MEM_70D_VDD	MEM_CLK	MEM B CLK P<1..0>
MEM_70D_VDD	MEM_CLK	MEM B CLK N<1..0>
MEM_70D_VDD	MEM_CLK	MEM B CLK P<4..3>
MEM_70D_VDD	MEM_CLK	MEM B CLK N<4..3>
MEM_40S_VDD	MEM_CTRL	MEM B CKE<3..0>
MEM_40S_VDD	MEM_CTRL	MEM B CS L<3..0>
MEM_40S_VDD	MEM_CTRL	MEM B ODT<3..0>
MEM_40S_VDD	MEM_CMD	MEM B A<14..0>
MEM_40S_VDD	MEM_CMD	MEM B BA<2..0>
MEM_40S_VDD	MEM_CMD	MEM B RAS L
MEM_40S_VDD	MEM_CMD	MEM B CAS L
MEM_40S_VDD	MEM_CMD	MEM B WE L
MEM_40S	MEM_DATA	MEM B DQ<7..0>
MEM_40S	MEM_DATA	MEM B DM<0>
MEM_40S	MEM_DATA	MEM B DQ<15..8>
MEM_40S	MEM_DATA	MEM B DM<1>
MEM_40S	MEM_DATA	MEM B DQ<23..16>
MEM_40S	MEM_DATA	MEM B DM<2>
MEM_40S	MEM_DATA	MEM B DQ<31..24>
MEM_40S	MEM_DATA	MEM B DM<3>
MEM_40S	MEM_DATA	MEM B DQ<39..32>
MEM_40S	MEM_DATA	MEM B DM<4>
MEM_40S	MEM_DATA	MEM B DQ<47..40>
MEM_40S	MEM_DATA	MEM B DM<5>
MEM_40S	MEM_DATA	MEM B DQ<55..48>
MEM_40S	MEM_DATA	MEM B DM<6>
MEM_40S	MEM_DATA	MEM B DQ<63..56>
MEM_40S	MEM_DATA	MEM B DM<7>
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP VDD
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP GND

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE	
	PHYSICAL	SPACING
MEM_70D	MEM_DQS	MEM B DQS P<0>
MEM_70D	MEM_DQS	MEM B DQS N<0>
MEM_70D	MEM_DQS	MEM B DQS P<1>
MEM_70D	MEM_DQS	MEM B DQS N<1>
MEM_70D	MEM_DQS	MEM B DQS P<2>
MEM_70D	MEM_DQS	MEM B DQS N<2>
MEM_70D	MEM_DQS	MEM B DQS P<3>
MEM_70D	MEM_DQS	MEM B DQS N<3>
MEM_70D	MEM_DQS	MEM B DQS P<4>
MEM_70D	MEM_DQS	MEM B DQS N<4>
MEM_70D	MEM_DQS	MEM B DQS P<5>
MEM_70D	MEM_DQS	MEM B DQS N<5>
MEM_70D	MEM_DQS	MEM B DQS P<6>
MEM_70D	MEM_DQS	MEM B DQS N<6>
MEM_70D	MEM_DQS	MEM B DQS P<7>
MEM_70D	MEM_DQS	MEM B DQS N<7>
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP VDD
MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP GND

SYNC MASTER=MASTER

SYNC DATE=N/A

Memory Constraints

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PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
CLK_PCIE_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	0.5 MM	?
MCP_PEX_COMP	*	0.2 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4X_DIELECTRIC	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=4X_DIELECTRIC	?
SATA_TERM	*	0.2 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=3X_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.7.1.

Electrical Constraints

ELECTRICAL_CONSTRAINT_SET	NET_TYPE	PHYSICAL	SPACING
PCIE GRAPHICS			
	PCIE_90D	PCIE	PEG_R2D_C_P<15..0>
	PCIE_90D	PCIE	PEG_R2D_C_N<15..0>
	PCIE_90D	PCIE	PEG_D2R_P<15..0>
	PCIE_90D	PCIE	PEG_D2R_N<15..0>
	PCIE_90D	PCIE	MXM_PCIE_R2D_P<15..0>
	PCIE_90D	PCIE	MXM_PCIE_R2D_N<15..0>
	PCIE_90D	PCIE	MXM_PCIE_D2R_P<15..0>
	PCIE_90D	PCIE	MXM_PCIE_D2R_N<15..0>
PCIE I/O			
	PCIE_90D	PCIE	PCIE_MINI_R2D_P
	PCIE_90D	PCIE	PCIE_MINI_R2D_N
	PCIE_90D	PCIE	PCIE_MINI_R2D_C_P
	PCIE_90D	PCIE	PCIE_MINI_R2D_C_N
	PCIE_90D	PCIE	PCIE_MINI_R2D_L_P
	PCIE_90D	PCIE	PCIE_MINI_R2D_L_N
	PCIE_90D	PCIE	PCIE_MINI_D2R_P
	PCIE_90D	PCIE	PCIE_MINI_D2R_N
	PCIE_90D	PCIE	PCIE_FW_R2D_P
	PCIE_90D	PCIE	PCIE_FW_R2D_N
	PCIE_90D	PCIE	PCIE_FW_R2D_C_P
	PCIE_90D	PCIE	PCIE_FW_R2D_C_N
	PCIE_90D	PCIE	PCIE_FW_D2R_P
	PCIE_90D	PCIE	PCIE_FW_D2R_N
	PCIE_90D	PCIE	PCIE_FW_D2R_C_P
	PCIE_90D	PCIE	PCIE_FW_D2R_C_N
PCIE REF CLOCKS			
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_P
	CLK_PCIE_100D	CLK_PCIE	GPU_CLK100M_PCIE_N
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_P
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_N
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CON_P
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_MINI_CON_N
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_P
	CLK_PCIE_100D	CLK_PCIE	PCIE_CLK100M_FW_N
SATA			
	SATA_100D	SATA	SATA_HDD_R2D_C_P
	SATA_100D	SATA	SATA_HDD_R2D_C_N
	SATA_100D	SATA	SATA_HDD_R2D_P
	SATA_100D	SATA	SATA_HDD_R2D_N
	SATA_100D	SATA	SATA_HDD_D2R_P
	SATA_100D	SATA	SATA_HDD_D2R_N
	SATA_100D	SATA	SATA_HDD_D2R_C_P
	SATA_100D	SATA	SATA_HDD_D2R_C_N
	SATA_100D	SATA	SATA_ODD_R2D_C_P
	SATA_100D	SATA	SATA_ODD_R2D_C_N
	SATA_100D	SATA	SATA_ODD_R2D_P
	SATA_100D	SATA	SATA_ODD_R2D_N
	SATA_100D	SATA	SATA_ODD_D2R_P
	SATA_100D	SATA	SATA_ODD_D2R_N
	SATA_100D	SATA	SATA_ODD_D2R_C_P
	SATA_100D	SATA	SATA_ODD_D2R_C_N
	MCP_50R	SATA_TERM	MCP_SATA_TERM
MISC			
	MCP_50R	MCP_PEX_COMP	MCP_PEX_CLK_COMP
	MCP_OV_COMP	MCP_PEX_COMP	MCP_IFPAB_RSET
	MCP_50R	MCP_PEX_COMP	MCP_IFPAB_VPROBE
			PM_SLP_S3_L
			PM_SLP_S4_L

SYNC MASTER=MASTER

SYNC DATE=N/A

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ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
	PCI_55S	PCI	PCI REQ0 L	19
	PCI_55S	PCI	PCI REQ1 L	19
	CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP R	19
	CLK_PCI_55S	CLK_PCI	PCI CLK33M MCP	19
	LPC_55S	LPC	LPC AD<3..0>	19 49 51
	LPC_55S	LPC	LPC AD R<3..0>	19
	LPC_55S	LPC	LPC FRAME L	19 49 51
	LPC_55S	LPC	LPC FRAME R L	19
	LPC_55S	LPC	LPC RESET L	9 19
	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC R	9 19
	CLK_LPC_55S	CLK_LPC	LPC CLK33M SMC	9 49
	CLK_LPC_55S	CLK_LPC	LPC CLK33M LPCPLUS	9 51
	CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK R	9 21
	CLK_LPC_55S	CLK_LPC	PM CLK32K SUSCLK	9 49
	MCP_USB_RB1AS	USB	MCP USB RB1AS_GND	20
	USB_90D	USB	USB EXTA P	20 46
	USB_90D	USB	USB EXTA N	20 46
	USB_90D	USB	USB PORT0 P	46
	USB_90D	USB	USB PORT0 N	46
	USB_90D	USB	USB EXTB P	20 46
	USB_90D	USB	USB EXTB N	20 46
	USB_90D	USB	USB PORT1 P	46
	USB_90D	USB	USB PORT1 N	46
	USB_90D	USB	USB EXTC P	20 46
	USB_90D	USB	USB EXTC N	20 46
	USB_90D	USB	USB PORT2 P	46
	USB_90D	USB	USB PORT2 N	46
	USB_90D	USB	USB EXTD P	20 46
	USB_90D	USB	USB EXTD N	20 46
	USB_90D	USB	USB D_MUXED P	46
	USB_90D	USB	USB D_MUXED N	46
	USB_90D	USB	USB PORT3 P	46
	USB_90D	USB	USB PORT3 N	46
	USB_90D	USB	USB CAMERA P	20 47
	USB_90D	USB	USB CAMERA N	20 47
	USB_90D	USB	USB CAMERA I_P	47 110
	USB_90D	USB	USB CAMERA I_N	47 110
	USB_90D	USB	USB BT P	20 47
	USB_90D	USB	USB BT N	20 47
	USB_90D	USB	USB BT L_P	47 110
	USB_90D	USB	USB BT L_N	47 110
	USB_90D	USB	USB IR_P	20 47
	USB_90D	USB	USB IR_N	20 47
	USB_90D	USB	USB IR L_P	47 110
	USB_90D	USB	USB IR L_N	47 110
	USB_90D	USB	USB SDCARD P	20 47
	USB_90D	USB	USB SDCARD N	20 47
	USB_90D	USB	USB SDCARD I_P	47 110
	USB_90D	USB	USB SDCARD I_N	47 110
	MCP_50S	SPI	SPI CLK R	21 51 61
	MCP_50S	SPI	SPI CLK	61
	MCP_50S	SPI	SPI MOSI R	21 51 61
	MCP_50S	SPI	SPI MOSI	61
	MCP_50S	SPI	SPI MISO	21 51 61
	MCP_50S	SPI	SPI MISO R	61
	MCP_50S	SPI	SPI CS0 R_L	21 51
	MCP_50S	SPI	SPI CS0 L	51
	HDA_55S	HDA	HDA BIT_CLK	21 62
	MCP_HDA_COMP	MCP_HDA_COMP	MCP_HDA_PULLDN_COMP	21
	HDA_55S	HDA	HDA BIT_CLK_R	21
	HDA_55S	HDA	HDA RST_L	21 62
	HDA_55S	HDA	HDA RST_R_L	21
	HDA_55S	HDA	HDA SDOUT	21 62
	HDA_55S	HDA	HDA SDOUT_R	21
	HDA_55S	HDA	HDA SYNC	21 62
	HDA_55S	HDA	HDA SYNC_R	21
	HDA_55S	HDA	HDA SDINO	21 62
	HDA_55S	HDA	AUD SDI_R	62
		HDA	AUD SPDIF_IN	9 66
		HDA	AUD SPDIF_OUT	62 66
		HDA	AUD SPDIF_CHIP	62
	CLK_MCP_XTAL	XTAL	MCP CLK25M_XTALOUT	21 28
	CLK_MCP_XTAL	XTAL	MCP CLK25M_XTALIN	21 28
	CLK_MCP_XTAL	XTAL	RTC CLK32K_XTALOUT	21 28
	CLK_MCP_XTAL	XTAL	RTC CLK32K_XTALIN	21 28

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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_MII_COMP	*	=STANDARD	0.2 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD
ENET_MII_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF0_CLK	*	=3:1_SPACING	?
ENET_MII	*	0.3 MM	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

RTL8211CLGR (ETHERNET PHY) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_MDI_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

NET_TYPE

ELECTRICAL_CONSTRAINT_SET		PHYSICAL	SPACING
	MCP_MII_COMP		MCP_MII_COMP_VDD
	MCP_MII_COMP		MCP_MII_COMP_GND
	ENET_MII_55S	MCP_BUF0_CLK	MCP_CLK25M_BUF0_R
	ENET_MII_55S	MCP_BUF0_CLK	RTL8211_CLK25M_CKXTAL1
	ENET_MII_55S	ENET_MII	ENET_MDIO
	ENET_MII_55S	ENET_MII	ENET_MDC
	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK
	ENET_MII_55S	ENET_MII	ENET_CLK125M_RXCLK_R
	ENET_MII_55S	ENET_MII	ENET_RXD<0>
	ENET_MII_55S	ENET_MII	ENET_RXD_R<0>
	ENET_MII_55S	ENET_MII	ENET_RXD<3..1>
	ENET_MII_55S	ENET_MII	ENET_RXD_R<3..1>
	ENET_MII_55S	ENET_MII	ENET_RX_CTRL
	ENET_MII_55S	ENET_MII	ENET_RXCTL_R
	ENET_MII_55S	ENET_MII	ENET_CLK125M_TXCLK
	ENET_MII_55S	ENET_MII	ENET_TXD<0>
	ENET_MII_55S	ENET_MII	ENET_TXD<3..1>
	ENET_MII_55S	ENET_MII	ENET_TX_CTRL
	ENET_MDI_100D	ENET_MDI	ENET_MDI_P<3..0>
	ENET_MDI_100D	ENET_MDI	ENET_MDI_N<3..0>
	ENET_MDI_100D	ENET_MDI	ENET_MDI_T_P<3..0>
	ENET_MDI_100D	ENET_MDI	ENET_MDI_T_N<3..0>

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ETHERNET CONSTRAINTS

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SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Section 2.10.1.

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x DIELECTRIC	?

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
	PHYSICAL	SPACING		
	SMB_55G	SMB	SMBUS_SMC_A_S3_SCL	52
	SMB_55G	SMB	SMBUS_SMC_A_S3_SDA	52
	SMB_55G	SMB	SMBUS_SMC_B_S0_SCL	52
	SMB_55G	SMB	SMBUS_SMC_B_S0_SDA	52
	SMB_55G	SMB	SMBUS_SMC_0_S0_SCL	52
	SMB_55G	SMB	SMBUS_SMC_0_S0_SDA	52
	SMB_55G	SMB	SMBUS_SMC_BSA_SCL	52
	SMB_55G	SMB	SMBUS_SMC_BSA_SDA	53
	SMB_55G	SMB	SMBUS_SMC_MGMT_SCL	52 106
	SMB_55G	SMB	SMBUS_SMC_MGMT_SDA	52 106
	SMB_55G	SMB	SMBUS_SMC_MGMT_SCL	52 106
	SMB_55G	SMB	SMBUS_SMC_MGMT_SDA	52 106
	SMB_55G	SMB	SMBUS_MCP_0_CLK	13 21 52
	SMB_55G	SMB	SMBUS_MCP_0_DATA	13 21 52

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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	0.08 MM	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
LVDS_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
MCP_DV_COMP	*	Y	0.5 MM	0.5 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?	DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?	LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface DG (DG-03328-001_v0D), Sections 2.5.3 & 2.5.4.

ELECTRICAL_CONSTRAINT_SET ASSIGNED IN CONT. MGR.	NET_TYPE			
	PHYSICAL	SPACING		
	DP_100D	DISPLAYPORT	MXM DP A ML P<3..0>	84 91
	DP_100D	DISPLAYPORT	DP IG ML P<3..0>	9 91
	DP_100D	DISPLAYPORT	MXM DP A ML N<3..0>	84 91
	DP_100D	DISPLAYPORT	DP IG ML N<3..0>	9 91
	DP_100D	DISPLAYPORT	DP ML P<3..0>	91 94
	DP_100D	DISPLAYPORT	DP ML N<3..0>	91 94
	DP_100D	DISPLAYPORT	DP ML CONN P<3..0>	94
	DP_100D	DISPLAYPORT	DP ML CONN N<3..0>	94
	DP_100D	DISPLAYPORT	DP IG AUX CH P	18 93
	DP_100D	DISPLAYPORT	DP IG AUX CH N	18 93
	DP_100D	DISPLAYPORT	DP AUXCH SW P	93
	DP_100D	DISPLAYPORT	DP AUXCH SW N	93
	DP_100D	DISPLAYPORT	DP AUX CH C P	93 94
	DP_100D	DISPLAYPORT	DP AUX CH C N	93 94
	DP_100D	DISPLAYPORT	MXM DP A AUX P	84 93
	DP_100D	DISPLAYPORT	MXM DP A AUX N	84 93
	LVDS_100D	LVDS	LVDS IG A CLK P	18 89
	LVDS_100D	LVDS	LVDS IG A CLK N	18 89
	LVDS_100D	LVDS	LVDS IG A DATA P<3..0>	18 89
	LVDS_100D	LVDS	LVDS IG A DATA N<3..0>	18 89
	LVDS_100D	LVDS	LVDS IG B CLK P	18 89
	LVDS_100D	LVDS	LVDS IG B CLK N	18 89
	LVDS_100D	LVDS	LVDS IG B DATA P<3..0>	18 89
	LVDS_100D	LVDS	LVDS IG B DATA N<3..0>	18 89
	LVDS_100D	LVDS	LVDS A DATA P<3..0>	89
	LVDS_100D	LVDS	LVDS A DATA N<3..0>	89
	LVDS_100D	LVDS	LVDS B DATA P<3..0>	89
	LVDS_100D	LVDS	LVDS B DATA N<3..0>	89
	LVDS_100D	LVDS	LCD CONN A DATA P<3..0>	89 90
	LVDS_100D	LVDS	LCD CONN A DATA N<3..0>	89 90
	LVDS_100D	LVDS	LCD CONN B DATA P<3..0>	89 90
	LVDS_100D	LVDS	LCD CONN B DATA N<3..0>	89 90
	LVDS_100D	LVDS	LVDS A CLK P	89
	LVDS_100D	LVDS	LVDS A CLK N	89
	LVDS_100D	LVDS	LVDS B CLK P	89
	LVDS_100D	LVDS	LVDS B CLK N	89
	LVDS_100D	LVDS	LCD CONN A CLK P	89 90
	LVDS_100D	LVDS	LCD CONN A CLK N	89 90
	LVDS_100D	LVDS	LCD CONN B CLK P	89 90
	LVDS_100D	LVDS	LCD CONN B CLK N	89 90
	MCP_DV_COMP		MCP_HDMI_RSET	18 26
	MCP_DV_COMP		MCP_HDMI_VPROBE	18 26

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GRAPHICS CONSTRAINTS

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SPACING _{min} RULE _{min} SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
PPDDR _{min} MEM	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DATA	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM
MEM_CLK	PDDDR_MEM	*	PWR_P2MM
MEM_CMD	PDDDR_MEM	*	PWR_P2MM
MEM_CTRL	PDDDR_MEM	*	PWR_P2MM
MEM_DATA	PDDDR_MEM	*	PWR_P2MM
MEM_DQS	PDDDR_MEM	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_FSB	GND	*	GND_P2MM
CPU_COMP	GND	*	GND_P2MM
CPU_OTLREF	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM
FSB_DSTB	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	PWR	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
SWITCHNODE	*	*	SWITCHNODE
THERMAL	PWR	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM
AUDIO	*	*	AUDIO

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S_VDD OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_70D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	600 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_90D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
USB_90D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MEM_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_MII_COMP OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_USB_RBIAS OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
MCP_DV_COMP OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.25 MM OVERRIDE	250 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

K50/K51 SPECIFIC NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	NET_TYPE		SPACING	
		FEDDR_MEM		=PPIV5_S3_MEM_A	6 30 31
		FEDDR_MEM		=PPIV5_S3_MEM_B	6 30 32
		SWITCHNODE		VR_CPU_SW1	72
		SWITCHNODE		VR_CPU_SW2	72
		SWITCHNODE		VR_CPU_SW3	72
		SWITCHNODE		1V8_SW	80
		SWITCHNODE		1V155_SW	79
		SWITCHNODE		PVTT50_PHASE	76
		SWITCHNODE		3V3S5_SW	76
		SWITCHNODE		5VS3_SW	73
		SWITCHNODE		MCPCORE0_PHASE	74
		THERM_DIFF	THERMAL	SNS_T_DP1_DN6	55
		THERM_DIFF	THERMAL	SNS_T_DN1_DP6	55
		THERM_DIFF	THERMAL	SNS_T_DP2_DN3	55
		THERM_DIFF	THERMAL	SNS_T_DN2_DP3	55
		THERM_DIFF	THERMAL	CPU_THERMD_P	11 55
		THERM_DIFF	THERMAL	CPU_THERMD_N	11 55
		THERM_DIFF	THERMAL	SNS_T_DP4_DN5	55
		THERM_DIFF	THERMAL	SNS_T_DN4_DP5	55
		THERM_DIFF	THERMAL	MCP_THMDIODE_P	21 55
		THERM_DIFF	THERMAL	MCP_THMDIODE_N	21 55
		THERM_DIFF	THERMAL	MMX_PWRSRC_SENSOR_P	53
		THERM_DIFF	THERMAL	MMX_PWRSRC_SENSOR_N	53
		THERM_DIFF	THERMAL	SENSE_1V5_S0_P	54
		THERM_DIFF	THERMAL	SENSE_1V5_S0_N	54
		THERM_DIFF	THERMAL	SNS_LCD_P	55 110
		THERM_DIFF	THERMAL	SNS_LCD_N	55 110
		THERM_DIFF	THERMAL	SNS_ODD_P	55 110
		THERM_DIFF	THERMAL	SNS_ODD_N	55 110
		THERM_DIFF	THERMAL	SNS_CPU_H_P	55
		THERM_DIFF	THERMAL	SNS_CPU_H_N	55
		THERM_DIFF	THERMAL	SNS_MCP_P	55
		THERM_DIFF	THERMAL	SNS_MCP_N	55
		THERM_DIFF	THERMAL	SNS_AMB_P	55 110
		THERM_DIFF	THERMAL	SNS_AMB_N	55 110
		THERM_DIFF	THERMAL	SNS_MXM_P	55
		THERM_DIFF	THERMAL	SNS_MXM_N	55
		SNS_DIFF	THERMAL	VR_CPU_ISNS1_P	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS1_N	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_P	71
		SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N	71
		SNS_DIFF	THERMAL	VR_CPU_ISNS2_P	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS2_N	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_P	71
		SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N	71
		SNS_DIFF	THERMAL	VR_CPU_ISNS3_P	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS3_N	71 72
		SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_P	71
		SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N	71
1240		THERMAL		SMC_CPU_ISENSE	49 53
1245		THERMAL		VR_CPU_IOUT	53 71
1247		THERMAL		VR_ISNS_CPU_P	53
1249		THERMAL		VR_ISNS_CPU_N	53
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1255		THERMAL		SMC_CPU_VSENSE	49 53
1257		THERMAL		CPU_VCC_SENSE	12 53
1259		THERMAL		SMC_GPU_VSENSE	49 53
1260		THERMAL		SMC_GPU_ISENSE	49 53
1265		THERMAL		SMC_1V5_S0_ISENSE	50 54
1266		THERMAL		SMC_1V5_S0_ISENSE_R	54
1267		THERMAL		SMC_1V5_S0_VSENSE	50 54
1267		THERMAL		SMC_MCP_CORE_ISENSE	50 54
1269		THERMAL		SMC_MCP_CORE_VSENSE	50 54
1280		THERMAL		MCPCORE0_IMON	54 74
1284		THERMAL		CPU_PECI_L	11 55
1285		THERMAL		SMB_PECI_L	55
1286		THERMAL		CPU_PECI_MCP	14 55
1290		THERMAL		HDD_OOB_TEMP_FILT	55
1293		THERMAL		HDD_OOB_TEMP	55
1294		THERMAL		HDD_OOB_TEMP_R	55
1296		THERMAL		SMC_HDD_OOB_TEMP	55

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K22/K23 SPECIFIC CONSTRAINTS			
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		SIZE D	
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K50/K51 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS						
BOARD LAYERS		BOARD AREAS		BOARD UNITS (MIL OF MM)	ALLEGRO VERSION	
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, BOTTOM		NO_TYPE, BGA, PINH		MM	15.5.1	

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.300 MM	0.085 MM	=STANDARD		
27P4_OHM_SE	*	Y	0.275 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.085 MM	*STANDARD		
40_OHM_SE	*	Y	0.15 MM	0.085 MM	*STANDARD	*STANDARD	*STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP,BOTTOM	Y	0.151 MM	0.085 MM	=STANDARD		
42_OHM_SE	*	Y	0.136 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3, ISL6	Y	0.155 MM	0.085 MM	=STANDARD	0.135 MM	0.1 MM
70_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	16L3,16L6	Y	0.099 MM	0.085 MM	12 MM	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	1SL3,1SL6	Y	0.081 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.085 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	*	POWER_WIDTH
VR_CTL_PHY	*	POWER_WIDTH

CONSTRAINTS ARE BASED ON MCP79 DESIGN GUIDE DG-03328-001_V06
PCI,LPC,SMB,HDA,SPI,RGMII,SMBUS ARE ROUTED AS 55 OHM SE SIGNALS

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SPACING_0.5MM	*	0.5 MM	?
CLK_SPACING_0.6MM	*	0.6 MM	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.6 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	0.2 MM	?
BGA_P3MM	*	0.3 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_FSB	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P1MM
FSB_DSTB	FSB_DSTB	BGA_P1MM	BGA_P1MM
CLK_LPC	*	BGA_P1MM	BGA_P1MM
CLK_PCI	*	BGA_P1MM	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MCP_FSB_COMP	*	BGA_P1MM	BGA_P2MM
MCP_MEM_COMP	*	BGA_P1MM	BGA_P2MM
MCP_PEX_COMP	*	BGA_P1MM	BGA_P2MM

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FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT							
J4700 USB CAMERA 103 47 USB_CAMERA_I_P FUNC_TEST=TRUE 103 47 USB_CAMERA_I_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4700 2 Ground Testpoints near J4700 J4750 USB CARD READER 103 47 USB_SDCARD_I_P FUNC_TEST=TRUE 103 47 USB_SDCARD_I_N FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4750 2 Ground Testpoints near J4750 J4720 USB BLUETOOTH 103 47 USB_BT_I_P FUNC_TEST=TRUE 103 47 USB_BT_I_N FUNC_TEST=TRUE 1 PP3V3_S3 Testpoint near J4720 2 Ground Testpoints near J4720 J4780 IR BOARD 103 47 USB_IR_I_P FUNC_TEST=TRUE 103 47 USB_IR_I_N FUNC_TEST=TRUE 1 PP5V_S3_REG Testpoint near J4780 2 Ground Testpoints near J4780 J4520 SATA ODD (HIGH SPEED) 102 45 SATA_ODD_R2D_P FUNC_TEST=TRUE 102 45 SATA_ODD_R2D_N FUNC_TEST=TRUE 102 45 SATA_ODD_D2R_C_N FUNC_TEST=TRUE 102 45 SATA_ODD_D2R_C_P FUNC_TEST=TRUE 49 45 SMC_ODD_DETECT FUNC_TEST=TRUE 1 PP5V_S0 Testpoint near J4520 5 Ground Testpoints near J4520 J4510 SATA HDD (HIGH SPEED) 102 45 SATA_HDD_R2D_P FUNC_TEST=TRUE 102 45 SATA_HDD_R2D_N FUNC_TEST=TRUE 102 45 SATA_HDD_D2R_C_N FUNC_TEST=TRUE 102 45 SATA_HDD_D2R_C_P FUNC_TEST=TRUE 3 Ground Testpoints near J4510							
J5520 ANALOG LCD TEMP SENSOR 108 55 SNS_LCD_P FUNC_TEST=TRUE 108 55 SNS_LCD_N FUNC_TEST=TRUE J5521 AMBIENT TEMP SENSOR 108 55 SNS_AMB_P FUNC_TEST=TRUE 108 55 SNS_AMB_N FUNC_TEST=TRUE J5551 ODD TEMP SENSOR 108 55 SNS_ODD_P FUNC_TEST=TRUE 108 55 SNS_ODD_N FUNC_TEST=TRUE J5600 ODD FAN 56 FAN_0_PWR_L FUNC_TEST=TRUE 56 FAN_TACH0_L FUNC_TEST=TRUE 56 PP12V_S0_FAN0_L FUNC_TEST=TRUE 56 FAN_0_GND FUNC_TEST=TRUE J5700 CPU FAN 57 FAN_2_PWR_L FUNC_TEST=TRUE 57 FAN_TACH2_L FUNC_TEST=TRUE 57 PP12V_S0_FAN2_L FUNC_TEST=TRUE 57 FAN_2_GND FUNC_TEST=TRUE J5601 HD FAN 56 FAN_1_PWR_L FUNC_TEST=TRUE 56 FAN_TACH1_L FUNC_TEST=TRUE 56 PP12V_S0_FAN1_L FUNC_TEST=TRUE 56 FAN_1_GND FUNC_TEST=TRUE							
J6601 AUDIO MICROPHONE 66 AUD_MIC_IN1_N_CONN FUNC_TEST=TRUE 66 GND_AUDIO_MIC1_CONN FUNC_TEST=TRUE 66 AUD_MIC_IN1_P_CONN FUNC_TEST=TRUE 1 Ground Testpoint near J6601 J6602 AUDIO RIGHT SPEAKER 66 AUD_SPKR_OUTLO2R_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO2R_N FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO1R_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO1R_N FUNC_TEST=TRUE J6603 AUDIO LEFT SPEAKER 66 AUD_SPKR_OUTLO2L_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO2L_N FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO1L_P FUNC_TEST=TRUE 66 AUD_SPKR_OUTLO1L_N FUNC_TEST=TRUE							
GND 16 TP16 FUNC_TEST=TRUE RIN_ALLOWED_TPS=16 78 6 PP3V3_S3 2 TP16 FUNC_TEST=TRUE RIN_ALLOWED_TPS=2 73 6 PP5V_S3_REG 2 TP16 FUNC_TEST=TRUE RIN_ALLOWED_TPS=2 78 6 PP5V_S0 FUNC_TEST=TRUE RIN_ALLOWED_TPS=1							

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